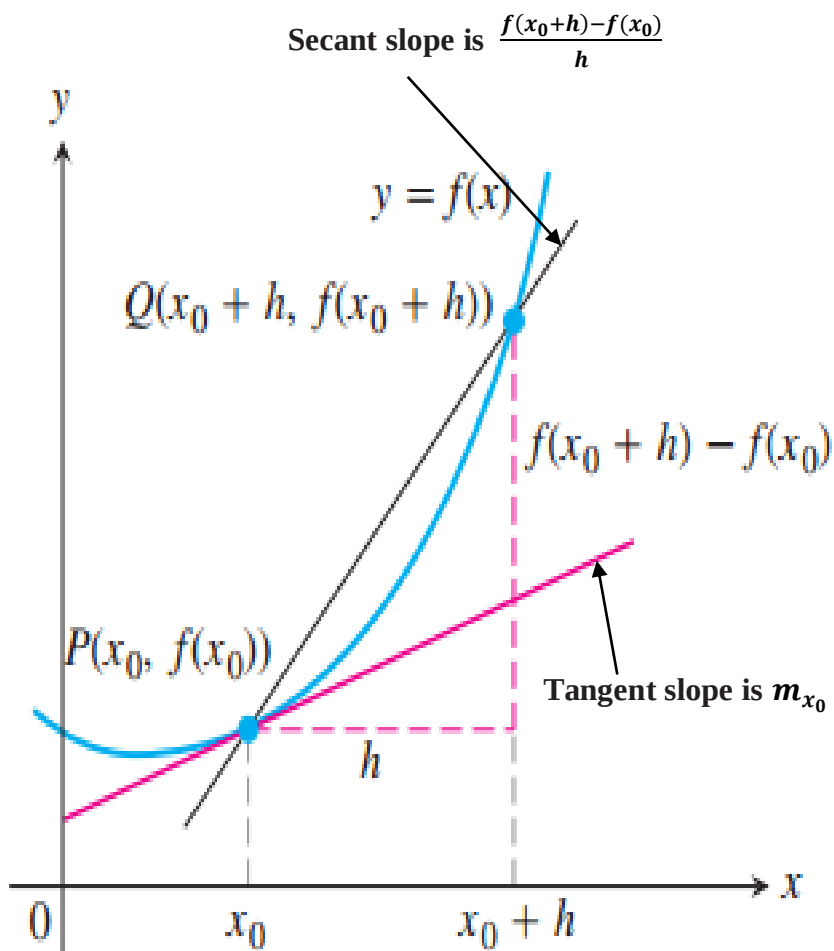




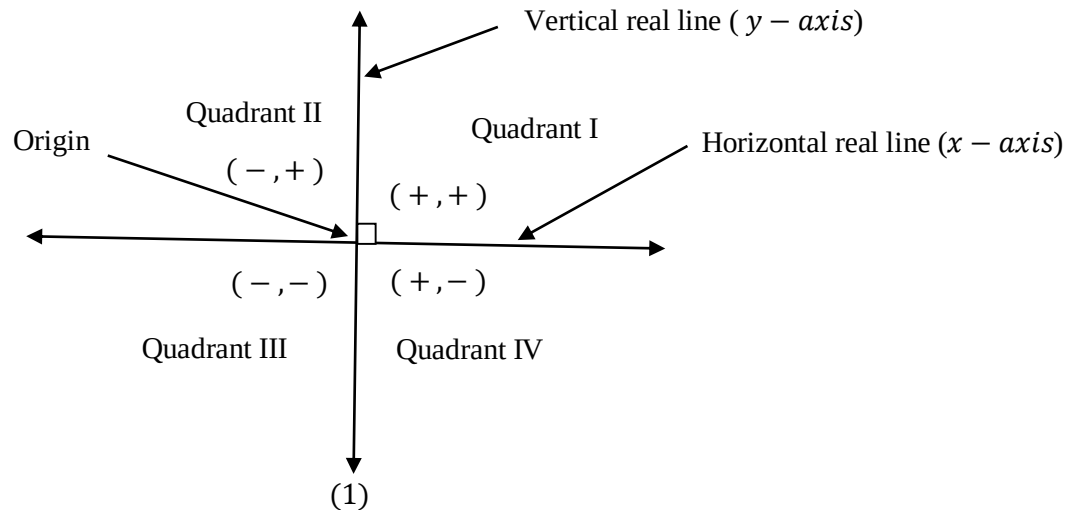
Differential Calculus



Lecturer: Mr. Mohammed Bafgeh & Mr. Rafat Bin Talib

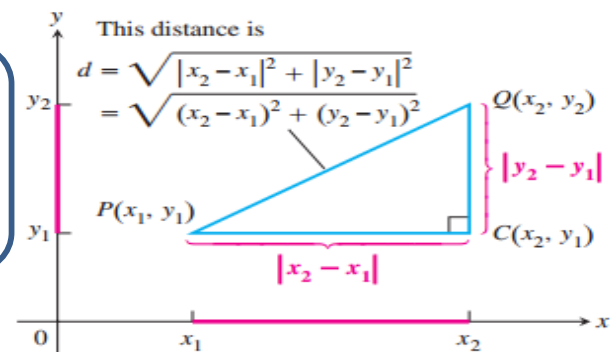
Chapter 1 : Introductory**Functions****The Cartesian plane**

The Cartesian plane is formed by using two real number lines intersection at right angles, as shown in Figure (1). The horizontal real number line is usually called the $x - axis$ and the vertical real number line is usually called the $y - axis$. The point of intersection of these two axes is the origin, and the two axes divide the plane into four parts called quadrants.

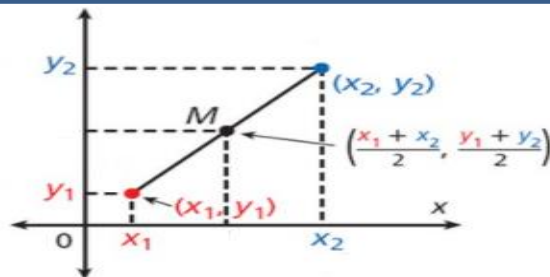
**Distance formula**

Suppose $p_1(x_1, y_1)$ and $p_2(x_2, y_2)$ are two points in the xy -plane or coordinate plane. Then the distance between p_1 and p_2

written $d(p_1, p_2)$ is $d(p_1, p_2) = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

**Midpoint formula**

The midpoint of the segment from $P_1(x_1, y_1)$ to $P_2(x_2, y_2)$ is $\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$



Example: Find the distance $d(A, B)$ and the coordinate of the midpoint of segment $\overline{AB}$ if

- 1) $A(-3, 2)$, $B(-1, -2)$, 2) $A(-7, -5)$, $B(5, 6)$

Solution:

$$1) d(A, B) = \sqrt{(-1 + 3)^2 + (-2 - 2)^2} = \sqrt{2^2 + (-4)^2} = \sqrt{4 + 16} = \sqrt{20} = 2\sqrt{5}$$

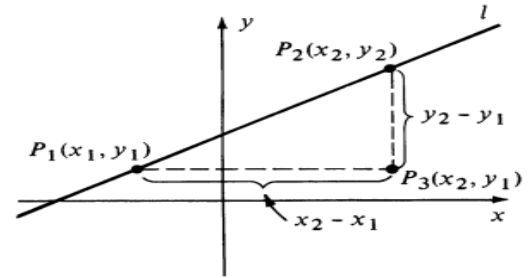
The midpoint of $\overline{AB} = \left(\frac{-3-1}{2}, \frac{2-2}{2}\right) = (-2, 0)$

$$2) \text{ Exercise (Ans: } d(A, B) = \sqrt{265} \text{ , The midpoint of } \overline{AB} \left(-1, \frac{1}{2}\right) \text{)}$$

Lines

Slope (m):

The slope of a non vertical line ($\vec{L}$) which passes through the points (x_1, y_1) and (x_2, y_2) is $m = \frac{y_2 - y_1}{x_2 - x_1}$ where $x_1 \neq x_2$



Notes:

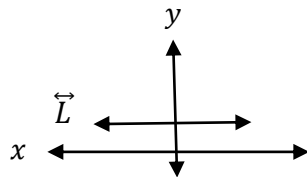
$$1) m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{y_1 - y_2}{x_1 - x_2} \neq \frac{y_2 - y_1}{x_1 - x_2} \neq \frac{y_1 - y_2}{x_2 - x_1}$$

Example : Find the slope of the line passes through the points $A(2, -4)$ and $B(1, 3)$

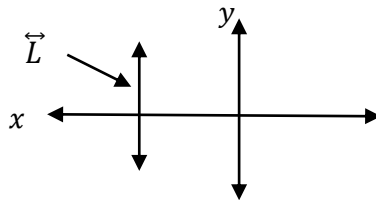
Solution :

$$\text{Since } m = \frac{y_2 - y_1}{x_2 - x_1}, \text{ then } m = \frac{3 - (-4)}{1 - 2} = \frac{7}{-1} = -7$$

2) The slope of any horizontal line $\vec{L}: y = b$ s.t. b is y -intercept equal to zero.

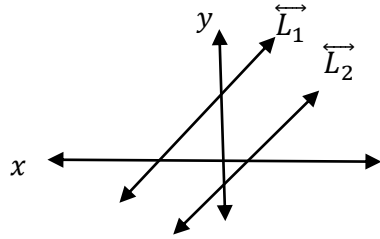


3) The slope of any vertical line $\vec{L}: x = a$ s.t. a is x -intercept undefined.

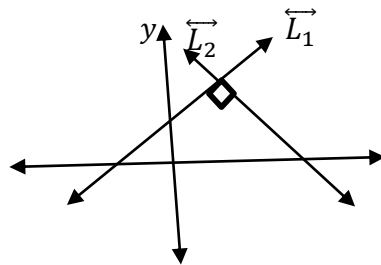


4) Let $\vec{L}_1, \vec{L}_2$ are two lines with slopes m_1, m_2 a respectively (على التوالي) then

a) $\vec{L}_1$ and $\vec{L}_2$ are parallel ($\vec{L}_1 // \vec{L}_2$) if and only if ($\Leftrightarrow$) $m_1 = m_2$.



b) $\vec{L}_1$ and $\vec{L}_2$ are perpendicular ($\vec{L}_1 \perp \vec{L}_2$) if and only if ($\Leftrightarrow$) $m_1 \cdot m_2 = -1$ or $m_1 = \frac{-1}{m_2}$.



Equations of a line

General form of the equation of a line:

The general equation or standard equation of straight line is given by

$$ax + by + c = 0 \quad , \quad \text{where } a, b, c \in \mathbb{R} \text{ and either } a \neq 0 \text{ or } b \neq 0. \dots \dots \dots (*)$$

Note

From (*) we get

- i. The slope : $\frac{-a}{b}$
- ii. y – intercept : $-\frac{c}{b}$
- iii. x – intercept : $-\frac{c}{a}$

Example : Find 1) The slope 2) y – intercept 3) x – intercept of the following equations

i) $5y - 3x = 7$ ii) $4x = 2y - 9$

Solution :

i) Since $5y - 3x = 7$, then $5y - 3x - 7 = 0$

$$a = -3, \quad b = 5, \quad c = -7$$

1) The slope $m = -\frac{a}{b} = \frac{3}{5}$ 2) y -intercept $= -\frac{c}{b} = \frac{7}{5}$ 3) x -intercept $= -\frac{c}{a} = -\frac{7}{3}$

ii) $4x = 2y - 9$ (Exercise)

How finding equation of the line

* There exist other forms to write an equation line.

1) The slope – intercept equation

An equation of the line with slope m and y -intercept b is

$$y = mx + b \quad \text{where } a, b, c \in \mathbb{R} \text{ and } a \neq 0$$

2) The point-slope equation

An equation of the line passes through the point (x_1, y_1) and has a slope m is

$$y = m(x - x_1) + y_1$$

Example: Find the equation of the line passes through the points $(1,7)$, $B(-3,2)$.

Solution:

Since the slope m of the line is $m = \frac{y_2 - y_1}{x_2 - x_1}$

Then we get $m = \frac{7-2}{1-(-3)} = \frac{5}{4}$

Now, we can use the coordinates of ether A or B in the points-slope equation . Using $A(1,7)$ we get

$$\begin{aligned} y &= \frac{5}{4}(x - 1) + 7 \\ \Rightarrow 4y &= 5x - 5 + 28 \\ \Rightarrow 4y &= 5x + 23 \\ \Rightarrow 5x - 4y + 23 &= 0 \end{aligned}$$

Example: Find the equation of a line passes through the points $A(5, -7)$ that is parallel to the line $6x + 3y = 4$.

Solution:

Let $\vec{L}_1 : 6x + 3y = 4$ and let $\vec{L}_2$: the required line

$$\text{Since } 3y = -6x + 4 \Rightarrow y = -2x + \frac{4}{3}$$

$$\Rightarrow m_1 = -2$$

$$\text{Since } \vec{L}_1 // \vec{L}_2 \Rightarrow m_2 = -2$$

Now using the point-slope equation $y = m(x - x_1) + y_1$ we get

$$y = -2(x - 5) - 7$$

$$\Rightarrow y = -2x + 3$$

$$\Rightarrow y = -2x + 3 \text{ or } 2x + y - 3 = 0$$

Exercise : Find the equation of a line passes through the points $(5, -7)$ that is perpendicular to the line $6x + 3y = 4$.

$$(\text{Ans : } 2y - x + 19 = 0)$$

Exercises

In exercises (1 – 3) find (a) the distance $d(A, B)$ between the points A and B , and (b) the midpoint of segment $\overline{AB}$

1) $A(6, -2), B(2, 1)$ (Ans: a) 5 b) $(4, -\frac{1}{2})$)

2) $A(0, -7), B(-1, -2)$ (Ans : a) $\sqrt{26}$, b) $(-\frac{1}{2}, -\frac{9}{2})$)

3) $A(-3, -2), B(-3, -2)$ (Ans : a) 0 , b) $(-3, -2)$)

4) Find the distance $d(A, B)$ of segment $\overline{AB}$ if $A(2, 3)$ and the midpoint between A and B is $(-1, 4)$

$$(Ans: d(A, B) = \sqrt{40} = 2\sqrt{10})$$

5) Prove that the triangle with vertices $A(-3, 4)$, $B(2, -1)$ and $C(9, 6)$ is a right triangle and find its area

(Hint: For any **triangle** with sides a, b, c, $a^2 + b^2 = c^2$, $area \text{ of right triangle} = \frac{1}{2} \times \text{base} \times \text{height}$)

$$(Ans : area = 35)$$

In exercises (6 – 10) find an equation for the line satisfying the given conditions.

6) Through $A(2, -6)$, $m = \frac{1}{2}$ (Ans : $x - 2y - 14 = 0$)

7) Through $A(-5, -7), B(3, -4)$ (Ans : $3x - 8y - 41 = 0$)

8) Through $A(8, -2)$, $y - \text{intercept} = 3$ (Ans : $x - 8y - 24 = 0$)

9) Through $A(10, -6)$, *parallel to* a) $y - \text{axis}$ b) $x - \text{axis}$ (Ans : a) $x = 10$ b) $y = -6$)

10) Through $A(7, -3)$ *perpendicular* to the line with equation $2x - 5y = 8$ (Ans : $2y = -5x + 29$)

In exercises(11 – 15) find the slope and y-intercept of the line with the given equation .

11) $3x - 4y + 8 = 0$ (Ans : $m = \frac{3}{4}$, $y - \text{intercept}: 2$)

12) $x + 2y = 0$ (Ans : $m = -\frac{1}{2}$, $y - \text{intercept}: 0$)

13) $y = 4$ (Ans : $m = 0$, $y - \text{intercept}: 4$)

14) $5x + 4y = 20$ (Ans : $m = -\frac{5}{4}$, $y - \text{intercept}: 5$)

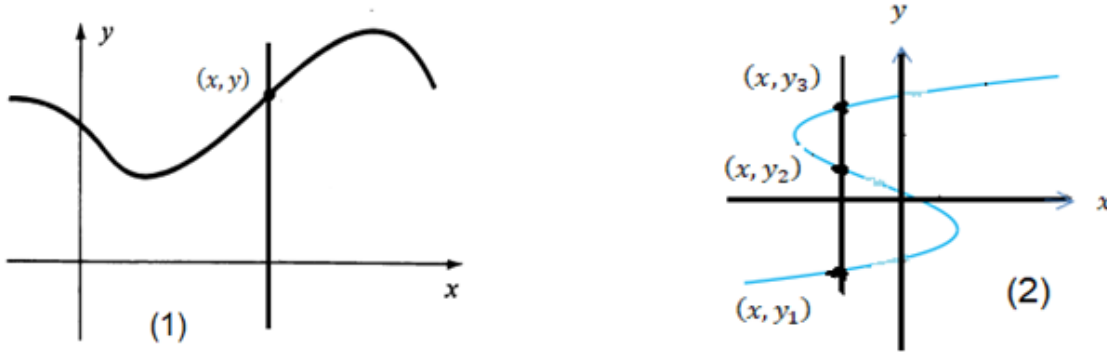
15) $x = 3y + 7$ (Ans : $m = \frac{1}{3}$, $y - \text{intercept}: -\frac{7}{3}$)

Functions

Definition: A set f of order pairs is said to be a function for a set X to a set Y if $f \subseteq X * Y$ and for each x in X , there exists a unique $y \in Y$ such that $(x, y) \in f$.

In sense that f is a relation from X to Y .

Thus, the graph in figure 1 represent a function but the graph in figure 2 cannot represent a function because more than order pair (x, y) have the first element x .



Note: Let f be a function form X to Y , if $(x, y) \in f$ then

- 1) y is called the image of x under f and denoted by $f(x)$.
- 2) The set X is called the domain of the function denoted by D_f .
- 3) The set Y is called the codomain of the function.
- 4) The all image of element of X called the range of the function.

Example : If $f(x) = 2x + 1$, $g(x) = \sqrt{3x^2 - 5}$ and $h(x) = \frac{7x-4}{x-1}$ find

- 1) $f(3)$ 2) $f(a + 1)$ 3) $g(-a)$ 4) $h(2)$ 5) $f(2) - 3$

Solution :

- 1) $f(3) = 2(3) + 1 = 6 + 1 = 7$
- 2) $f(a + 1) = 2(a + 1) + 1 = 2a + 2 + 1 = 2a + 3$
- 3) $g(-a) = \sqrt{3(-a)^2 - 5} = \sqrt{3a^2 - 5}$
- 4) $h(2) = \frac{7(2)-4}{2-1} = \frac{14-4}{1} = 10$
- 5) $f(2) - 3$ Exercise (Ans : 2)

Remarks:

- a. If the domain of the function is not given , it will be assumed to be the largest possible set of the real numbers. Then $D_f = \{x \in \mathbb{R} : f(x) \in \mathbb{R}\}$ thus to find the Domain of given function we must keep the following two restrictions in mind
- The denominator $\neq 0$.
 - The quantity under to the n^{th} root ($\sqrt[n]{\quad}$) must be greater than or equal to zero such that n is an even integer.
- b. If the function has not denominator and the n^{th} root ($\sqrt[n]{\quad}$) such that n is an even integer , then the domain equal the all real numbers ($D_f = \mathbb{R}$)

Example: Find the domain of the following functions

$$1) f(x) = -6 \quad 2) f(x) = x^2 \quad 3) f(x) = 2x^3 + 6x - 1 \quad 4) f(x) = \sqrt[3]{x^2 - 3x + 2} \quad 5) f(x) = \sqrt{2x - 7}$$

$$6) f(x) = \sqrt{-x^2 - 2x + 3} \quad 7) f(x) = \frac{2x - 1}{x^2 - 2x} \quad 8) f(x) = \sqrt{x^2 - 4} + \sqrt{2x + 6}$$

Solution :

$$1) f(x) = -6 \quad D_f = \mathbb{R} \quad \dots \dots \dots \text{Why ?}$$

$$2) f(x) = x^2 \quad D_f = \mathbb{R} \quad \dots \dots \dots \text{Why ?}$$

$$3) f(x) = 2x^3 + 6x - 1 \quad D_f = \mathbb{R} \quad \dots \dots \dots \text{Why ?}$$

Note: Any a polynomial function as following :

$$f(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0 : a_n, a_{n-1}, \dots, a_1, a_0 \in \mathbb{R}, n \in \mathbb{Z}^+ \text{ and } a_n \neq 0 \text{ Then } D_f = \mathbb{R}.$$

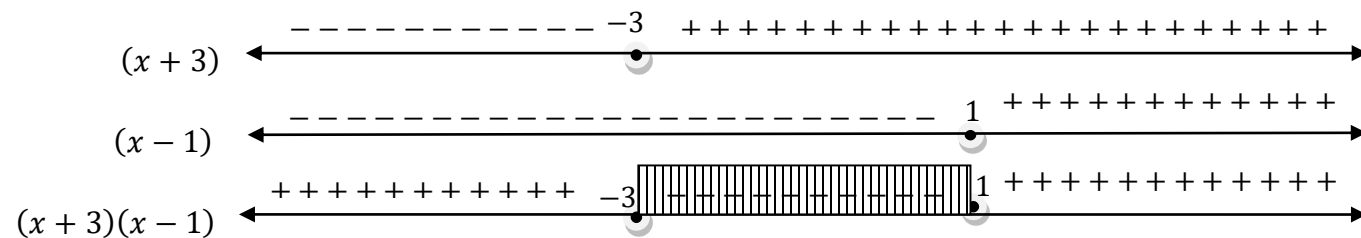
$$4) f(x) = \sqrt[3]{x^2 - 3x + 2} \quad D_f = \mathbb{R} \quad \dots \dots \dots \text{Why ?}$$

$$5) f(x) = \sqrt{2x - 7}$$

$$\text{Let } 2x - 7 \geq 0 \Rightarrow 2x \geq 7 \Rightarrow x \geq \frac{7}{2} \Rightarrow D_f = \left[\frac{7}{2}, \infty \right)$$

$$6) f(x) = \sqrt{-x^2 - 2x + 3}$$

$$\text{To find the domain let } -x^2 - 2x + 3 \geq 0 \Rightarrow x^2 + 2x - 3 \leq 0 \Rightarrow (x + 3)(x - 1) \leq 0$$



$$\Rightarrow D_f = [-3, 1]$$

$$7) f(x) = \frac{2x-1}{x^2-2x}$$

$$\text{Let } x^2 - 2x = 0 \Rightarrow x(x-2) = 0$$

$$\Rightarrow \text{either } x = 0 \text{ or } x - 2 = 0 \Rightarrow x = 2$$

$$\text{Finally, } D_f = \mathbb{R} \setminus \{0, 2\}$$

Note :

If $f(x) = \frac{P_1}{P_2}$, Where P_1 and P_2 are two polynomials, then $D_f = \mathbb{R} \setminus \{P_2 \neq 0\}$

$$8) f(x) = \sqrt{x^2 - 4} + \sqrt{2x + 6}$$

First, we will find the domain of $(\sqrt{x^2 - 4})$ thus let

$$x^2 - 4 \geq 0$$

$$\Rightarrow x^2 \geq 4$$

$$\Rightarrow |x| \geq 2 \Rightarrow x \geq 2 \text{ or } x \leq -2$$

Then the domain of $(\sqrt{x^2 - 4}) = (-\infty, -2] \cup [2, \infty)$

Second, we will find the domain of $(\sqrt{2x + 6})$ thus let

$$2x + 6 \geq 0$$

$$\Rightarrow 2x \geq -6 \Rightarrow x \geq -3$$

Then the domain of $(\sqrt{2x + 6}) = [-3, \infty)$

Finally, $D_f = ((-\infty, -2] \cup [2, \infty)) \cap [-3, \infty)$

$$= [-3, -2] \cup [2, \infty)$$

Note: The Domain of

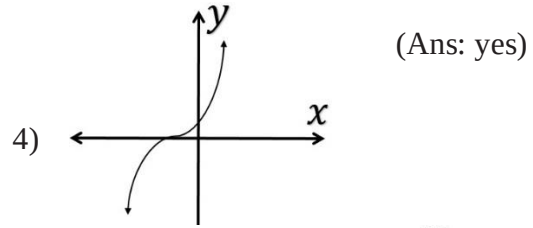
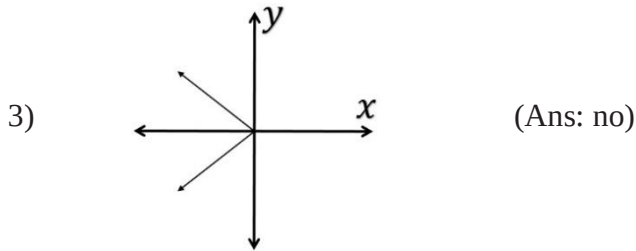
$$1) f(x) \pm g(x) \quad , \quad 2) \frac{f(x)}{g(x)} : g(x) \neq 0 \quad , \quad 3) f(x) \cdot g(x) \text{ is } D_f \cap D_g$$

Exercises

In exercises (1- 4) decide whether the following relations is a function:

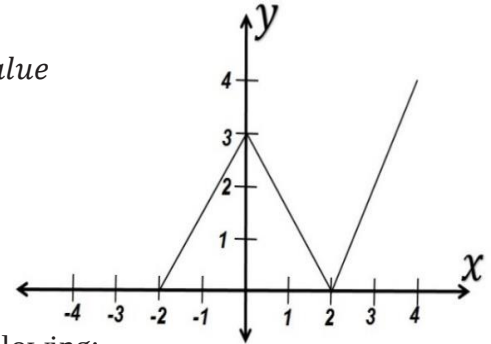
1) $\{(x, y) | 2x^2 = y + 3\}$ (Ans: yes)

2) $\{(x, y) | x^2 + y^2 = 4\}$ (Ans: no)



5) for of the following graph (a) find $f(-2)$, $f(0)$ and $f(4)$, (b) find the x - value that correspond to the following function values: 3, 0 and 2

(Ans: $\left\{ \begin{array}{l} (a) f(-2) = 0, f(0) = 3, f(4) = 4 \\ (b) \{0, 3.5\}, \{-2, 2\}, \{-1, 1, 3\} \end{array} \right.$)



6) let $f(x) = 3 - 2x^2$, $g(x) = \sqrt{4x - 2}$ and $h(x) = \frac{x}{2-x}$. Find each of the following:

(a) $h(3)$ (b) $g(5)$ (c) $f(4) - 5$ (d) $[g(1)]^2$

(e) $f\left(\frac{1}{a}\right)$ (f) $\frac{1}{f(a)}$ (g) $\frac{f(5a)}{g(a)}$

(Ans: (a) -3 , (b) $\sqrt{18}$, (c) -34 , (d) 2 , (e) $3 - \frac{2}{a^2}$, (f) $\frac{1}{3-2a^2}$, (g) $\frac{3-50a^2}{\sqrt{4a-2}}$)

In exercises (7-12) find the domain of the given function:

7) $f(x) = 2x - 7$ (Ans : $\mathbb{R}$)

8) $f(x) = \sqrt[5]{x^2 - 5x + 6}$ (Ans : $\mathbb{R}$)

9) $f(x) = \frac{x}{x-2}$ (Ans: $\mathbb{R} \setminus \{2\}$)

10) $f(x) = \frac{1}{x^2-1}$ (Ans: $\mathbb{R} \setminus \{\pm 1\}$)

11) $f(x) = \sqrt{x^2 + x - 6}$ (Ans: $(-\infty, -3] \cup [2, \infty)$)

12) $f(x) = \sqrt{\frac{x-1}{x}}$ (Ans: $(-\infty, 0) \cup [1, \infty)$)

13) $f(x) = \frac{x}{\sqrt{x^2-2x-3}}$ (Ans: $(-\infty, -1) \cup (3, \infty)$)

14) $f(x) = \sqrt{9 - x^2} - \sqrt{2x + 4}$ (Ans : $[-2, 3]$)

15) $f(x) = \frac{5}{3-\sqrt{x-1}}$ (Ans: $[1, \infty) \setminus \{10\}$)

16) $f(x) = \frac{\sqrt{2x+6}}{x^2-16}$ (Ans : $[-3, \infty) \setminus \{4\}$)

Chapter 2 : Limits of functionsThe LimitsIntroduction:

The notation of a limit is fundamental concept of calculus in interval University. You will learn how to evaluate limits and how they are used in the basic problems of calculus:

- 1) the tangent line problem (The derivative)
- 2) the area problem (Integral)

Discussion of the idea of limit

Let $f(x)$ be a function defined on open interval containing areal number 'a' except possibly at 'a' itself we may ask the following questions :

- 1) As x gets closer to 'a' (but $x \neq a$) , does the function value $f(x)$ get closer to some real number L ?
- 2) Can we make the function value $f(x)$ as closer L as desired by choosing x sufficiently close to 'a' (but $x \neq a$) ?

If the answers to the previous questions are yes we write $\lim_{x \rightarrow a} f(x) = L$ (we say the limit of $f(x)$ as approaches to 'a' equals 'L')

Illustration

Let $f(x) = 3x - 2$. What happens to $f(x)$ as x is chosen closer and closer to 2 ?

First we will construct a table that shows values of $f(x)$ for two sets of x -values . One set that approaches 2 from **the left** and one set that approaches 2 from **the right** .

		The left set					The right set				
x	...	1.9	1.99	1.999	1.9999	$\dots \rightarrow 2 \leftarrow \dots$	2.0001	2.001	2.01	2.1	...
$f(x)$	...	3.7000	3.9700	3.9970	3.9997	$\dots \rightarrow 4 \leftarrow \dots$	4.0003	4.003	4.030	4.300	...

From the table , it approaches that the closer x gets to 2 , the closer $f(x)$ gets to 4 .
be 4 . Figure (1) adds further support to this conclusion. Then $\lim_{x \rightarrow 2} (3x - 2) = 4$

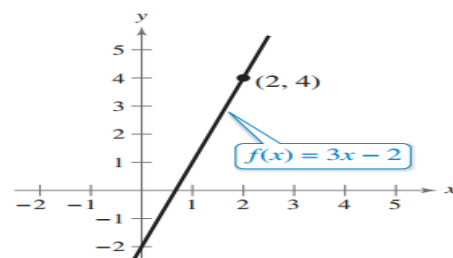


Figure (1)

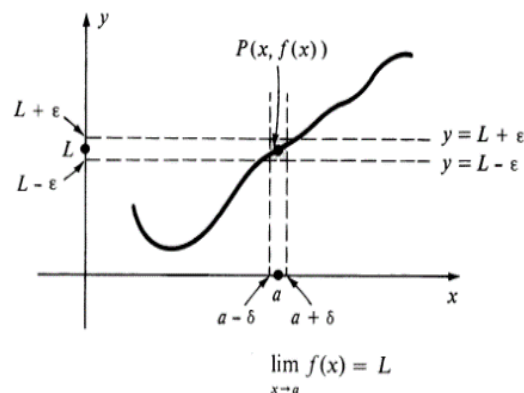
Exercise : Discuss the limit of the function $f(x) = \frac{\sin x}{x}$ as x gets closer to zero

Using the table (ans : 1)

Definition of the limit

: Let $f(x)$ be a function that is defined one open interval containing 'a' except possibly at 'a' itself, and let 'L' be a real number . Then statement $\lim_{x \rightarrow a} f(x) = L$ means that

$\forall \epsilon > 0, \exists \delta > 0$ such that
If $0 < |x - a| < \delta$ then $|f(x) - L| < \epsilon$



Example: Prove that $\lim_{x \rightarrow 4} \frac{1}{2}(3x - 1) = \frac{11}{2}$ by the definition.

Solution:

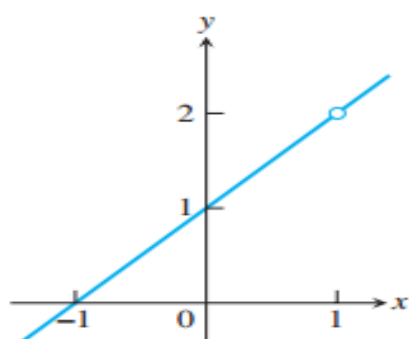
Let $f(x) = \frac{1}{2}(3x - 1)$, $a = 4$ and $L = \frac{11}{2}$

$$\begin{aligned} \text{As } |f(x) - L| < \varepsilon &\Leftrightarrow \left| \frac{1}{2}(3x - 1) - \frac{11}{2} \right| < \varepsilon \\ &\Leftrightarrow \left| \frac{3}{2}x - \frac{12}{2} \right| < \varepsilon \\ &\Leftrightarrow \frac{3}{2}|x - 4| < \varepsilon \\ &\Leftrightarrow |x - 4| < \frac{2\varepsilon}{3} \end{aligned}$$

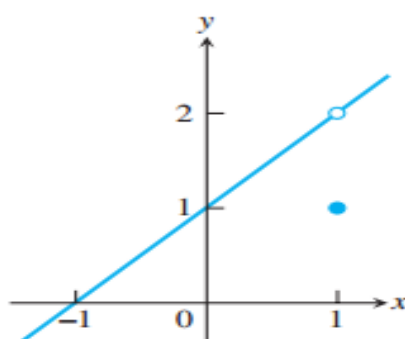
Thus , we can take $\delta = \frac{2\varepsilon}{3}$, then if $0 < |x - 4| < \delta$, the last inequality in the list is true and consequently so is the first. Hence by Definition $\lim_{x \rightarrow 4} \frac{1}{2}(3x - 1) = \frac{11}{2}$.

Note : In the previous example we can take $\delta = \frac{2\varepsilon}{4}$ or $\delta = \frac{2\varepsilon}{7}$ or ... provided $\delta \leq \frac{2\varepsilon}{3}$

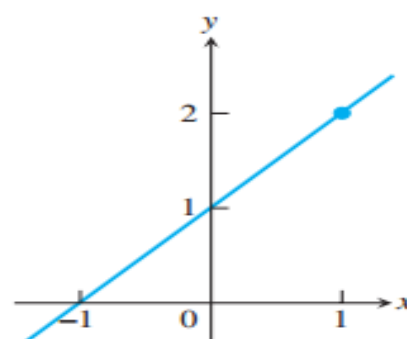
Example: From the following graphs , find (if they exist) 1) $\lim_{x \rightarrow 1} f(x)$ 2) $f(1)$



(a) $f(x) = \frac{x^2 - 1}{x - 1}$



(b) $g(x) = \begin{cases} \frac{x^2 - 1}{x - 1}, & x \neq 1 \\ 1, & x = 1 \end{cases}$



(c) $h(x) = x + 1$

Solution:

a) 1) $\lim_{x \rightarrow 1} f(x) = 2$ and $f(1)$ is undefined

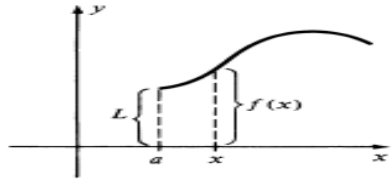
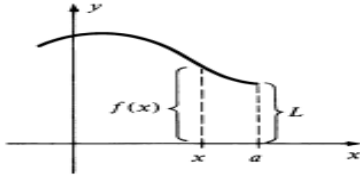
b) 1) $\lim_{x \rightarrow 1} f(x) = 2$ and $f(1) = 1$

c) 1) $\lim_{x \rightarrow 1} f(x) = 2$ and $f(1) = 2$

Note :

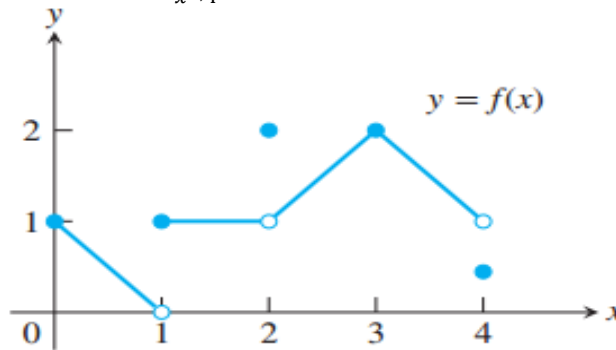
The limit value does not depend on how the function is defined at the same point .

One sided limits

Notation	Intuitive meaning	Graph
$\lim_{x \rightarrow a^+} f(x) = L$ Right - hand limit	We can make $f(x)$ as closer to 'L' as desired by choosing x sufficiently close to 'a', and $x > a$	 (i) $\lim_{x \rightarrow a^+} f(x) = L$
$\lim_{x \rightarrow a^-} f(x) = L$ Left - hand limit	We can make $f(x)$ as closer to 'L' as desired by choosing x sufficiently close to 'a', and $x < a$	 (ii) $\lim_{x \rightarrow a^-} f(x) = L$

Example : From the graph find the following (if they exist).

- 1) $\lim_{x \rightarrow 0^-} f(x)$ 2) $\lim_{x \rightarrow 0^+} f(x)$ 3) $f(0)$ 4) $\lim_{x \rightarrow 1^-} f(x)$ 5) $\lim_{x \rightarrow 1^+} f(x)$ 6) $f(1)$ 7) $\lim_{x \rightarrow 2} f(x)$ 8) $f(2)$
9) $\lim_{x \rightarrow 3} f(x)$ 10) $f(3)$ 11) $f(4)$ 12) $\lim_{x \rightarrow 4} f(x)$



Solution :

- 1) $\lim_{x \rightarrow 0^-} f(x)$ does not exist because $f(x)$ is undefined if $x < 0$ 2) $\lim_{x \rightarrow 0^+} f(x) = 1$ 3) $f(0) = 1$
4) $\lim_{x \rightarrow 1^-} f(x) = 0$ 5) $\lim_{x \rightarrow 1^+} f(x) = 1$ 6) $f(1) = 1$
7) $\lim_{x \rightarrow 2} f(x) = 1$ because $\lim_{x \rightarrow 2^-} f(x) = 1 = \lim_{x \rightarrow 2^+} f(x)$ 8) $f(2) = 2$
9) $\lim_{x \rightarrow 3} f(x) = 2$ because $\lim_{x \rightarrow 3^-} f(x) = 2 = \lim_{x \rightarrow 3^+} f(x)$ 10) $f(3) = 2$
11) $f(4) = \frac{1}{2}$
12) $\lim_{x \rightarrow 4} f(x)$ does not exist ... why ?

Theorem :

$$\lim_{x \rightarrow a} f(x) = L \text{ if and only if } \lim_{x \rightarrow a^+} f(x) = \lim_{x \rightarrow a^-} f(x) = L$$

Example : From the graph find the following (if they exist).

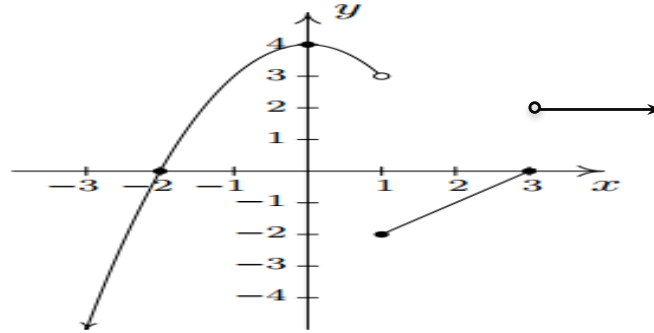
1) $\lim_{x \rightarrow 1^-} f(x)$

2) $\lim_{x \rightarrow 1^+} f(x)$

3) $\lim_{x \rightarrow 1} f(x)$

4) $\lim_{x \rightarrow 3} f(x)$

Solution :



1) $\lim_{x \rightarrow 1^-} f(x) = 3$

2) $\lim_{x \rightarrow 1^+} f(x) = -2$

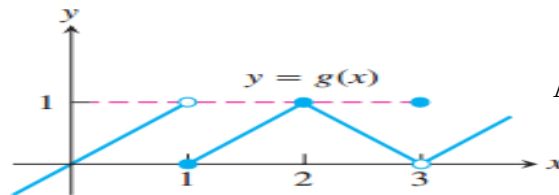
3) $\lim_{x \rightarrow 1} f(x)$ does not exist ... why? 4) Exercise

Example : For the function $g(x)$ graphed here, find the following

a. $\lim_{x \rightarrow 1} g(x)$

b. $\lim_{x \rightarrow 2} g(x)$

c. $\lim_{x \rightarrow 3} g(x)$



Ans {
a. does not exist
b. 1
c. 0

Techniques for finding Limits

Basic Limits:

Let $a, b \in \mathbb{R}$ and $n \in \mathbb{Z}^+$, then

1) $\lim_{x \rightarrow a} b = b$, 2) $\lim_{x \rightarrow a} x = a$, 3) $\lim_{x \rightarrow a} x^n = a^n$

4) $\lim_{x \rightarrow a} \sqrt[n]{x} = \sqrt[n]{a}$ provided either n is an odd or n is even and $a > 0$.

Example: Find 1) $\lim_{x \rightarrow 4} 5$, 2) $\lim_{x \rightarrow -3} \sqrt[5]{x}$, 3) $\lim_{x \rightarrow 4} \sqrt{x}$ 4) 3) $\lim_{x \rightarrow -4} \sqrt{x}$

Solution:

1) $\lim_{x \rightarrow 4} 5 = 5$

2) $\lim_{x \rightarrow -3} \sqrt[5]{x} = \sqrt[5]{-3}$

3) $\lim_{x \rightarrow 4} \sqrt{x} = \sqrt{4} = 2$

3) $\lim_{x \rightarrow 4} \sqrt{x}$ does not exist ... why?

Properties of the Limits

Let $a, b \in \mathbb{R}$, and $n \in \mathbb{Z}^+$ and $\lim_{x \rightarrow a} f(x) = L$, $\lim_{x \rightarrow a} g(x) = K$

$$1) \lim_{x \rightarrow a} [bf(x)] = bL$$

$$2) \lim_{x \rightarrow a} [f(x) \pm g(x)] = L \pm K$$

$$3) \lim_{x \rightarrow a} [f(x) \cdot g(x)] = L \cdot K$$

$$4) \lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \frac{L}{K} \text{ provided } K \neq 0$$

$$5) \lim_{x \rightarrow a} [f(x)]^n = L^n$$

Limits of Polynomial and Rational functions

1- If p is a polynomial function and $a \in \mathbb{R}$ then

$$\lim_{x \rightarrow a} p(x) = p(a)$$

2- If $r(x)$ is a rational function given by $r(x) = \frac{p(x)}{q(x)}$ and $a \in \mathbb{R}$, then

$$\lim_{x \rightarrow a} r(x) = \lim_{x \rightarrow a} \frac{p(x)}{q(x)} = \frac{p(a)}{q(a)}, \text{ provided } q(a) \neq 0$$

Example: Find each limit (if they exist)

$$1) \lim_{x \rightarrow -1} (x^2 + x - 6) \quad , \quad 2) \lim_{x \rightarrow 2} \frac{x^2 + x - 6}{x + 3}$$

Solution:

$$1) \lim_{x \rightarrow -1} (x^2 + x - 6) = (-1)^2 + (-1) - 6 = -6$$

$$2) \lim_{x \rightarrow 2} \frac{x^2 + x - 6}{x + 3} = \frac{(2)^2 + (2) - 6}{(2) + 3} = \frac{0}{5} = 0$$

Example: Find the following Limits (if they exist)

$$1) \lim_{x \rightarrow 2} \begin{cases} 2x + 1, & x < 2 \\ x^2 + 1, & x > 2 \end{cases} \quad , \quad 2) \lim_{x \rightarrow 3} \begin{cases} 2x + 3, & x < 0 \\ x^2 - 3x + 1, & 0 \leq x < 1 \\ \sqrt{3x + 7}, & x \geq 1 \end{cases}$$

Solution:

$$1) \text{ Since } \lim_{x \rightarrow 2^+} \begin{cases} 2x + 1, & x < 2 \\ x^2 + 1, & x > 2 \end{cases} = (2)^2 + 1 = 5 \text{ and } \lim_{x \rightarrow 2^-} \begin{cases} 2x + 1, & x < 2 \\ x^2 + 1, & x > 2 \end{cases} = 2(2) + 1 = 5$$

$$\text{Then } \lim_{x \rightarrow 2} \begin{cases} 2x + 1, & x < 2 \\ x^2 + 1, & x > 2 \end{cases} = 5$$

2) Exercise (Ans : 4)

Indeterminate Limit

Let $\lim_{x \rightarrow a} f(x) = 0$ and $\lim_{x \rightarrow a} g(x) = 0$, the situation $\lim_{x \rightarrow a} \frac{f(x)}{g(x)}$ called indeterminate Limit (zero over zero) .

In this case it takes some work on our part to find the limit.

Example: Evaluate the following limits (if they exist)

$$1) \lim_{x \rightarrow 1} \frac{x^2+x-2}{x^2-x} \quad 2) \lim_{x \rightarrow 0} \frac{\sqrt{x^2+100}-10}{x^2} \quad 3) \lim_{x \rightarrow 3} \frac{x^3-27}{x-3} \quad 4) \lim_{x \rightarrow 3} \frac{|x-3|}{x-3} \quad 5) \lim_{x \rightarrow 4} \frac{\sqrt{x-3}-1}{x^2-16}$$

Solution:

$$1) \lim_{x \rightarrow 1} \frac{x^2+x-2}{x^2-x} = \frac{0}{0} \text{ (I . F)}$$

$$\text{Thus } \lim_{x \rightarrow 1} \frac{x^2+x-2}{x^2-x} = \lim_{x \rightarrow 1} \frac{(x-1)(x+2)}{x} = \lim_{x \rightarrow 1} \frac{x+2}{x} = 3$$

$$2) \lim_{x \rightarrow 0} \frac{\sqrt{x^2+100}-10}{x^2} = \frac{0}{0} \text{ (I . F)}$$

$$\text{thus } \lim_{x \rightarrow 0} \frac{\sqrt{x^2+100}-10}{x^2} = \lim_{x \rightarrow 0} \frac{\sqrt{x^2+100}-10}{x^2} \cdot \frac{\sqrt{x^2+100}+10}{\sqrt{x^2+100}+10}$$

$$= \lim_{x \rightarrow 0} \frac{x^2+100-100}{x^2\sqrt{x^2+100}+10}$$

$$= \lim_{x \rightarrow 0} \frac{x^2}{x^2\sqrt{x^2+100}+10} = \lim_{x \rightarrow 0} \frac{1}{\sqrt{x^2+100}+10} = \frac{1}{20}$$

$$3) \lim_{x \rightarrow 3} \frac{x^3-27}{x-3} = \frac{0}{0} \text{ (I . F)}$$

$$\begin{aligned} \text{Thus } \lim_{x \rightarrow 3} \frac{x^3-27}{x-3} &= \lim_{x \rightarrow 3} \frac{(x-3)(x^2+3x+9)}{x-3} \\ &= \lim_{x \rightarrow 3} (x^2+3x+9) \\ &= 27 \end{aligned}$$

$$4) \lim_{x \rightarrow 3} \frac{|x-3|}{x-3} = \frac{0}{0} \text{ (I . F)}$$

$$\text{Since } |x-3| = \begin{cases} x-3, & x \geq 3 \\ x-3, & x < 3 \end{cases} \quad , \quad \text{then } \frac{|x-3|}{x-3} = \begin{cases} 1, & x \geq 3 \\ -1, & x < 3 \end{cases}$$

$$\Rightarrow \lim_{x \rightarrow 3^+} \frac{|x-3|}{x-3} = \lim_{x \rightarrow 3^+} 1 = 1 \text{ and } \lim_{x \rightarrow 3^-} \frac{|x-3|}{x-3} = \lim_{x \rightarrow 3^-} (-1) = -1 \Rightarrow \lim_{x \rightarrow 3} \frac{|x-3|}{x-3} \text{ does not exist}$$

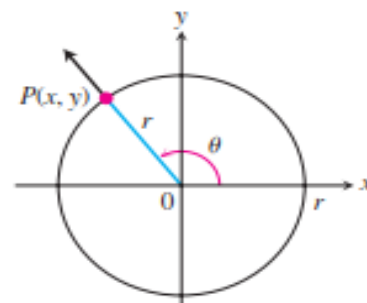
$$5) \lim_{x \rightarrow 4} \frac{\sqrt{x-3}-1}{x^2-16} \quad \text{Exercise (Ans: } \frac{1}{16} \text{)}$$

Limits of trigonometric functions

Definition of trigonometric functions: Let (x, y) be a point other than the origin on the terminal side of an angle θ in standard position. Let r be the distance the origin to (x, y) . Then the trigonometric function of θ are defined as follows (Assume no denominators are zero)

$$\sin \theta = \frac{y}{r} \quad , \quad \cos \theta = \frac{x}{r} \quad , \quad \tan \theta = \frac{y}{x}$$

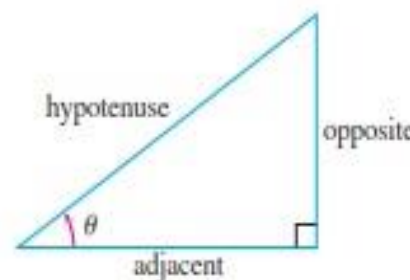
$$\cot \theta = \frac{x}{y} \quad , \quad \sec \theta = \frac{r}{x} \quad , \quad \csc \theta = \frac{r}{y}$$

**Remark:**

If θ is an acute angle of a right triangle, then

$$\sin \theta = \frac{\text{opp}}{\text{hyp}} \quad , \quad \cos \theta = \frac{\text{adj}}{\text{hyp}} \quad , \quad \tan \theta = \frac{\text{opp}}{\text{adj}}$$

$$\cot \theta = \frac{\text{adj}}{\text{opp}} \quad , \quad \sec \theta = \frac{\text{hyp}}{\text{adj}} \quad , \quad \csc \theta = \frac{\text{hyp}}{\text{opp}}$$

**Fundamental Trigonometric Identities****Reciprocal Identities**

$$\sin \theta = \frac{1}{\csc \theta} \quad , \quad \cos \theta = \frac{1}{\sec \theta} \quad , \quad \tan \theta = \frac{1}{\cot \theta}$$

$$\csc \theta = \frac{1}{\sin \theta} \quad , \quad \sec \theta = \frac{1}{\cos \theta} \quad , \quad \cot \theta = \frac{1}{\tan \theta}$$

Quotient Identities

$$\tan \theta = \frac{\sin \theta}{\cos \theta} \quad , \quad \cot \theta = \frac{\cos \theta}{\sin \theta}$$

Pythagorean Identities

$$\sin^2 \theta + \cos^2 \theta = 1 \quad , \quad 1 + \tan^2 \theta = \sec^2 \theta \quad , \quad 1 + \cot^2 \theta = \csc^2 \theta \quad .$$

Double Angle Identities

$$\begin{aligned} \sin 2\theta &= 2 \sin \theta \cos \theta \quad , \quad \cos 2\theta = \cos^2 \theta - \sin^2 \theta \\ &= 1 - 2 \sin^2 \theta \\ &= 2 \cos^2 \theta - 1 \end{aligned}$$

nth Angle Identities

$$1 - \cos(n\theta) = 2 \sin^2\left(\frac{n}{2}\theta\right)$$

Theorem:

$$1) \lim_{x \rightarrow 0} \frac{\sin x}{x} = 1 \quad , \quad 2) \lim_{x \rightarrow 0} \frac{\tan x}{x} = 1$$

Remarks:

$$1) \lim_{x \rightarrow 0} \frac{\sin^n bx}{x^n} = b^n \qquad 2) \lim_{x \rightarrow 0} \frac{x^n}{\sin^n bx} = \frac{1}{b^n}$$

$$3) \lim_{x \rightarrow 0} \frac{\tan^n bx}{x^n} = b^n \qquad 4) \lim_{x \rightarrow 0} \frac{x^n}{\tan^n bx} = \frac{1}{b^n}$$

Example: Find the following Limits (if they exist)

$$1) \lim_{x \rightarrow 0} \frac{4x}{\tan 3x + \sin 2x}$$

$$2) \lim_{x \rightarrow 0} \frac{1 - \cos x}{x \sin x}$$

$$3) \lim_{x \rightarrow 1} \frac{\sin(3x-3)}{1-x^3}$$

$$4) \lim_{x \rightarrow 1} \frac{\sin^2(\pi x)}{1-2x+x^2}$$

Solution:

$$1) \lim_{x \rightarrow 0} \frac{4x}{\tan 3x + \sin 2x} = \frac{0}{0} \text{ (I . F)}$$

$$\begin{aligned} \text{Thus } \lim_{x \rightarrow 0} \frac{4x}{\tan 3x + \sin 2x} &= \lim_{x \rightarrow 0} \frac{\frac{4x}{x}}{\frac{\tan 3x}{x} + \frac{\sin 2x}{x}} \\ &= \frac{\lim_{x \rightarrow 0} 4}{\lim_{x \rightarrow 0} \frac{\tan 3x}{x} + \lim_{x \rightarrow 0} \frac{\sin 2x}{x}} = \frac{4}{3+2} = \frac{4}{5} \end{aligned}$$

$$2) \lim_{x \rightarrow 0} \frac{1 - \cos x}{x \sin x} = \frac{0}{0} \text{ (I . F)}$$

$$\begin{aligned} \text{Thus } \lim_{x \rightarrow 0} \frac{1 - \cos x}{x \sin x} &= \lim_{x \rightarrow 0} \frac{2 \sin^2\left(\frac{x}{2}\right)}{x \sin x} \quad \left(\because 1 - \cos x = 2 \sin^2\left(\frac{x}{2}\right) \right) \\ &= \lim_{x \rightarrow 0} \frac{2 \sin^2\left(\frac{x}{2}\right)}{x^2 \frac{\sin x}{x}} = \lim_{x \rightarrow 0} \frac{2 \sin^2\left(\frac{1}{2}x\right)}{\frac{x^2}{\frac{\sin x}{x}}} = \frac{\lim_{x \rightarrow 0} 2 \sin^2\left(\frac{1}{2}x\right)}{\lim_{x \rightarrow 0} \frac{x^2}{\frac{\sin x}{x}}} = \frac{2\left(\frac{1}{2}\right)^2}{1} = \frac{1}{2} \end{aligned}$$

$$3) \lim_{x \rightarrow 1} \frac{\sin(3x-3)}{1-x^3} = \frac{0}{0} \text{ (I . F)}$$

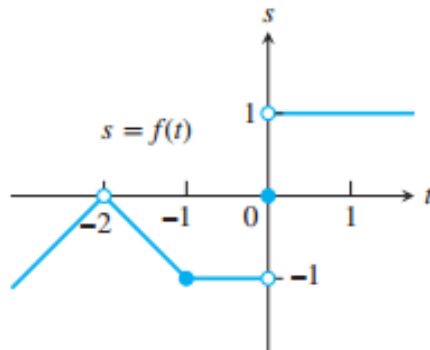
$$\begin{aligned} \text{Thus } \lim_{x \rightarrow 1} \frac{\sin(3x-3)}{1-x^3} &= \lim_{x \rightarrow 1} \frac{\sin(3(x-1))}{(1-x)(1+x+x^2)} \\ &= \lim_{x \rightarrow 1} \frac{\sin(3(x-1))}{-(x-1)} \lim_{x \rightarrow 1} \frac{1}{(1+x+x^2)} = -3 \left(\frac{1}{3} \right) = -1 \end{aligned}$$

$$4) \lim_{x \rightarrow 1} \frac{\sin^2(\pi x)}{1-2x+x^2} \quad \text{Exercise (Ans: } \pi^2 \text{)}$$

Exercises

1) For the function $f(t)$ graphed here, find the following limits or explain why they do not exist.

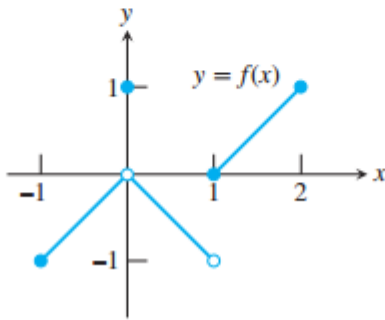
a. $\lim_{t \rightarrow -2} f(t)$ b. $\lim_{t \rightarrow -1} f(t)$ c. $\lim_{t \rightarrow 0} f(t)$



(Ans { (a) 0
(b) -1
(c) Does not exist. As t approaches 0 from the left, $f(t)$ approaches -1. As t approaches 0 from the right, $f(t)$ approaches 1. There is no single number L that $f(t)$ gets arbitrarily close to as $t \rightarrow 0$.)

2) Which of the following statements about the function $y = f(x)$ graphed here are true, and which are false?

- a. $\lim_{x \rightarrow 0} f(x)$ exists.
 b. $\lim_{x \rightarrow 0} f(x) = 0$.
 c. $\lim_{x \rightarrow 0} f(x) = 1$.
 d. $\lim_{x \rightarrow 1} f(x) = 1$.
 e. $\lim_{x \rightarrow 1} f(x) = 0$.
 f. $\lim_{x \rightarrow x_0} f(x)$ exists at every point x_0 in $(-1, 1)$.



(Ans { (a) True (b) True (c) False
(d) False (e) False (f) True)

3) Find each limit and why (if they exist)

i) $\lim_{x \rightarrow 1} f(x)$ ii) $\lim_{x \rightarrow -1} f(x)$ iii) $\lim_{x \rightarrow 2} f(x)$

$$a) f(x) = \begin{cases} x^2 - 3 & \text{if } x > 1 \\ 2 & \text{if } x = 1 \\ 4 - x & \text{if } x < 1 \end{cases} \quad b) f(x) = \begin{cases} x + 2 & \text{if } x < -1 \\ x^2 & \text{if } -2 > x \geq -1 \\ x + 1 & \text{if } x > 2 \end{cases}$$

Ans : a) i) does not exist ii) 5 iii) 1 b) i) 1 ii) 1 iii) does not exist

4) Why

$$a) \lim_{x \rightarrow 3} \sqrt{2x-6} \text{ does not exist} \quad b) \lim_{x \rightarrow 5} \frac{2x-10}{|x-5|} \text{ does not exist} \quad c) \lim_{x \rightarrow 1} f(x) \text{ does not exist if } f(x) = \begin{cases} x+3 & \text{if } x < 1 \\ x & \text{if } x > 1 \end{cases}$$

5) Using the definition of the limit to prove that

$$a) \lim_{x \rightarrow 2} 3x - 1 = 5 \quad b) \lim_{x \rightarrow 3} \frac{2}{5}x + \frac{1}{2} = \frac{17}{10} \quad c) \lim_{x \rightarrow 1} \frac{2x^2 - 2}{x - 1} = 4$$

In exercises (6 – 18) find the following limits (if they exist)

$$6) \lim_{x \rightarrow 1} \frac{x^2 + 2x + 3}{x - 6} \quad \left(\text{ans: } -\frac{6}{5} \right) \quad 7) \lim_{x \rightarrow 2} \frac{2x+4}{x-7} \quad \left(\text{ans: } -\frac{8}{5} \right) \quad 8) \lim_{x \rightarrow 6} \sqrt{2x-5} \quad \left(\text{ans: } \sqrt{7} \right)$$

$$9) \lim_{x \rightarrow 1} \frac{x^3 - 1}{x - 1} \quad \left(\text{ans: } 3 \right) \quad 10) \lim_{x \rightarrow 2} \frac{x^2 + 3x - 10}{3x^2 - 5x - 2} \quad \left(\text{ans: } 1 \right) \quad 11) \lim_{x \rightarrow 0} \frac{\sqrt{1+x} - 1}{x} \quad \left(\text{ans: } \frac{1}{2} \right)$$

$$12) \lim_{x \rightarrow 1} \begin{cases} 2x + 1 & \text{if } x > 1 \\ x^2 - 1 & \text{if } x < 1 \end{cases} \quad \left(\text{ans: does not exist} \right) \quad 13) \lim_{x \rightarrow 0} \frac{4x}{|x+1| - |x-1|} \quad \left(\text{ans: } 2 \right)$$

$$14) \lim_{x \rightarrow 4} \begin{cases} 6 & \text{if } x = 4 \\ \frac{4-x}{5-\sqrt{x^2+9}} & \text{if } x \neq 4 \end{cases} \quad \left(\text{ans: } \frac{10}{8} \right) \quad 15) \lim_{x \rightarrow 2} \frac{x^3 + x^2 - 2x - 8}{6x - 12} \quad \left(\text{ans: } \frac{7}{3} \right)$$

$$16) \lim_{x \rightarrow 0} x - |x| \quad \left(\text{ans: } 0 \right) \quad 17) \lim_{x \rightarrow 1} |x+1| + x^2 + 7 \quad \left(\text{ans: } 10 \right) \quad 18) \lim_{x \rightarrow 0} \frac{\sin 6x}{\cos 4x} \quad \left(\text{ans: } 0 \right)$$

$$19) \lim_{x \rightarrow 0} \frac{x^3}{\tan^3 2x} \quad \left(\text{ans: } \frac{1}{8} \right) \quad 20) \lim_{x \rightarrow 0} \frac{\tan x - \sin x}{x \cos x} \quad \left(\text{ans: } 0 \right) \quad 21) \lim_{x \rightarrow 0} \frac{\sin^2 x \cos x}{1 - \cos x} \quad \left(\text{ans: } 2 \right)$$

$$22) \lim_{x \rightarrow 0} \frac{\cos\left(\frac{\pi}{2} - x\right)}{x} \quad \left(\text{ans: } 1 \right) \quad 23) \lim_{x \rightarrow 0} \frac{\cos 2x - 1}{2x^2} \quad \left(\text{ans: } -1 \right) \quad 24) \lim_{x \rightarrow \frac{\pi}{2}} (\sec x - \tan x) \quad \left(\text{ans: } 0 \right)$$

$$25) \lim_{x \rightarrow 1} \frac{\tan(\pi x)}{x^2 - x} \quad \left(\text{ans: } \pi \right) \quad 26) \lim_{x \rightarrow 1} \frac{\cos\left(\frac{\pi x}{2}\right)}{x^2 - 1} \quad \left(\text{ans: } -\frac{\pi}{4} \right) \quad 27) \lim_{x \rightarrow 0} \frac{1 - 2x^2 - 2 \cos x + \cos^2 x}{x^2} \quad \left(\text{ans: } -2 \right)$$

$$28) \lim_{x \rightarrow -2} \frac{\sin^2(\pi x)}{x^2 + 4x + 4} \quad \left(\text{ans: } \pi^2 \right) \quad 29) \lim_{x \rightarrow 0} \frac{\sin x}{\sqrt{x+1} - \sqrt{1-x}} \quad \left(\text{ans: } 1 \right) \quad 30) \lim_{x \rightarrow 0} \frac{1 - \sqrt{\cos x}}{x^2} \quad \left(\text{ans: } \frac{1}{4} \right)$$

Continuity**Definition:**

A function f is continuity at a number ' a ' if the following three conditions are satisfied

- 1) f is defined on an open interval containing ' a ' ($f(a)$ exist)
- 2) $\lim_{x \rightarrow a} f(x)$ exist
- 3) $f(a) = \lim_{x \rightarrow a} f(x)$

Example: If $f(x) = 2x + 1$ discuss the continuity of $f(x)$ at $x = 1$.

Solution:

$$\text{Since } f(1) = 2(1) + 1 = 3$$

$$\text{Also } \lim_{x \rightarrow 1} (2x + 1) = 3, \text{ then } \lim_{x \rightarrow a} f(x) = f(a)$$

Thus f is continuous at $= 1$.

Notes:

- 1) The polynomial function is continuous $\forall x \in \mathbb{R}$.
- 2) $f(x) = \sin x$ and $f(x) = \cos x$ are continuous $\forall x \in \mathbb{R}$.

Example: If $f(x) = \begin{cases} \frac{x^2-9}{x-3} & \text{if } x \neq 3 \\ x^2 + 1 & \text{if } x = 3 \end{cases}$ discuss the continuity of $f(x)$ at $= 3$.

Solution:

$$f(3) = 3^2 + 1 = 10$$

$$\text{As } \lim_{x \rightarrow 3} f(x) = \lim_{x \rightarrow 3} \frac{x^2-9}{x-3} \text{ (I.F)}$$

$$\Rightarrow \lim_{x \rightarrow 3} f(x) = \lim_{x \rightarrow 3} \frac{x^2-9}{x-3} = \lim_{x \rightarrow 3} \frac{(x-3)(x+3)}{x-3} \lim_{x \rightarrow 3} x + 3 = 6$$

$$\text{Since } \lim_{x \rightarrow 3} f(x) \neq f(3)$$

Then f is not continuous at $= 3$.

Exercise: Find the value of the constant K s.t. the function $f(x)$ is continuous at $x = 0$,

$$f(x) = \begin{cases} \frac{1-\cos 6x}{x^2} & , x \neq 0 \\ K + 2 & , x = 0 \end{cases} \quad (\text{Ans : } 16)$$

Limit at infiniteDefinition :

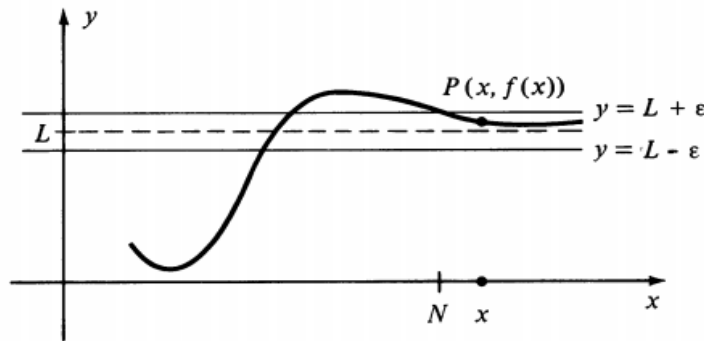
Let f be defined on an infinite interval (c, ∞) . The statement

$$\lim_{x \rightarrow \infty} f(x) = L$$

means that for every $\varepsilon > 0$ there corresponds a positive number N

such that

if $x > N$, then $|f(x) - L| < \varepsilon$



Note: In the previous figure above the graph of a function f approaches the line $y = L$. This line $y = L$ called a **horizontal asymptote**.

Definition :

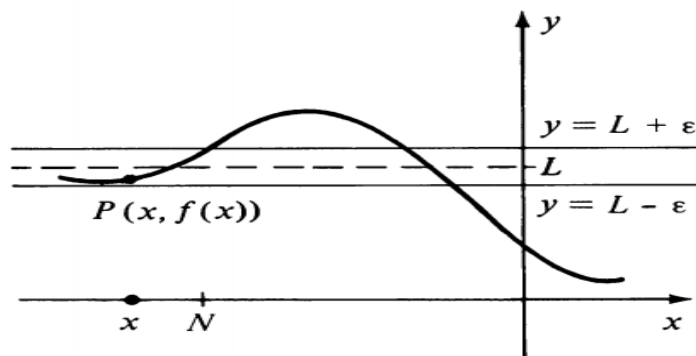
Let f be defined on an infinite interval $(-\infty, c)$. The statement

$$\lim_{x \rightarrow -\infty} f(x) = L$$

means that for every $\varepsilon > 0$ there corresponds a negative number N

such that

if $x < N$, then $|f(x) - L| < \varepsilon$



Note: Also in the previous figure above the graph of a function f approaches the line $x = L$. This line $x = L$ called a **horizontal asymptote**

Theorem:

If k is a positive rational number and c is any real number, then

$$\lim_{x \rightarrow \infty} \frac{c}{x^k} = 0 \quad \text{and} \quad \lim_{x \rightarrow -\infty} \frac{c}{x^k} = 0 \quad \text{provided } x^k \text{ is always defined}$$

Remark: Let $P_1(x), P_2(x)$ are two polynomials s.t.

$$P_1(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0; a_0, a_1, \dots, a_{n-1}, a_n \in \mathbb{R}, n \in \mathbb{Z}^+, a_n \neq 0$$

$$P_2(x) = b_m x^m + b_{m-1} x^{m-1} + \dots + b_1 x + b_0; b_0, b_1, \dots, b_{m-1}, b_m \in \mathbb{R}, m \in \mathbb{Z}^+, b_m \neq 0$$

Then

$$1) \lim_{x \rightarrow \pm\infty} P_1(x) = \lim_{x \rightarrow \pm\infty} a_n x^n$$

$$2) \lim_{x \rightarrow \pm\infty} \frac{P_1(x)}{P_2(x)} = \lim_{x \rightarrow \pm\infty} \frac{a_n x^n}{b_m x^m}$$

Example: Find the following limits (if they exist)

$$1) \lim_{x \rightarrow -\infty} (3x^3 - 10x^2 + 1) \quad , \quad 2) \lim_{x \rightarrow \infty} \frac{3x^2 + 3x - 2}{-2x^2 + 5} \quad , \quad 3) \lim_{x \rightarrow -\infty} \frac{x^2 - 3x + 7}{x^3 + 4x - 9}$$

$$4) \lim_{x \rightarrow \infty} \frac{7x^2 + x + 11}{4 - x} \quad , \quad 5) \lim_{x \rightarrow -\infty} \frac{x + 3}{\sqrt{16x^2 + 5x}}$$

Solution:

$$1) \lim_{x \rightarrow -\infty} (3x^3 - 10x^2 + 1) = \lim_{x \rightarrow -\infty} 3x^3 = -\infty.$$

$$2) \lim_{x \rightarrow \infty} \frac{3x^2 + 3x - 2}{-2x^2 + 5} = \lim_{x \rightarrow \infty} \frac{3x^2}{-2x^2} = \lim_{x \rightarrow \infty} \frac{3}{-2} = -\frac{3}{2}.$$

$$3) \lim_{x \rightarrow -\infty} \frac{x^2 - 3x + 7}{x^3 + 4x - 9} = \lim_{x \rightarrow -\infty} \frac{x^2}{x^3} = \lim_{x \rightarrow -\infty} \frac{1}{x} = 0.$$

$$4) \lim_{x \rightarrow \infty} \frac{7x^2 + x + 11}{4 - x} = \lim_{x \rightarrow \infty} \frac{7x^2}{-x} = \lim_{x \rightarrow \infty} -7x = -\infty.$$

$$5) \lim_{x \rightarrow -\infty} \frac{x+3}{\sqrt{16x^2+5x}} = \lim_{x \rightarrow -\infty} \frac{x+3}{\sqrt{x^2(16+\frac{5}{x})}} = \lim_{x \rightarrow -\infty} \frac{x+3}{|x|\sqrt{16+\frac{5}{x}}} \\ = \lim_{x \rightarrow -\infty} \frac{x}{-x\sqrt{16}} = \lim_{x \rightarrow -\infty} \frac{1}{-4} = -\frac{1}{4}$$

Notes:

1) In the previous example $y = 0$ is a **horizontal asymptote** of $f(x) = \frac{3}{x^2+5}$ and $f(x) = \frac{x^2-3x+7}{x^3+4x-9}$

also $y = -\frac{1}{4}$ is a **horizontal asymptote** of $f(x) = \frac{x+3}{\sqrt{16x^2+5x}}$

2) The following forms also represent indeterminate forms

$$1) \frac{\pm\infty}{\pm\infty} \quad , \quad 2) 0 \pm \infty \quad , \quad 3) \infty - \infty$$

Infinite Limit

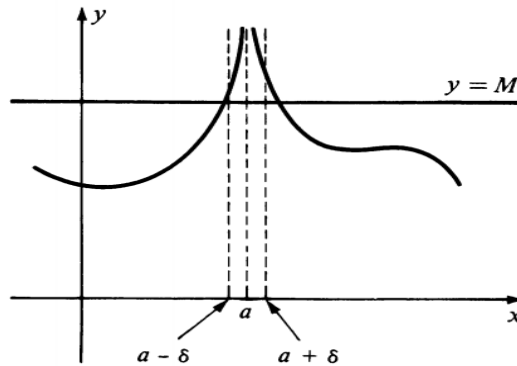
Definition:

Let $f(x)$ be defined on an open interval containing 'a' except possibly at a . The statement

$$\lim_{x \rightarrow a} f(x) = \infty$$

Means that for every $M > 0$, there corresponds $\delta > 0$ such that

$$\text{If } 0 < |x - a| < \delta, \text{ then } f(x) > M$$



Note:

The line $x = a$ is called a **vertical asymptote** for the graph of $f(x)$

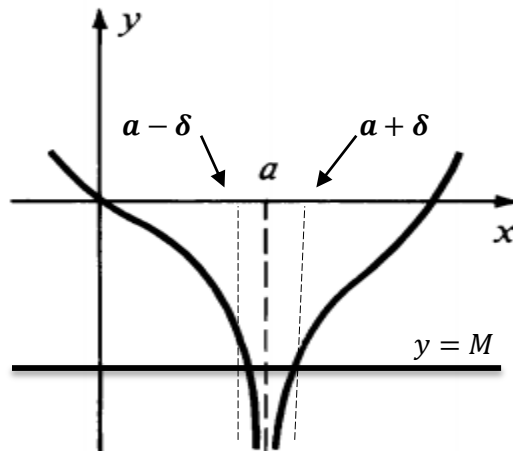
Definition:

Let $f(x)$ be defined on an open interval containing 'a' except possibly at a . The statement

$$\lim_{x \rightarrow a} f(x) = -\infty$$

Means that for every $M < 0$, there corresponds $\delta > 0$ such that

$$\text{If } 0 < |x - a| < \delta, \text{ then } f(x) < M$$



Note:

The line $x = a$ is called a **vertical asymptote** for the graph of $f(x)$

Example: Find the following limits (if they exist)

$$1) \lim_{x \rightarrow 3} \frac{2x^2}{9 - x^2} \quad , \quad 2) \lim_{x \rightarrow 1} \frac{5x^2}{(x - 1)^2}$$

Solution:

$$1) \text{ Since } \lim_{x \rightarrow 3} (9 - x^2) = 0 \text{ and } \lim_{x \rightarrow 3} 2x^2 \neq 0, \text{ then } \lim_{x \rightarrow 3} \frac{2x^2}{9 - x^2} \begin{cases} \lim_{x \rightarrow 3^-} \frac{2x^2}{9 - x^2} = \infty \\ \lim_{x \rightarrow 3^+} \frac{2x^2}{9 - x^2} = -\infty \end{cases} \text{ does not exist}$$

$$2) \text{ Since } \lim_{x \rightarrow 1} (x - 1)^2 = 0, \text{ and } \lim_{x \rightarrow 1} 5x^2 \neq 0 \text{ then } \lim_{x \rightarrow 1} \frac{5x^2}{(x - 1)^2} \begin{cases} \lim_{x \rightarrow 1^-} \frac{5x^2}{(x - 1)^2} = \infty \\ \lim_{x \rightarrow 1^+} \frac{5x^2}{(x - 1)^2} = \infty \end{cases} = \infty$$

Notes:

1) in the previous example note that $x = 3$ is a vertical asymptote of $f(x) = \frac{2x^2}{9 - x^2}$ and $x = 1$ is a vertical asymptote of $f(x) = \frac{5x^2}{(x - 1)^2}$

2) If the denominator $\neq 0 \forall x \in \mathbb{R}$, then the function has not a vertical asymptotes to $f(x)$

3) If $x = a$ is zero for the denominator, then $x = a$ can be a vertical asymptote if $\lim_{x \rightarrow a} f(x) = \pm \infty$

Example: Find the vertical and horizontal asymptotes of the following functions

$$1) f(x) = \frac{2x + 6}{9 - x^2} \quad 2) f(x) = \frac{2x^4}{x^4 + 1}$$

Solution:

$$1) \text{ Let } 9 - x^2 = 0 \Rightarrow x^2 = 9 \Rightarrow x = \pm 3$$

Since $x = -3, x = 3$ are two zeros for the denominator, then $x = -3, x = 3$ are two lines candidates can be a vertical asymptotes

Since $\lim_{x \rightarrow 3^-} \frac{2x + 6}{9 - x^2} = \infty$ more over $\lim_{x \rightarrow 3^+} \frac{2x + 6}{9 - x^2} = -\infty$, then $x = 3$ is vertical asymptote

Also

$$\text{Since } \lim_{x \rightarrow -3} \frac{2x + 6}{9 - x^2} = \frac{0}{0} \text{ (I.F.)}$$

$$\text{Thus } \lim_{x \rightarrow -3} \frac{2x + 6}{9 - x^2} = \lim_{x \rightarrow -3} \frac{2(x + 3)}{(3 + x)(3 - x)} = \lim_{x \rightarrow -3} \frac{2}{(3 - x)} = \frac{2}{6} = \frac{1}{3}$$

Then $x = -3$ is not vertical asymptote

To find the horizontal asymptotes, we consider

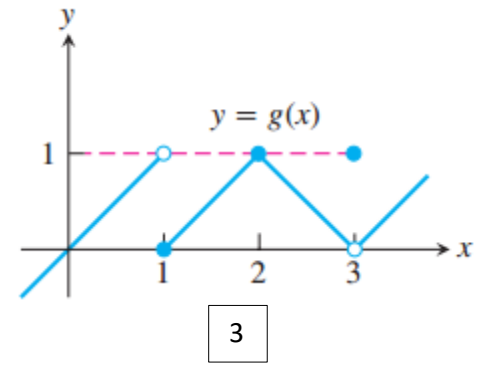
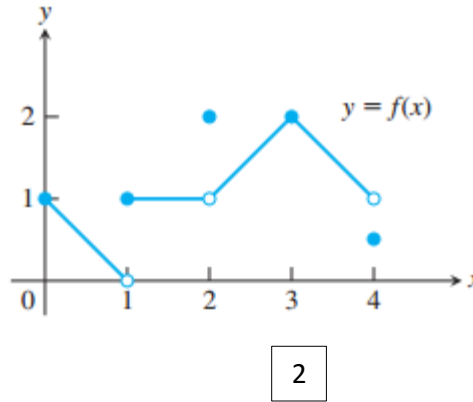
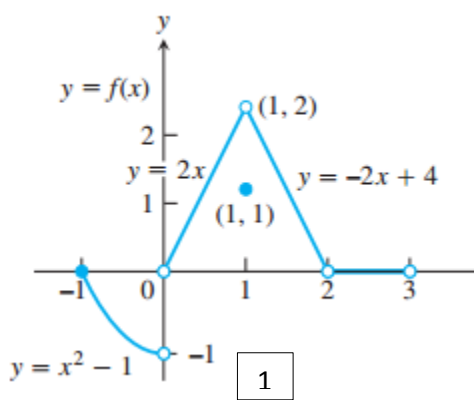
$$\lim_{x \rightarrow \infty} \frac{2x^2}{9 - x^2} = \lim_{x \rightarrow \infty} \frac{2x^2}{-x^2} = -2, \text{ also } \lim_{x \rightarrow -\infty} \frac{2x^2}{9 - x^2} = \lim_{x \rightarrow -\infty} \frac{2x^2}{-x^2} = -2$$

Then $y = -2$ is horizontal asymptote

2) Exercise (Ans: does not exist a vertical asymptote, $y = 2$ is horizontal)

Exercises

In exercises (1 – 3) discuss the continuity of the following graphs at $x = 1$, $x = 2$ and $x = 3$



In exercises(4 – 6) discuss continuity of the following functions at $x = a$.

$$4) f(x) = \begin{cases} x - x^2 & \text{if } x < -2 \\ 10 & \text{if } x = -2 \\ -5x & \text{if } x > -2 \end{cases} \quad a = -2 \quad (\text{ans : not continuous})$$

$$5) f(x) = \begin{cases} \frac{3x + \sin 2x}{\tan 3x} & \text{if } x \neq 0 \\ \frac{5}{3} & \text{if } x = 0 \end{cases} \quad a = 0 \quad (\text{ans : continuous})$$

$$6) f(x) = \begin{cases} \frac{1 - \cos 2x}{1 - \cos x} & \text{if } x \neq 0 \\ 4 & \text{if } x = 0 \end{cases} \quad a = 0 \quad (\text{ans : continuous})$$

In exercises (7 – 10) find the values of the constant K so that the function $f(x)$ is continuous at a

$$7) f(x) = \begin{cases} \frac{x^2 - 2x - 3}{x^3 + 1} & \text{if } x \neq -1 \\ K & \text{if } x = -1 \end{cases} \quad a = -1 \quad (\text{ans : } -\frac{4}{3})$$

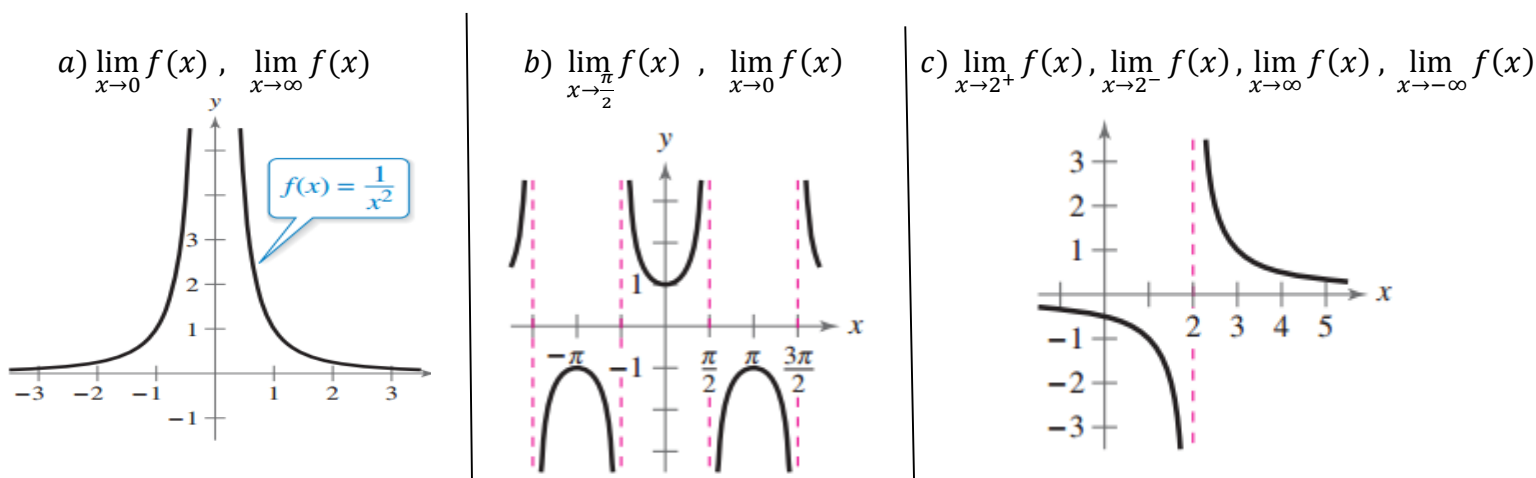
$$8) f(x) = \begin{cases} \frac{1 - \cos 2Kx}{x \sin x} & \text{if } x \neq 0 \\ K + 6 & \text{if } x = 0 \end{cases} \quad a = 0 \quad (\text{ans : } -\frac{3}{2}, 2)$$

$$9) f(x) = \begin{cases} \frac{\sqrt{7x+2} - \sqrt{6x+4}}{x-2} & \text{if } x \neq 2 \\ K & \text{if } x = 2 \end{cases} \quad a = 2 \quad (\text{ans : } \frac{1}{8})$$

$$10) f(x) = \begin{cases} \frac{1 - \sin x}{(2x - \pi)^2} & , x \neq \frac{\pi}{2} \\ K^3 & , x = \frac{\pi}{2} \end{cases} \quad (\text{ans : } \frac{1}{2})$$

$$11) f(x) = \begin{cases} \frac{1 - \sin 2x}{\cos^2 2x} & , x \neq \frac{\pi}{4} \\ c \sin x & , x = \frac{\pi}{4} \end{cases} \quad (\text{ans : } \frac{1}{\sqrt{2}})$$

12) From the following graph find (if they exist)



In exercises (13 – 24) find the following limits (if they exist)

13) $\lim_{x \rightarrow 4} \frac{5}{x-4}$ (ans : does not exist) 14) $\lim_{x \rightarrow -8} \frac{3x}{(x+8)^2}$ (ans : $-\infty$) 15) $\lim_{x \rightarrow 3} \frac{1}{x(x-3)^2}$ (ans : ∞)

16) $\lim_{x \rightarrow 1} \frac{4x^2}{x^2-4x+3}$ (ans : does not exist) 17) $\lim_{x \rightarrow \infty} \frac{5x^2-3x+1}{2x^2+4x-7}$ (ans : $\frac{5}{2}$) 18) $\lim_{x \rightarrow -\infty} \frac{4-7x}{2+3x}$ (ans : $-\frac{7}{3}$)

19) $\lim_{x \rightarrow -\infty} \frac{-3x^3-5x+7}{x^2-1}$ (ans : ∞) 20) $\lim_{x \rightarrow -\infty} |x-3| + x^3$ (ans : $-\infty$) 21) $\lim_{x \rightarrow \infty} \frac{x-3}{x^2+4}$ (ans : 0)

22) $\lim_{x \rightarrow \infty} \frac{\sqrt{16x^2-3}}{x+4}$ (ans : 4) 23) $\lim_{x \rightarrow \infty} \sqrt{x^2+5x} - x$ (ans : $\frac{5}{2}$)

24) $\lim_{x \rightarrow 2} (\frac{4}{x} + \frac{3}{x-2})$ (ans : does not exist)

In exercises (25 – 30) find the vertical and horizontal asymptotes of the graph of the following functions

25) $f(x) = \frac{1}{x^2-4}$ (ans $\begin{cases} v: x = 2, x = -2 \\ h: y = 0 \end{cases}$)

26) $f(x) = \frac{2x^2}{x^2+1}$ (ans $\begin{cases} v: \text{does not exist} \\ h: y = 2 \end{cases}$)

27) $f(x) = \frac{1}{x^3+x^2-6x}$ (ans $\begin{cases} v: x = 0, x = 2, x = -3 \\ h: y = 0 \end{cases}$)

28) $f(x) = \frac{x^2+2x+2}{x^2+2x-3}$ (ans $\begin{cases} v: x = -3, x = 1 \\ h: y = 1 \end{cases}$)

29) $f(x) = \frac{x+4}{x^2-16}$ (ans $\begin{cases} v: x = 4 \\ h: y = 0 \end{cases}$)

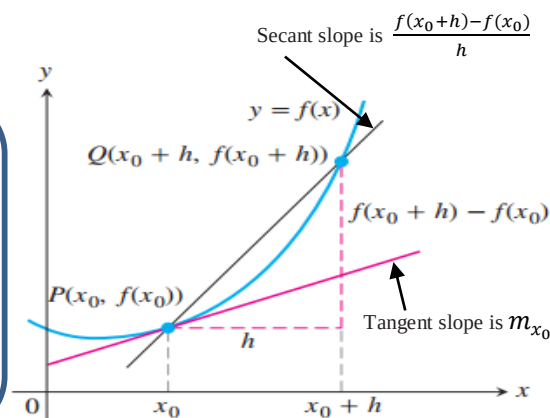
30) $f(x) = \frac{x^2+25}{x-5}$ (ans $\begin{cases} v: x = 5 \\ h: \text{does not exist} \end{cases}$)

The Derivative**Tangent lines****Definition:**

The slope of the tangent line to the graph of a function f

at $(x_0, f(x_0))$ is $m_{x_0} = \lim_{h \rightarrow 0} \frac{f(x_0+h)-f(x_0)}{h}$

provided the limit exist



Note: The tangent line of the curve $y = f(x)$ at the point $(x_0, f(x_0))$ is the line passes through the point $(x_0, f(x_0))$ with the slope m_{x_0} . It's equation

$$y = m_{x_0}(x - x_0) + f(x_0)$$

Example:

- 1) Find the slope of the function $f(x) = x^2$ at the point $(-2,4)$.
- 2) Find the equation of the tangent line at $(-2,4)$ to the graph of $f(x) = x^2$.
- 3) Find the equation of the normal line at $(-2,4)$ to the graph of $f(x) = x^2$

Solution:

- 1) The slope of the tangent line at $(-2,4)$ is m_{-2} s.t.

$$\begin{aligned} m_{-2} &= \lim_{h \rightarrow 0} \frac{f(-2+h) - f(-2)}{h} \\ &= \lim_{h \rightarrow 0} \frac{(-2+h)^2 - (-2)^2}{h} \\ &= \lim_{h \rightarrow 0} \frac{4 - 4h + h^2 - 4}{h} \\ &= \lim_{h \rightarrow 0} \frac{-4h + h^2}{h} = \lim_{h \rightarrow 0} (-4 + h) = -4 \end{aligned}$$

- 2) The equation of the tangent line at $(-2,4)$ is

$$\begin{aligned} y &= m_{-2}(x - (-2)) + f(-2) \\ &= -4(x + 2) + 4 \\ \Rightarrow y &= -4x - 4 \\ \text{or } y + 4x + 4 &= 0 \end{aligned}$$

- 3) Exercises (Ans : $4y - x - 18 = 0$)

The derivative of the function

At the previous subject, we defined the slope of a curve at the point where $x = x_0$ to be $\lim_{h \rightarrow 0} \frac{f(x_0+h)-f(x_0)}{h}$. We called this limit, when it existed, the derivative of f at x_0 . We now investigate the derivative as a *function* derived from f by considering the limit at each point of the domain of f .

Definition:

The **derivative** of the function $f(x)$ with respect to the variable x is the function $f'(x)$ whose value at x is

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

provided the limit exists.

Note: Now we can write the equation of the tangent line in the form $y = f'(x_0)(x - x_0) + f(x_0)$

Example: If $f(x) = 3x^2 - 12x + 8$ find 1) $f'(x)$, 2) $f'(3)$

3) The point on the graph at which the tangent line is a horizontal.

Solution:

1) Since $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h)-f(x)}{h}$

$$\begin{aligned} \text{Then } f'(x) &= \lim_{h \rightarrow 0} \frac{3(x+h)^2 - 12(x+h) + 8 - 3x^2 + 12x - 8}{h} \\ &= \lim_{h \rightarrow 0} \frac{3x^2 + 6xh + 3h^2 - 12x - 12h + 8 - 3x^2 + 12x - 8}{h} \\ &= \lim_{h \rightarrow 0} \frac{6xh + 3h^2 - 12h}{h} \\ &= \lim_{h \rightarrow 0} \frac{h(6x + 3h - 12)}{h} = \lim_{h \rightarrow 0} (6x + 3h - 12) = 6x - 12 \end{aligned}$$

2) $f'(3) = 6(3) - 12 = 6$

3) If the tangent line is horizontal , then $f'(x) = 0 \Rightarrow 6x - 12 = 0 \Rightarrow x = 2$
then the tangent line is horizontal at $(2, f(2)) = (2, -4)$.

Remark:

We denoted of $f'(x)$ by 1) $\frac{dy}{dx}$, 2) $\frac{df}{dx}$, 3) $D_x y$, 4) y' and $f'(x)$ at $x = x_0$ is denoted by

$$1) f'(x_0) , 2) \left. \frac{df}{dx} \right|_{x=x_0} , 3) \left. \frac{dy}{dx} \right|_{x=x_0} , 4) y'(x_0).$$

Example: Let $f(x) = |x|$ 1) Discuss the continuity of $f(x)$ at $x = 0$

2) Discuss the differentiable of $f(x)$ at $x = 0$

Solution:

$$\text{As } f(x) = |x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases}$$

1)

$$\text{Since } \lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^-} -x = 0 \quad \text{and} \quad \lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} x = 0$$

Then $\lim_{x \rightarrow 0} f(x) = 0 \dots (*)$ why?

$$f(0) = 0 \dots (*,*)$$

From $(*)$ and $(*,*)$ we get $\lim_{x \rightarrow 0} f(x) = f(0)$, then $f(x)$ is continuous at $x = 0$

2)

Right – hand derivative of $f(x) = |x|$ at $x = 0$ is

$$f'_+(0) = \lim_{h \rightarrow 0^+} \frac{(f(0+h)-f(0))}{h} = \lim_{h \rightarrow 0^+} \frac{|0+h|-|0|}{h} = \lim_{h \rightarrow 0^+} \frac{|h|}{h} = \lim_{h \rightarrow 0^+} \frac{h}{h} = 1 \dots (*)$$

Left – hand derivative of $f(x) = |x|$ at $x = 0$ is

$$f'_-(0) = \lim_{h \rightarrow 0^-} \frac{(f(0+h)-f(0))}{h} = \lim_{h \rightarrow 0^-} \frac{|0+h|-|0|}{h} = \lim_{h \rightarrow 0^-} \frac{|h|}{h} = \lim_{h \rightarrow 0^-} \frac{-h}{h} = -1 \dots (*,*)$$

From $(*)$ and $(*,*)$ we not that $f(x) = |x|$ has not derivative at $x = 0$

Theorem :

If $f(x)$ has a derivative at $x = a$, then $f(x)$ is continuous at $x = a$

Note:

The converse of the above theorem is false, that is there exist continuous function that is not differentiable

Techniques of differentiation**Rules for finding derivative**

Let $f(x)$, $g(x)$ and $h(x)$ are functions differentiable, n any real number and a is a constant value, then

$$1) \text{ If } f(x) = a \Rightarrow f'(x) = 0.$$

$$2) \text{ If } f(x) = ax \Rightarrow f'(x) = a$$

$$3) \text{ If } f(x) = ax^n \Rightarrow f'(x) = n \times a x^{n-1}.$$

$$4) \text{ If } f(x) = a g(x) \Rightarrow f'(x) = a g'(x).$$

$$5) \text{ If } f(x) = g(x) \pm h(x) \Rightarrow f'(x) = g'(x) \pm h'(x).$$

$$6) \text{ If } f(x) = g(x) \cdot h(x) \Rightarrow f'(x) = g'(x) \cdot h(x) + g(x) \cdot h'(x).$$

$$7) \text{ If } f(x) = \frac{g(x)}{h(x)} \text{ provided } h(x) \neq 0 \Rightarrow f'(x) = \frac{g'(x) \cdot h(x) - g(x) \cdot h'(x)}{[h(x)]^2}.$$

Example: Find the derivative of the following functions

$$1) f(x) = 2x^3 + x^2 - 2, \quad 2) f(x) = (3x^2 - 5)(2x^4 - x), \quad 3) f(x) = \frac{5x-5}{x^2+7}.$$

Solution:

$$1) f'(x) = 6x^2 + 2x.$$

$$2) f'(x) = (3x^2 - 5)'(2x^4 - x) + (3x^2 - 5)(2x^4 - x)'$$

$$= 6x \cdot (2x^4 - x) + (3x^2 - 5)(8x^3 - 1)$$

$$= 12x^5 - 6x^2 + 24x^5 - 3x^2 - 40x^3 + 5$$

$$= 36x^5 - 40x^3 - 9x^2 + 5$$

$$3) f'(x) = \frac{(5x - 5)'(x^2 + 7) - (5x - 5)(x^2 + 7)'}{(x^2 + 7)^2}$$

$$= \frac{5 \cdot (x^2 + 7) - (5x - 5) \cdot 2x}{(x^2 + 7)^2}$$

$$= \frac{5x^2 + 35 - 10x^2 + 10x}{(x^2 + 7)^2} = \frac{-5x^2 - 10x + 35}{(x^2 + 7)^2}$$

Exercise: Find an equation of the tangent line to the graph of $y = 2x^3 + 4x^2 - 5x - 3$ at the point $(1, -3)$.

(Ans: $y = 9x - 12$)

Chain Rule

Theorem: If $y = f(u)$, $u = g(x)$ and the derivatives $\frac{dy}{du}$ and $\frac{du}{dx}$ both exist, then the composite function defined by $y = f(g(x))$ has a derivative given by

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx} = f'(u) \cdot g'(x) = f'(g(x)) \cdot g'(x)$$

Example: Find $\frac{dy}{dx}$ for each of the following

1) 1) If $y = u^5$ and $u = x^3 + 8$, 2) $y = \frac{1}{(4x^2 + 6x - 7)^3}$, 3) $y = \sqrt{x^3 + x + 1}$, 4) $y = \sqrt[5]{7x^3 - 2x + 9}$

Solution:

1)

Since $\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$ and $\frac{dy}{du} = 5u^4$, $\frac{du}{dx} = 3x^2$

Then $\frac{dy}{dx} = 5u^4 \cdot 3x^2 = 15x^2(x^3 + 8)^4$

Note :

$$\text{If } f(x) = [g(x)]^n \Rightarrow f'(x) = n[g(x)]^{n-1} \cdot g'(x)$$

2) $y = \frac{1}{(4x^2 + 6x - 7)^3} = (4x^2 + 6x - 7)^{-3}$

$$\Rightarrow \frac{dy}{dx} = -3(8x + 6)(4x^2 + 6x - 7)^{-4}$$

$$= \frac{-3(8x + 6)}{(4x^2 + 6x - 7)^4}$$

3) $y = \sqrt{x^3 + x + 1} = (x^3 + x + 1)^{\frac{1}{2}}$

$$\Rightarrow \frac{dy}{dx} = \frac{1}{2} (x^3 + x + 1)^{-\frac{1}{2}} (3x^2 + 1) = \frac{3x^2 + 1}{2\sqrt{x^3 + x + 1}}$$

Note :

$$\text{If } f(x) = \sqrt[n]{g(x)} \Rightarrow f'(x) = \frac{g'(x)}{n \sqrt[n]{(g(x))^{n-1}}}$$

3) Exercise (Ans : $y' = \frac{21x^2 - 2}{5^5 \sqrt[5]{(7x^3 - 2x + 9)^4}}$))

Derivative of trigonometric functions

Theorem: If $g(x)$ is a differentiable function . then by chain rule , we get the following

$$1) \frac{d}{dx}(\sin(g(x))) = g'(x) \cos(g(x)) \quad 2) \frac{d}{dx}(\cos(g(x))) = -g'(x) \sin(g(x))$$

$$3) \frac{d}{dx}(\tan(g(x))) = g'(x) \sec^2(g(x)) \quad 4) \frac{d}{dx}(\cot(g(x))) = -g'(x) \csc^2(g(x))$$

$$5) \frac{d}{dx}(\sec(g(x))) = g'(x) \sec(g(x)) \tan(g(x)) \quad 6) \frac{d}{dx}(\csc(g(x))) = -g'(x) \csc(g(x)) \cot(g(x))$$

Example: Find the derivative of the following functions

$$1) y = 3x + \cot x \quad , \quad 2) y = \sec^6 x \quad , \quad 3) y = \frac{\cot x}{1+\csc x} \quad 4) f(x) = \sec^2(x^3 - 5) \quad 5) f(x) = \tan(\sin x)$$

$$6) f(x) = 3 \tan(\sqrt{x})$$

Solution:

$$1) \frac{dy}{dx} = 3 - \csc^2 x$$

$$2) \frac{dy}{dx} = 6 \sec^5 x \cdot (\sec x)' = 6 \sec^5 x \cdot (\sec x \cdot \tan x) = 6 \sec^6 x \cdot \tan x$$

$$\begin{aligned} 3) \frac{dy}{dx} &= \frac{(\cot x)'(1+\csc x) - (\cot x)(1+\csc x)'}{[1+\csc x]^2} = \frac{-\csc^2 x(1+\csc x) - (\cot x)(-\csc x \cot x)}{[1+\csc x]^2} \\ &= \frac{-\csc^2 x - \csc^3 x + \csc x \cot^2 x}{[1+\csc x]^2} \end{aligned}$$

$$\begin{aligned} 4) f'(x) &= 2 \sec(x^3 - 5) (\sec(x^3 - 5))' \\ &= 2 \sec(x^3 - 5) \sec(x^3 - 5) \tan(x^3 - 5) (3x^2) \\ &= 6x^2 \sec^2(x^3 - 5) \tan(x^3 - 5) \end{aligned}$$

$$5) f'(x) = \sec^2(\sin x) \cdot \cos x = \cos x \cdot \sec^2(\sin x)$$

$$6) \text{ Exercise (Ans: } f'(x) = \frac{3 \sec^2(\sqrt{x})}{2\sqrt{x}} \text{)}$$

Derivative of logarithmic functions**Definition of Logarithmic function**

If $a > 0$, $a \neq 1$ and $x > 0$ then the function $f(x)$ defined by $f(x) = \log_a x$ called the Logarithmic function with base a .

Note: The Logarithmic function $f(x) = \log_a x$ is called

- 1) The natural Logarithmic function in case $a = e$ s.t. $e \cong 2.72$ and denoted by **$\ln x$** .
- 2) The common Logarithmic function in case $a = 10$ and denoted by **$\log x$** .

Theorem: If $a > 0$, $a \neq 1$ then

$$1) \log_a 1 = 0 \quad , \quad 2) \log_a a = 1 \quad 3) \text{ if } x > 0 \text{ then } a^{\log_a x} = x \text{ and } \log_a a^x = x.$$

Some properties of Logarithmic

If $a, x, y \in \mathbb{R}^+$ and $a \neq 1$ then

- 1) $\log_a(xy) = \log_a x + \log_a y$.
- 2) $\log_a\left(\frac{x}{y}\right) = \log_a x - \log_a y$.
- 3) $\log_a x^n = n \log_a x$, $\forall n \in \mathbb{R}$.

Theorem: If $g(x)$ is differentiable function with respect to x and $g(x) > 0$ then by chain rule

$$1) \frac{d}{dx} [\log_a g(x)] = \frac{g'(x)}{g(x) \ln a} \qquad 2) \frac{d}{dx} [\ln(g(x))] = \frac{g'(x)}{g(x)}$$

Example : Find the derivative of the following functions

$$1) y = \ln(x^2 + 1)$$

$$2) y = \ln |\sin x|$$

$$3) y = \log_3(\sqrt{x})$$

$$4) y = \ln \left[\frac{|\cos x|}{\sqrt{4-3x^2}} \right]$$

Solution :

$$1) \frac{dy}{dx} = \frac{(x^2+1)'}{x^2+1} = \frac{2x}{x^2+1}$$

$$2) \frac{dy}{dx} = \frac{(\sin x)'}{\sin x} = \frac{\cos x}{\sin x} = \cot x$$

$$3) \text{ As } y = \log_3(\sqrt{x}) = \log_3(x^{\frac{1}{2}}) = \frac{1}{2} \log_3(x) \Rightarrow \frac{dy}{dx} = \frac{1}{(2 \ln 3)x}$$

$$4) \text{ As } y = \ln \left[\frac{|\cos x|}{\sqrt{4-3x^2}} \right] = \ln|\cos x| - \ln\sqrt{4-3x^2} = \ln|\cos x| - \frac{1}{2}\ln(4-3x^2)$$

$$\begin{aligned} \Rightarrow \frac{dy}{dx} &= \frac{d}{dx}(\ln|\cos x|) - \frac{d}{dx}\left(\frac{1}{2}\ln(4-3x^2)\right) = -\frac{\sin x}{\cos x} - \frac{-6x}{2(4-3x^2)} \\ &= -\tan x + \frac{3x}{(4-3x^2)} \end{aligned}$$

Derivative of Exponential functions

Definition of exponential function

The function $f(x)$ defined by $f(x) = a^x$, $a > 0$ and $a \neq 1$ called the exponential function with base a

Note: The exponential function defined by $f(x) = e^x$ s.t. $e \cong 2.72$ is called the natural exponential function.

Theorem: If $g(x)$ is differentiable function with respect to x and $g(x) > 0$ then by chain rule

$$1) \frac{d}{dx}(a^{g(x)}) = (g'(x) \ln a) a^{g(x)} \qquad 2) \frac{d}{dx}(e^{g(x)}) = g'(x) e^{g(x)}$$

Example : Find $\frac{dy}{dx}$ for each the following functions

$$\begin{array}{ll} 1) y = e^{x^3} & 2) y = e^x(7 - \sqrt{x}) \\ 3) y = 2^{\cot x} & 4) y = 3^{\sec x^2} \end{array}$$

Solution :

$$1) \frac{dy}{dx} = e^{x^3} (x^3)' = 3x^2 e^{x^3}$$

$$\begin{aligned} 2) \frac{dy}{dx} &= (e^x)'(7 - \sqrt{x}) + e^x(7 - \sqrt{x})' = e^x(7 - \sqrt{x}) + e^x\left(-\frac{1}{2\sqrt{x}}\right) \\ &= e^x(7 - \sqrt{x}) - \frac{e^x}{2\sqrt{x}} \end{aligned}$$

$$3) \frac{dy}{dx} = (2^{\cot x} \ln 2)(\cot x)' = (2^{\cot x} \ln 2)(-\csc^2 x) = (-2^{\cot x} \ln 2)$$

$$4) \text{ Exercise (Ans : } \frac{dy}{dx} = 2x \sec x^2 \tan x^2 (\ln 3) 3^{\sec x^2} \text{)}$$

Implicit Differentiation

Introduction:

Suppose that y is a function of x that satisfies the equation $f(x, y) = 0$ (e.g : $3x^3 - 4y - 2x + 1 = 0$)

One way to find $\frac{dy}{dx}$ is to solve first for y and then differentiate . Is also possible to find $\frac{dy}{dx}$ by implicit differentiation

Implicit Differentiation :

Is a method without first solving the equation for y as follows:

- 1) Apply $\frac{dy}{dx}$ to both sides of the equation
- 2) Solve for y'

Notes :

1) **General Power Rule :** $\frac{d}{dx} y^n = n y^{n-1} y'$

2) **Chain Rule :** $\frac{d}{dx} g(y) = g'(y) y'$

3) **Implicit differentiation is particularly useful where it is impossible to first solve the given equation for y**

Example : Find y' if $3x^3 - 4y - 2x + 1 = 0$

Solution :

First , we will solve the equation for y

$$\text{Since } 3x^3 - 4y - 2x + 1 = 0$$

$$\text{Then } y = \frac{3}{4}x^3 - \frac{1}{2}x + \frac{1}{4} \Rightarrow y' = \frac{9}{4}x^2 - \frac{1}{2}$$

Second , we will use implicit differentiation

$$\frac{d}{dx}(3x^3 - 4y - 2x + 1) = \frac{d}{dx}(0)$$

$$\Rightarrow 9x^2 - 4y' - 2 = 0$$

$$\Rightarrow -4y' = -9x^2 + 2 \Rightarrow y' = \frac{9}{4}x^2 - \frac{1}{2}$$

Example: Use implicit differentiation to find $\frac{dy}{dx}$

$$1) x^2y + 3y^2 = 6 \quad , \quad 2) x + \sin y = xy \quad 3) e^{xy} = e^{4x} - e^{5y}$$

Solution:

$$1) \frac{d}{dx}(x^2y) + \frac{d}{dx}(3y^2) = \frac{d}{dx}(6) \Rightarrow y \frac{d}{dx}(x^2) + x^2 \frac{d}{dx}(y) + \frac{d}{dx}(3y^2) = \frac{d}{dx}(6)$$

$$\Rightarrow 2xy + x^2 \frac{dy}{dx} + 6y \frac{dy}{dx} = 0$$

$$\Rightarrow (x^2 + 6y) \frac{dy}{dx} = -2xy \Rightarrow \frac{dy}{dx} = \frac{-2xy}{x^2 + 6y}$$

$$\begin{aligned}
 2) \quad \frac{d}{dx}(x) + \frac{d}{dx}(\sin y) &= \frac{d}{dx}(xy) \Rightarrow 1 + \cos y \frac{dy}{dx} = y \frac{d}{dx}(x) + x \frac{d}{dx}(y) \\
 &\Rightarrow (\cos y - x) \frac{dy}{dx} = y - 1 \\
 &\Rightarrow \frac{dy}{dx} = \frac{y - 1}{\cos y - x}
 \end{aligned}$$

Remark: It is possible to solve the previous example using the following rule

$$y' = \frac{-\frac{df(x,y)}{dx}\big|_{y \text{ constant}}}{\frac{df(x,y)}{dy}\big|_{x \text{ constant}}} \quad \text{Provided we make } f(x, y) = 0.$$

$$3) \text{ Exercise (Ans : } y' = \frac{4e^{4x} - ye^{xy}}{xe^{xy} + 5e^{5y}} \text{)}$$

Example: Find the equations of the tangent and normal line to the curve $x^2 - xy + y^2 = 7$ at the point $(-1, 2)$

Solution:

$$\text{Since } y' = \frac{-\frac{df(x,y)}{dx}\big|_{y \text{ constant}}}{\frac{df(x,y)}{dy}\big|_{x \text{ constant}}} \quad \text{and } x^2 - xy + y^2 - 7 = 0$$

$$\text{Then } y' = \frac{y-2x}{2y-x} \Rightarrow y'(-1, 2) = \frac{2-2(-1)}{2(2)-(-1)} = \frac{4}{5}$$

Since the equation the tangent line is $y = m(x - a) + f(a)$

$$\text{Then } y = \frac{4}{5}(x + 1) + 2 \Rightarrow 4x - 5y + 14 = 0$$

Since the equation of the normal line is $y = \frac{-1}{m}(x - a) + f(a)$

$$\text{Then } y = \frac{-5}{4}(x + 1) + 2 \Rightarrow 5x + 4y - 3 = 0$$

Exercises

In exercises(1 – 4) find the slope of the tangent line at a on the graph of f **by the definition** .

$$1) f(x) = 2 - x^2, a = 2 \quad (ans : -4) \quad 2) f(x) = \sqrt{x} + 1, a = 9 \quad (ans : \frac{1}{6})$$

$$3) f(x) = \frac{1}{2+x}, a = 1 \quad (ans : -\frac{1}{9})$$

In exercises(4 – 7) differentiate the following functions **by the definition of the derivative** .

$$4) f(x) = 37 \quad (ans : 0) \quad 5) f(x) = 9x - 2 \quad (ans: 9)$$

$$6) f(x) = 2 + 8x - 5x^2 \quad (ans : 8 - 10x) \quad 7) f(x) = \sqrt{3x + 1} \quad (ans : \frac{3}{2\sqrt{3x+1}})$$

$$8) \text{ Let } f(x) = \begin{cases} x^2, & x \geq 0 \\ 2x, & x < 0 \end{cases}$$

i) Show that f is continuous at $x = 0$ ii) Does the function f is differentiable at $x = 0$

In exercises(9 – 11) differentiate the following functions

$$9) f(x) = (x^3 - 7)(2x^2 + 3) \quad (ans : 10x^4 + 9x^2 - 28x) \quad 10) f(x) = \frac{(2x^3 - 7x^2 + 4x + 3)}{x^2} \quad (ans : 2 - \frac{4}{x^2} - \frac{6}{x^3})$$

$$11) f(x) = \frac{(x^3 - 1)}{(x^3 + 1)} \quad (ans : \frac{6x^2}{(x^3 + 1)^2})$$

12) Find an equation of the tangent line to the graph of $y = \frac{5}{1+x^2}$ at each point

$$a) p(0,5), b) p(1, \frac{5}{2}), c) p(-2, 1) \quad (ans \ a)y = 5, b) 5x + 2y - 10 = 0, c) 4x - 5y + 13 = 0)$$

13) Find the coordinates of all points on the graph of $y = x^3 + 2x^2 - 4x + 3$ at which the tangent line

$$\text{is horizontal} \quad (ans: \frac{2}{3}, -2)$$

14) Find the equations of all lines having slope -1 that are tangent to the curve $y = \frac{1}{x-1}$

$$(ans : y = -x + 3, y = -x - 1)$$

15) Show that the tangent line to the curve $y = x^3$ at $(1, 1)$ passes through the point $(2, 4)$

16) Suppose the functions $F(x)$ and $G(x)$ satisfy the following properties:

$$\begin{aligned} F(3) &= 2, & G(3) &= 4, & G(0) &= 3 \\ F'(3) &= -1, & G'(3) &= 0, & G'(0) &= 0 \end{aligned}$$

(a) If $S(x) = \frac{F(x)}{G(x)}$, find $S'(3)$. Simplify your answer. (ans : a) $-\frac{1}{4}$ b) 0 c) $\frac{-1}{2}$)

(b) If $T(x) = F(G(x))$, find $T'(0)$. Simplify your answer.

(c) If $U(x) = \ln(F(x))$, find $U'(3)$. Simplify your answer.

In exercises(17 – 35) differentiate the following functions

17) $f(x) = (x^2 - 3x + 8)^3$ (ans : $(6x - 9)(x^2 - 3x + 8)^2$)

18) $f(x) = (17x - 5)^{1000}$ (ans : $17000(17x - 5)^{999}$)

19) $f(x) = [(2x + 1)^{10} + 1]^{10}$ (ans : $200(2x + 1)^9[(2x + 1)^{10} + 1]^9$)

20) $f(x) = \sin(x^2 + 2)$ (ans : $2x \cos(x^2 + 2)$)

21) $f(x) = x^3 \cos\left(\frac{1}{x}\right)$ (ans : $x \sin\left(\frac{1}{x}\right) + 3x^2 \cos\left(\frac{1}{x}\right)$)

22) $f(x) = \frac{\cos 4x}{1 - \sin 4x}$ (ans : $\frac{-4(1 - \sin 4x) \sin 4x + 4 \cos^2 4x}{(1 - \sin 4x)^2}$)

23) $f(x) = \frac{\csc x}{\sqrt{x}}$ (ans : $-\frac{\csc x \cot x}{\sqrt{x}} - \frac{\csc x}{2x^{\frac{3}{2}}}$)

24) $f(x) = x^2 \sec\left(\frac{1}{x}\right)$ (ans : $2x \sec\left(\frac{1}{x}\right) - \sec\left(\frac{1}{x}\right) \tan\left(\frac{1}{x}\right)$)

25) $f(x) = \left(\frac{1 + \cos x}{\sin x}\right)^{-1}$ (ans : $\frac{1}{1 + \cos x}$)

26) $f(x) = \frac{1}{\ln x} + \ln\left(\frac{1}{x}\right)$ (ans : $-\frac{1}{x}\left(\frac{1}{(\ln x)^2} + 1\right)$)

27) $f(x) = \ln \sqrt[3]{\frac{x^2 - 1}{x^2 + 1}}$ (ans : $\frac{2x}{3(x^2 - 1)} - \frac{2x}{3(x^2 + 1)}$)

28) $f(x) = \sqrt{1 + e^{\tan x}}$ (ans : $\frac{\sec^2 x e^{\tan x}}{2\sqrt{1 + e^{\tan x}}}$)

29) $f(x) = e^{-2x} \ln x$ (ans : $e^{-2x}\left(\frac{1}{x} - 2 \ln x\right)$)

30) $f(x) = \sqrt{\ln(e^{2x} - e^{-2x})}$ (ans : $\frac{e^{2x} + e^{-2x}}{(e^{2x} - e^{-2x})(\sqrt{\ln(e^{2x} - e^{-2x})})}$)

31) $f(x) = \log(x^4 + 3x + 1)$ (ans : $\frac{4x^3 + 3}{(\ln 10)(x^4 + 3x + 4)}$)

$$32) f(x) = 8^{x^2+1} \quad (ans: 8^{x^2+1}(2x \ln 8))$$

$$33) f(x) = 7^{\sqrt{x^4+9}} \quad \left(ans: 2x^3 \frac{(\ln 7) 7^{\sqrt{x^4+9}}}{\sqrt{x^4+9}} \right)$$

$$34) f(x) = \log_5 \left| \frac{6x+4}{2x-3} \right| \quad (ans: \left(\frac{\left(\frac{6}{6x+4} - \frac{2}{2x-3} \right)}{\ln 5} \right))$$

$$35) f(x) = \log(\cos^3 x) \quad \left(ans: \frac{-3 \sin x}{\ln 10 \cos x} \right)$$

In exercises (36 – 43) use implicit differentiable to find $\frac{dy}{dx}$

$$36) 3x - 2y + 4 = 2x^2 + 3y - 7x \quad \left(ans : \frac{dy}{dx} = \frac{10-4x}{5} \right)$$

$$37) 2x^3 + x^2y + y^3 = 1 \quad \left(ans : \frac{dy}{dx} = -\frac{6x^2+2xy}{x+3y} \right)$$

$$38) 5x^2 + xy - 4y^2 = 0 \quad \left(ans : \frac{dy}{dx} = \frac{10x-y}{x+8y} \right)$$

$$39) y^2 = x \cos y \quad \left(ans \frac{dy}{dx} = \frac{\cos y}{2y+x \sin y} \right)$$

$$40) y = \sin(3x + 4y) \quad \left(ans : \frac{dy}{dx} = \frac{3 \cos(3x+4y)}{1-4 \cos(3x+4y)} \right)$$

$$41) e^x \cot y = xe^{2y} \quad \left(ans : y' = \frac{e^x \cot y - e^{2y}}{2xe^{2y} + e^x \csc^2 x} \right)$$

$$42) 3y - x^2 + \ln xy = 2 \quad \left(ans : y' = \frac{2x^2y-y}{3yx+x} \right)$$

$$43) e^{xy} - x^3 + 3y^2 = 11 \quad \left(ans : y' = \frac{3x^2 - ye^{xy}}{xe^{xy} + 6y} \right)$$

$$44) \text{ Find an equation of the tangent line to the curve } \sin x + \cos y = x \text{ at } \left(0, \frac{\pi}{2} \right) \quad (ans : y = \frac{\pi}{2})$$

$$45) \text{ Find an equation of the normal line to the curve } x^2 - e^y = xy^2 \text{ at } (-1, 0) \quad (ans : y + 2x + 2 = 0)$$

Derivatives of Higher order

If a function f is a differentiable, then we can find a new function f' called first derivative. Also if f' is itself a differentiable, then we can find a new function f'' called second derivative. So long as we have differentiability, we can continue this more forming $f''' \dots$ etc. The prime notation is not used beyond the third order. Thus the fourth derivative we write $f^{(4)}$ and more generally for the n^{th} derivative $f^{(n)}$

Example :

Find the first four derivative of $f(x) = 3 - x + 5x^2$

Solution :

$$f'(x) = -1 + 10x \quad , \quad f''(x) = 10 \quad , \quad f'''(x) = 0 \quad , \quad f^{(4)} = 0$$

Note :

If $f(x)$ is a polynomial of degree n , then the higher derivatives of degree greater than n are equals zero

Example:

Find $y^{(2)}$ if $y^4 + 3y - 4x^3 = 6x + 1$

Solution:

$$\begin{aligned} \frac{d}{dx}(y^4 + 3y - 4x^3) &= \frac{d}{dx}(6x + 1) \\ 4y^3y' + 3y' - 12x^2 &= 6 \quad \Rightarrow \quad 4y^3y' + 3y' = 6 + 12x^2 \\ &\Rightarrow y'(4y^3 + 3) = 6 + 12x^2 \\ &\Rightarrow y' = \frac{6+12x^2}{(4y^3+3)} \end{aligned}$$

$$\begin{aligned} \text{Now } y'' &= \frac{(6+12x^2)'(4y^3+3) - (6+12x^2)(4y^3+3)'}{(4y^3+3)^2} = \frac{24x(4y^3+3) - (6+12x^2)12y^2y'}{(4y^3+3)^2} \\ &= \frac{24x(4y^3+3) - (6+12x^2)12y^2 \frac{(6+12x^2)}{(4y^3+3)}}{(4y^3+3)^2} \\ &= \frac{\frac{24x(4y^3+3)^2 - (6+12x^2)^2 12y^2}{(4y^3+3)}}{(4y^3+3)^2} \\ &= \frac{24x(4y^3+3)^2 - (6+12x^2)^2 12y^2}{(4y^3+3)^3} \end{aligned}$$

The n^{th} derivative

In the Leibniz notation the derivatives of higher order of the function $y = f(x)$ are written as:

$$\frac{d^2y}{dx^2} = \frac{d}{dx} \left(\frac{dy}{dx} \right) , \quad \frac{d^3y}{dx^3} = \frac{d}{dx} \left(\frac{d^2y}{dx^2} \right) , \quad \dots , \quad \frac{d^ny}{dx^n} = \frac{d}{dx} \left(\frac{d^{n-1}y}{dx^{n-1}} \right)$$

Or

$$\frac{d^2f(x)}{dx^2} = \frac{d}{dx} \left(\frac{df(x)}{dx} \right) , \quad \frac{d^3y}{dx^3} = \frac{d}{dx} \left(\frac{d^2f(x)}{dx^2} \right) , \quad \dots , \quad \frac{d^ny}{dx^n} = \frac{d}{dx} \left(\frac{d^{n-1}f(x)}{dx^{n-1}} \right)$$

Below we calculate the n^{th} derivative of some functions

1) $y = (ax + b)^m$, $a \neq 0$ and $m \in \mathbb{Z}^+$

Case (1) if $n < m$

Differentiating the given function successively w.r.t x we get

$$y' = m(ax + b)^{m-1}a = am(ax + b)^{m-1}$$

$$y'' = am(m-1)(ax + b)^{m-2}a = a^2m(m-1)(ax + b)^{m-2}$$

$$y''' = a^2m(m-1)(m-2)(ax + b)^{m-3}a = a^3m(m-1)(m-2)(ax + b)^{m-3}$$

.

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$$y^{(n)} = a^n m(m-1)(m-2) \dots (m-(n-1)) (ax + b)^{m-n} \quad \dots (*)$$

Case (2) if $n = m$ then from (*) we get

$$y^{(n)} = a^n n(n-1)(n-2) \dots (n-(n-1)) (ax + b)^{n-n}$$

$$y^{(n)} = a^n n(n-1)(n-2) \dots 3 \cdot 2 \cdot 1 = a^n n! \quad \text{because } n! = n(n-1)(n-2) \dots 3 \cdot 2 \cdot 1 \text{ e.g } 3! = 3 \cdot 2 \cdot 1 = 6$$

Case (3) if $n > m$ by (*) we get

$$y^{(n)} = 0 \quad \text{why?}$$

$$\text{Thus } y^{(n)} = \begin{cases} a^n m(m-1)(m-2) \dots (m-(n-1)) (ax + b)^{m-n} & \text{if } n < m \\ a^n n! & \text{if } n = m \\ 0 & \text{if } n > m \end{cases}$$

2) $y = (ax + b)^{-m}$, $a \neq 0$ and $m \in \mathbb{Z}^+$

Replace m by $-m$ in the form (*), we get

$$y^{(n)} = a^n (-m)(-m-1)(-m-2) \dots (-m-(n-1)) (ax + b)^{-m-n}$$

$$y^{(n)} = (-1)^n a^n m(m+1)(m+2) \dots (m+(n-1)) (ax + b)^{-(m+n)}$$

$$y^{(n)} = \frac{(-1)^n a^n m(m+1)(m+2) \dots (m+(n-1))}{(ax + b)^{m+n}} = \frac{(-1)^n a^n (m+n-1)!}{(m-1)! (ax + b)^{m+n}} \dots \quad (*, *)$$

Remark: If $m = 1$ in the form (*, *) then

$$y^{(n)} = \frac{(-1)^n a^n (n)!}{(ax+b)^{1+n}}$$

$$3) y = \ln(ax + b)$$

$$\text{Since } y' = \frac{a}{ax+b} = a(ax+b)^{-1}$$

$$\begin{aligned} \text{Then } y^{(n)} &= \frac{d^{n-1}y'}{dx^{n-1}} = \frac{d^{n-1}}{dx^{n-1}} a(ax+b)^{-1} = a \frac{d^{n-1}(ax+b)^{-1}}{dx^{n-1}} \\ &= a \frac{(-1)^{n-1} a^{n-1} (n-1)!}{(ax+b)^n} \quad (\text{by the above remark}) \\ &= \frac{(-1)^{n-1} a^n (n-1)!}{(ax+b)^n} \end{aligned}$$

$$4) y = a^x, \quad a \in \mathcal{R}^+$$

$$\text{Since } y = a^x, \text{ then } y = e^{x \ln a}$$

$$\Rightarrow y^{(1)} = \ln a \, e^{x \ln a}$$

$$\Rightarrow y^{(2)} = (\ln a)^2 \, e^{x \ln a}$$

$$\Rightarrow y^{(3)} = (\ln a)^3 \, e^{x \ln a}$$

$$\cdot$$

$$\cdot$$

$$\cdot$$

$$\Rightarrow y^{(n)} = (\ln a)^n \, e^{x \ln a} = (\ln a)^n \, a^x$$

$$5) y = \sin(ax + b), \quad a \neq 0$$

$$y' = a \cos(ax + b) = a \sin\left(\frac{\pi}{2} + ax + b\right)$$

$$\begin{aligned} y'' &= a^2 \cos\left(\frac{\pi}{2} + ax + b\right) = a^2 \sin\left(\frac{\pi}{2} + \frac{\pi}{2} + ax + b\right) \\ &= a^2 \sin\left(\frac{2\pi}{2} + ax + b\right) \end{aligned}$$

$$y^{(3)} = a^2 \sin\left(\frac{3\pi}{2} + ax + b\right)$$

$$\cdot$$

$$\cdot$$

$$\cdot$$

$$y^{(n)} = a^n \sin\left(\frac{n\pi}{2} + ax + b\right)$$

$$6) y = \cos(ax + b), \quad a \neq 0$$

$$\text{Exercise (Ans: } y^{(n)} = a^n \cos\left(\frac{n\pi}{2} + ax + b\right)$$

$$7) y = \sin^4 x$$

We first use some elementary trigonometric formulas to reduce the given function into form (6) so that

$$\begin{aligned} \sin^4 x &= \frac{1}{4} (2\sin^2 x)^2 = \frac{1}{4} (1 - \cos 2x)^2 \quad (\because 1 - \cos nx = 2\sin^2\left(\frac{n}{2}x\right)) \\ &= \frac{1}{4} (1 - 2\cos 2x + \cos^2 2x) \\ &= \frac{1}{4} - \frac{1}{2} \cos 2x + \frac{1}{8} (2\cos^2 2x) \\ &= \frac{1}{4} - \frac{1}{2} \cos 2x + \frac{1}{8} (1 + \cos 4x) \\ &= \frac{1}{8} (3 - 4\cos 2x + \cos 4x) \end{aligned}$$

Now we obtain $y^{(n)}$ as in (6)

$$\begin{aligned} y^{(n)} &= \frac{1}{8} \left(D^{(n)}(3) - 4D^{(n)}(\cos 2x) + D^{(n)}(\cos 4x) \right) \\ &= \frac{1}{8} \left(0 - 42^n \cos\left(\frac{n\pi}{2} + 2x\right) + 4^n \cos\left(\frac{n\pi}{2} + 4x\right) \right) \\ &= \frac{1}{8} \left(4^n \cos\left(\frac{n\pi}{2} + 4x\right) - 42^n \cos\left(\frac{n\pi}{2} + 2x\right) \right) \end{aligned}$$

$$7) y = \frac{1}{1-5x+6x^2}$$

$$\text{Since } 1 - 5x + 6x^2 = (1 - 3x)(1 - 2x)$$

$$\text{Then } y = \frac{1}{(1-3x)(1-2x)}$$

$$\begin{aligned} \text{Now let } y &= \frac{1}{(1-3x)(1-2x)} = \frac{A}{1-3x} + \frac{B}{1-2x} \\ \Rightarrow \frac{1}{(1-3x)(1-2x)} &= \frac{A(1-2x)+B(1-3x)}{(1-3x)(1-2x)} \\ \Rightarrow 1 &= A(1-2x) + B(1-3x) \end{aligned}$$

$$\text{At } x = \frac{1}{2} \Rightarrow B = -2$$

$$\text{At } x = \frac{1}{3} \Rightarrow A = 3$$

$$\begin{aligned} \Rightarrow y &= \frac{3}{1-3x} - \frac{2}{1-2x} \Rightarrow y^{(n)} = 3D^n\left(\frac{1}{1-3x}\right) - 2D^n\left(\frac{1}{1-2x}\right) \\ &\Rightarrow y^{(n)} = 3 \frac{(-1)^n(-3)^nn!}{(1-3x)^{n+1}} - 2 \frac{(-1)^n(-2)^nn!}{(1-2x)^{n+1}} \\ &\Rightarrow y^{(n)} = 3 \frac{(3)^nn!}{(1-3x)^{n+1}} - 2 \frac{(2)^nn!}{(1-2x)^{n+1}} \\ &\Rightarrow y^{(n)} = n! \left[\frac{(3)^{n+1}}{(1-3x)^{n+1}} - \frac{(2)^{n+1}}{(1-2x)^{n+1}} \right] \end{aligned}$$

Leibniz's Theorem:

Let u and v the two functions of x both differentiable as many as we wish, then the n^{th} derivative of the product is given by

$$D^n(uv) = D^n u \cdot v + C_1^n \cdot D^{n-1} u \cdot Dv + C_2^n \cdot D^{n-2} u \cdot D^2 v + \dots + C_r^n \cdot D^{n-r} u \cdot D^r v + \dots + u \cdot D^n v$$

$$\text{Where } C_r^n = \frac{n!}{r!(n-r)!}$$

Example: Find the n^{th} derivative of $y = x^4 \sin 3x$

Solution:

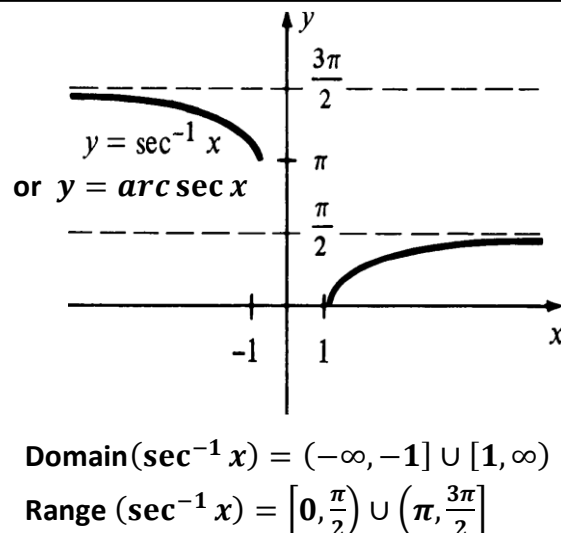
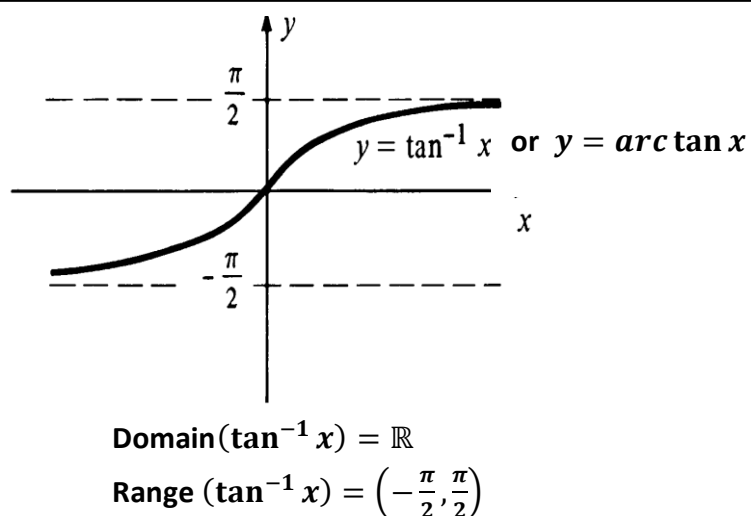
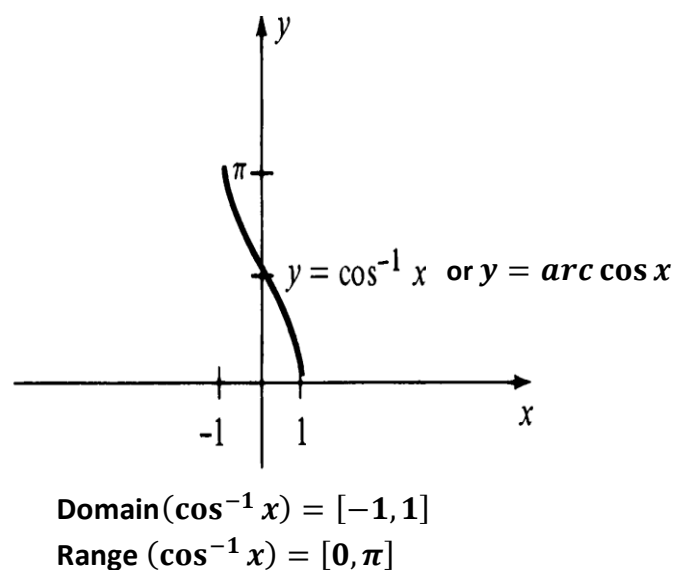
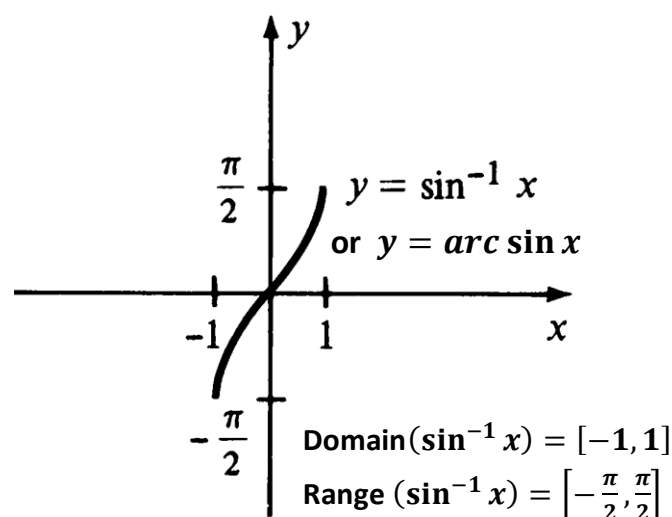
Let $u = \sin 3x$ and $v = x^4$ then

$$\begin{aligned} y^{(n)} &= D^n(x^4 \sin 3x) = x^4 \cdot D^n(\sin 3x) + C_1^n \cdot D x^4 \cdot D^{n-1}(\sin 3x) + C_2^n \cdot D^2 x^4 \cdot D^{n-2}(\sin 3x) + \\ &C_3^n \cdot D^3 x^4 \cdot D^{n-3}(\sin 3x) + C_4^n \cdot D^4 x^4 \cdot D^{n-4}(\sin 3x) + 0 + 0 + \dots \end{aligned}$$

$$\begin{aligned} \Rightarrow y^{(n)} &= x^4 3^n \sin\left(3x + \frac{n\pi}{2}\right) + 4x^3 3^{n-1} \sin\left(3x + \frac{(n-1)\pi}{2}\right) + 6n(n-1)x^2 3^{n-2} \sin\left(3x + \frac{(n-1)\pi}{2}\right) + \\ &4n(n-1)(n-2)x 3^{n-3} \sin\left(3x + \frac{(n-3)\pi}{2}\right) + n(n-1)(n-2)(n-3) 3^{n-4} \sin\left(3x + \frac{(n-4)\pi}{2}\right) \end{aligned}$$

The inverse trigonometric functions

Graphs of Trigonometric functions



Derivative of Inverse Trigonometric function

Theorem : If $g(x)$ is a differentiable function . then by chain rule , we get the following

$$\begin{array}{ll}
 1) \frac{d}{dx} [\sin^{-1}(g(x))] = \frac{g'(x)}{\sqrt{1-(g(x))^2}} & , \quad 2) \frac{d}{dx} [\cos^{-1}(g(x))] = -\frac{g'(x)}{\sqrt{1-(g(x))^2}} \\
 3) \frac{d}{dx} [\tan^{-1}(g(x))] = \frac{g'(x)}{1+(g(x))^2} & , \quad 4) \frac{d}{dx} [\cot^{-1}(g(x))] = -\frac{g'(x)}{1+(g(x))^2} \\
 5) \frac{d}{dx} [\sec^{-1}(g(x))] = \frac{g'(x)}{g(x)\sqrt{(g(x))^2-1}} & , \quad 6) \frac{d}{dx} [\csc^{-1}(g(x))] = -\frac{g'(x)}{g(x)\sqrt{(g(x))^2-1}}
 \end{array}$$

Note that:

$$1) \frac{d}{dx} [\cos^{-1}(g(x))] = -\frac{d}{dx} [\sin^{-1}(g(x))]$$

$$2) \frac{d}{dx} [\cot^{-1}(g(x))] = -\frac{d}{dx} [\tan^{-1}(g(x))]$$

$$3) \frac{d}{dx} [\csc^{-1}(g(x))] = -\frac{d}{dx} [\sec^{-1}(g(x))]$$

Example: Find $\frac{dy}{dx}$ of the following function

$$1) y = \sin^{-1}(x^3) \quad , \quad 2) y = \sec^{-1}(e^x) \quad , \quad 3) y = \sin^{-1}(3x) - \cos^{-1}(3x) \quad , \quad 4) y = \ln(\cos^{-1} x)$$

$$5) x^3 + x \tan^{-1} y = e^y$$

Solution:

$$1) y' = \frac{1}{\sqrt{1-(x^3)^2}} \frac{d}{dx} (x^3) = \frac{3x^2}{\sqrt{1-x^6}}$$

$$2) y' = \frac{1}{e^x \sqrt{(e^x)^2 - 1}} \frac{d}{dx} (e^x) = \frac{e^x}{e^x \sqrt{e^{2x} - 1}}$$

$$= \frac{1}{\sqrt{e^{2x} - 1}}$$

$$3) y' = \frac{1}{\sqrt{1-(3x)^2}} \frac{d}{dx} (3x) - 1 \sqrt{1-(3x)^2} \frac{d}{dx} (3x) = \frac{3}{\sqrt{1-9x^2}} + \frac{3}{\sqrt{1-9x^2}}$$

$$= \frac{6}{\sqrt{1-9x^2}}$$

$$4) y' = \frac{1}{\cos^{-1} x} \frac{d}{dx} (\cos^{-1} x) = \frac{1}{\cos^{-1} x} \left(-\frac{1}{\sqrt{1-x^2}} \frac{dx}{dx} \right)$$

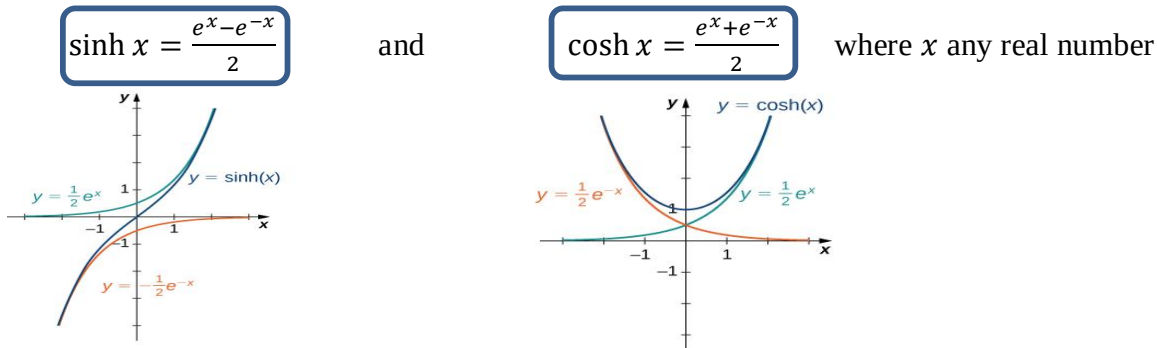
$$= -\frac{1}{(\cos^{-1} x) \sqrt{1-x^2}}$$

$$5) \text{ Exercise } \quad (\text{Ans: } y' = \frac{-(\tan^{-1} y + 3x^2)(1+y^2)}{x - (1+y^2)e^y})$$

The hyperbolic functions

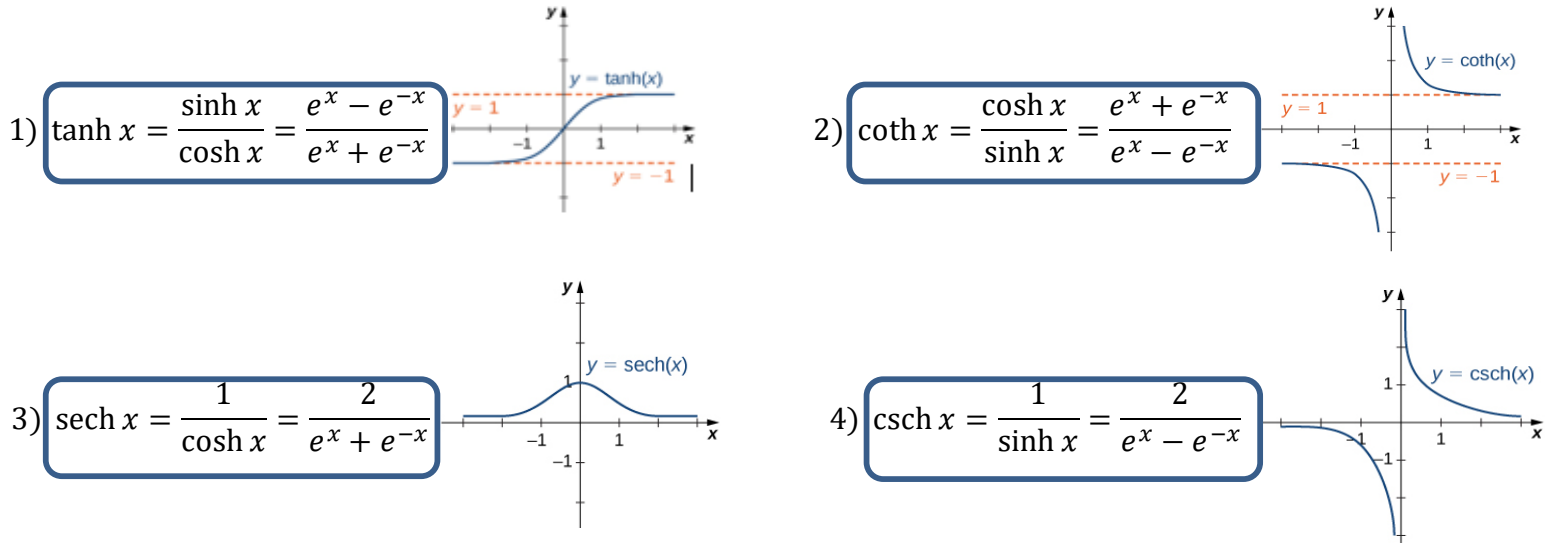
Definition :

The hyperbolic sine function denoted by **sinh** , and the hyperbolic cosine function denoted **cosh** are defined by



Def:

The hyperbolic tangent , cotangent , secant and cosecant functions denoted by **tanh** , **coth** , **sech** and **csch** respectively , are defined as follows



Identities for hyperbolic functions

$$1) \cosh^2 x - \sinh^2 x = 1$$

$$2) \sinh 2x = 2 \sinh x \cosh x$$

$$3) \cosh 2x = \cosh^2 x + \sinh^2 x$$

$$4) \cosh^2 x = \frac{\cosh 2x + 1}{2}$$

$$5) \sinh^2 x = \frac{\cosh 2x - 1}{2}$$

$$6) \tanh^2 x = 1 - \operatorname{sech}^2 x$$

$$7) \coth^2 x = 1 + \operatorname{csch}^2 x$$

Derivative of the hyperbolic functions

Theorem : If $g(x)$ is a differentiable function . then by chain rule , we get the following

$$1) \frac{d}{dx}(\sinh(g(x))) = g'(x) \cosh(g(x))$$

$$2) \frac{d}{dx}(\cosh(g(x))) = g'(x) \sinh(g(x))$$

$$3) \frac{d}{dx}(\tanh(g(x))) = g'(x) \operatorname{sech}^2(g(x))$$

$$4) \frac{d}{dx}(\coth(g(x))) = -g'(x) \operatorname{csch}^2(g(x))$$

$$5) \frac{d}{dx}(\operatorname{sech}(g(x))) = -g'(x) \operatorname{sech}(g(x)) \tanh(g(x))$$

$$6) \frac{d}{dx}(\operatorname{csch}(g(x))) = -g'(x) \operatorname{csc} h(g(x)) \cot h(g(x))$$

Example: Find the derivative of the following functions

$$1) f(x) = \cosh(x^2 + 1)$$

$$2) f(x) = x^3 \tanh^3 \sqrt{x}$$

$$3) f(x) = \operatorname{csch}(e^{2x})$$

$$4) f(x) = \ln(\tanh 2x)$$

Solution :

$$1) f'(x) = (x^2 + 1)' \sinh(x^2 + 1) = 2x \sinh(x^2 + 1)$$

$$\begin{aligned} 2) f'(x) &= (x^3)' \tanh^3 \sqrt{x} + x^3 (\tanh^3 \sqrt{x})' \\ &= 3x^2 \tanh^3 \sqrt{x} + 3 \tanh^2 \sqrt{x} (\tanh \sqrt{x})' \\ &= 3x^2 \tanh^3 \sqrt{x} + \frac{3 \tanh^2 \sqrt{x} \operatorname{sech}^2 \sqrt{x}}{2\sqrt{x}} \end{aligned}$$

$$\begin{aligned} 3) f'(x) &= -(e^{2x})' \operatorname{csc} h(e^{2x}) \cot h(e^{2x}) \\ &= -2e^{2x} \operatorname{csc} h(e^{2x}) \cot h(e^{2x}) \end{aligned}$$

$$4) \text{ Exercise (Ans: } y' = \frac{2 \operatorname{sech}^2 2x}{\tanh 2x} \text{)}$$

Derivative of the inverse hyperbolic functions**Theorem :** If $g(x)$ is a differentiable function . then by chain rule , we get the following

$$\begin{aligned}
1) \frac{d}{dx}(\sinh^{-1}(g(x))) &= \frac{g'(x)}{\sqrt{1+(g(x))^2}} & , & & 2) \frac{d}{dx}(\cosh^{-1}(g(x))) &= \frac{g'(x)}{\sqrt{(g(x))^2-1}} \\
3) \frac{d}{dx}(\tanh^{-1}(g(x))) &= \frac{g'(x)}{1-(g(x))^2} & , & & 4) \frac{d}{dx}(\coth^{-1}(g(x))) &= \frac{g'(x)}{1-(g(x))^2} \\
5) \frac{d}{dx}(\operatorname{sech}^{-1}(g(x))) &= \frac{-g'(x)}{g(x)\sqrt{1-(g(x))^2}} & , & & 6) \frac{d}{dx}(\operatorname{csch}^{-1}(g(x))) &= \frac{-g'(x)}{|g(x)|\sqrt{1+(g(x))^2}}
\end{aligned}$$

Example: Find y' for each the following functions

$$1) y = \sinh^{-1}\left(\frac{1}{x}\right) \quad 2) y = \cosh^{-1}(\sinh^{-1}x) \quad 3) y = \ln(\cosh^{-1}x) \quad 4) y = e^{\operatorname{sech}^{-1}(\sqrt{x})}$$

Solution:

$$\begin{aligned}
1) y' &= \frac{\left(\frac{1}{x}\right)'}{\sqrt{1+\left(\frac{1}{x}\right)^2}} = \frac{-\frac{1}{x^2}}{\sqrt{\frac{x^2+1}{x^2}}} \\
&= -\frac{1}{x^2} \frac{|x|^2}{\sqrt{x^2+1}} \\
&= -\frac{1}{|x|\sqrt{x^2+1}} \quad (\because x^2 = |x|^2)
\end{aligned}$$

$$2) y' = \frac{(\sinh^{-1}x)'}{\sqrt{(\sinh^{-1}x)^2-1}} = \frac{1}{\sqrt{1+x^2}\sqrt{(\sinh^{-1}x)^2-1}}$$

$$\begin{aligned}
3) y' &= \frac{(\cosh^{-1}x)'}{(\cosh^{-1}x)} = \frac{\frac{1}{\sqrt{x^2-1}}}{(\cosh^{-1}x)} \\
&= \frac{1}{(\cosh^{-1}x)\sqrt{x^2-1}}
\end{aligned}$$

$$4) \text{ Exercise (Ans: } y' = -\frac{e^{\operatorname{sech}^{-1}(\sqrt{x})}}{2x\sqrt{1-x}} \text{)}$$

Exercises

In exercises (1 – 4) find the third derivative of the following functions

- 1) $y = xe^x$ (ans : $y^{(3)} = 3e^x + xe^x$) 2) $y = x^2 \cos x$ (ans : $y^{(3)} = -6 \sin x - 2x \cos x - x^2 \cos x$)
 3) $y = \frac{x+2}{x-1}$ (ans : $y^{(3)} = (-1)^3 3 \frac{3!}{(x-1)^4}$) 4) $y = e^x + e^{-x}$ (ans : $y^{(3)} = e^x - e^{-x}$)

In exercises (5 – 10) find the indicated derivatives of the following

- 5) $\frac{d}{dx} \left[x \frac{d}{dx} (x - x^2) \right]$ (ans : $1 - 4x$) 6) $\frac{d^4}{dx^4} [3x - x^4]$ (ans : -24)
 7) $\frac{d^2}{dx^2} \left[(x^2 - 3x) \frac{d}{dx} (x - x^{-1}) \right]$ (ans : $2 + 12x^{-3}$) 8) $\frac{d^2 y}{dx^2} \Big|_{x=1}$ s.t $y = 4t^{-3} + 3t^2$ (ans : 54)
 9) $g^{(2)}(0)$ s.t $g(x) = \frac{e^x}{x+1}$ (ans : 1) 10) $\frac{d^6}{dx^6} x^{-1} \Big|_{x=-1}$ (ans : -720)

In exercises (11 – 15) prove the following

- 11) If $y = e^{3x}$ then $y'' - 5y' + 6y = 0$ 12) If $x = \sin y$ then $xy'' = \sec^3 y \sin^2 y$
 13) If $y = -xy - x$ then $x^3 y'' - 2y^3 = 0$ 14) If $e^{xy} = 2$ then $y'' = \frac{2y}{x^2}$
 15) If $\ln y + \ln x = 1$ then $y'' = \frac{2y}{x^2}$

In exercises (16 – 21) find the n^{th} derivative

- 16) $y = \frac{x+3}{(x-1)(x+2)}$ (ans : $y^{(n)} = \frac{1}{3}(-1)^n n! \left[\frac{4}{(x-1)^{n+1}} - \frac{1}{(x+2)^{n+1}} \right]$
 17) $y = \frac{2x+3}{x^2-4}$ (ans : $y^{(n)} = \frac{1}{4}(-1)^n n! \left[\frac{5}{(x-2)^{n+1}} + \frac{3}{(x+2)^{n+1}} \right]$
 18) $y = \ln(4x^2 - 9)$ (ans : $y^{(n)} = (-1)^{n-1} 2^n (n-1)! \left[\frac{1}{(2x-3)^n} + \frac{1}{(2x+3)^n} \right]$
 19) $y = \sin 5x \sin 3x$ (ans : $y^{(n)} = \frac{1}{2} \left[2^n \cos(2x + \frac{n}{2}\pi) - 8^n \cos(8x + \frac{n}{2}\pi) \right]$

Hint ($2 \sin ax \sin bx = \cos((b-a)x) - \cos((a+b)x)$)

- 20) $y = x \ln x$ (ans : $y^{(n)} = \frac{(-1)^{n-2} (n-2)!}{x^{n-1}}$
 21) $y = x^2 e^{3x}$ (ans : $y^{(n)} = 3^{n-2} e^{3x} [9x^2 + 6nx + n(n-1)]$

In exercises (21 – 25) find the third derivative of the following functions

- 21) $y = e^{-x} \operatorname{arcsec} e^{-x}$ (ans : $y' = -\frac{e^{-x}}{\sqrt{e^{-2x}-1}} - e^{-x} \sec e^{-x}$
 22) $y = (1 + \cos^{-1} 3x)^3$ (ans : $y' = -\frac{9}{\sqrt{1-9x^2}} (1 + \cos^{-1} 3x)^2$
 23) $y = \ln(\operatorname{arctan} x^2)$ (ans : $y' = \frac{2x}{(1+x^4) \operatorname{arctan}^{-1} x^2}$) 24) $y = \frac{1}{\sin^{-1} x}$ (ans : $y' = -\frac{1}{(\sin^{-1} x)^2 \sqrt{1-x^2}}$
 25) $x^2 + x \sin^{-1} y = ye^x$ (ans : $y' = \frac{ye^x - \sin^{-1} y 2x}{x\sqrt{1-y^2} - e^x}$) 26) $f(x) = \sinh 5x$ (ans : $5 \cosh 5x$)
 27) $f(x) = \frac{\operatorname{sech} x^2}{x^2+1}$ (ans : $-\frac{2x \operatorname{sech} x^2 ((x^2+1) \tanh x^2 + 1)}{(x^2+1)^2}$) 28) $f(x) = \ln(\sinh 2x)$ (ans : $2 \coth 2x$)
 29) $f(x) = \frac{1}{\tanh x+1}$ (ans : $-\operatorname{sech}^2 x (\tanh x + 1)^2$) 30) $f(x) = \sinh^{-1} 5x$ (ans : $\frac{5}{(\sqrt{25x^2+1})}$)
 31) $f(x) = \cosh^{-1} \sqrt{x}$ (ans : $\frac{1}{2\sqrt{x}\sqrt{x-1}}$) 32) $f(x) = \tanh^{-1}(x^2 - 1)$ (ans : $\frac{2x}{2x^2 - x^4}$)
 32) $f(x) = x \sinh^{-1} \frac{1}{x}$ (ans : $-\frac{|x|}{x\sqrt{x^2+1}} + \sinh^{-1} \frac{1}{x}$) 33) $f(x) = \ln(\cosh^{-1} 4x)$ (ans : $\frac{4}{\cosh^{-1}(4x) \sqrt{16x^2-1}}$)

Applications of the derivative**Indeterminate forms and i'hôpital's rule****The forms 0/0 and ∞/∞**

Indeterminate form	Limit form: $\lim_{x \rightarrow c} \frac{f(x)}{g(x)}$
$\frac{0}{0}$	$\lim_{x \rightarrow c} f(x) = 0 \text{ and } \lim_{x \rightarrow c} g(x) = 0$
$\frac{\infty}{\infty}$	$\lim_{x \rightarrow c} f(x) = \infty \text{ or } -\infty \text{ and } \lim_{x \rightarrow c} g(x) = \infty \text{ or } -\infty$

I'Hôpital's Rule:

Suppose that f and g are differentiable on an open interval (a, b) containing c , except possibly at c itself. If $f(x)/g(x)$ has the Indeterminate form $0/0$ or ∞/∞ at $x = c$ and if $g'(x) \neq 0$ for $x \neq c$, then

$$\lim_{x \rightarrow c} \frac{f(x)}{g(x)} = \lim_{x \rightarrow c} \frac{f'(x)}{g'(x)},$$

provided either

$$\lim_{x \rightarrow c} \frac{f'(x)}{g'(x)} \text{ exists or } \lim_{x \rightarrow c} \frac{f'(x)}{g'(x)} = \infty$$

Example: Find

$$1) \lim_{x \rightarrow 0} \frac{\cos x + 2x - 1}{3x} \quad 2) \lim_{x \rightarrow \infty} \frac{\ln x}{\sqrt{x}} \quad 3) \lim_{x \rightarrow \infty} \frac{e^{3x}}{x^2} \text{ (if it exists)}$$

Solution:

1) The indeterminate form is $0/0$. By I'Hôpital's Rule,

$$\lim_{x \rightarrow 0} \frac{\cos x + 2x - 1}{3x} = \lim_{x \rightarrow 0} \frac{-\sin x + 2}{3} = \frac{2}{3}$$

2) The indeterminate form is ∞/∞ . By I'Hôpital's Rule,

$$\lim_{x \rightarrow \infty} \frac{\ln x}{\sqrt{x}} = \lim_{x \rightarrow \infty} \frac{1/x}{1/(2\sqrt{x})} = \lim_{x \rightarrow \infty} \frac{2\sqrt{x}}{x} = \lim_{x \rightarrow \infty} \frac{2}{\sqrt{x}} = 0$$

3) The indeterminate form is ∞/∞ . By I'Hôpital's Rule,

$$\lim_{x \rightarrow \infty} \frac{e^{3x}}{x^2} = \lim_{x \rightarrow \infty} \frac{3e^{3x}}{2x}$$

The last quotient has the indeterminate form ∞/∞ , so we apply I'Hôpital's Rule a second time, obtaining

$$\lim_{x \rightarrow \infty} \frac{3e^{3x}}{2x} = \lim_{x \rightarrow \infty} \frac{9e^{3x}}{2} = \infty$$

Thus, $\frac{e^{3x}}{x^2}$ has no limit, increasing without bound as $x \rightarrow \infty$.

The forms $0 \cdot \infty$, 0^0 , ∞^0 , 1^∞ , and $\infty - \infty$

Indeterminate form	Limit form: $\lim_{x \rightarrow c} [f(x)g(x)]$
$0 \cdot \infty$	$\lim_{x \rightarrow c} f(x) = 0$ and $\lim_{x \rightarrow c} g(x) = \infty$ or $-\infty$

Guidelines for Investigating $\lim_{x \rightarrow c} [f(x)g(x)]$ for the Form $0 \cdot \infty$

- 1 Write $f(x)g(x)$ as

$$\frac{f(x)}{1/g(x)} \quad \text{or} \quad \frac{g(x)}{1/f(x)}.$$

- 2 Apply I'Hôpital's Rule to the resulting indeterminate form $0/0$ or ∞/∞ .

Example: Find 1) $\lim_{x \rightarrow 0^+} x^2 \ln x$ 2) $\lim_{x \rightarrow (\pi/2)^-} (2x - \pi) \sec x$.

Solution:

- 1) The indeterminate form is $0 \cdot \infty$. Applying guideline (1), we write

$$x^2 \ln x = \frac{\ln x}{1/x^2}$$

Because the quotient on the right has the indeterminate form ∞/∞ at $x = 0$, we may apply I'Hôpital's rule:

$$\lim_{x \rightarrow 0^+} x^2 \ln x = \lim_{x \rightarrow 0^+} \frac{\ln x}{1/x^2} = \lim_{x \rightarrow 0^+} \frac{1/x}{-2/x^3} = \lim_{x \rightarrow 0^+} \frac{x^3}{-2x} = \lim_{x \rightarrow 0^+} \frac{x^2}{-2} = 0$$

- 2) The indeterminate form is $0 \cdot \infty$. Applying guideline (1), we write

$$(2x - \pi) \sec x = \frac{2x - \pi}{1/\sec x} = \frac{2x - \pi}{\cos x}$$

Because the quotient on the right has the indeterminate form $0/0$ at $x = \pi/2$, we may apply I'Hôpital's rule:

$$\lim_{x \rightarrow (\pi/2)^-} (2x - \pi) \sec x = \lim_{x \rightarrow (\pi/2)^-} \frac{2x - \pi}{\cos x} = \lim_{x \rightarrow (\pi/2)^-} \frac{2}{-\sin x} = \frac{2}{-1} = -2$$

Indeterminate form	Limit form: $\lim_{x \rightarrow c} f(x)^{g(x)}$
0^0	$\lim_{x \rightarrow c} f(x) = 0 \text{ and } \lim_{x \rightarrow c} g(x) = 0$
∞^0	$\lim_{x \rightarrow c} f(x) = \infty \text{ or } -\infty \text{ and } \lim_{x \rightarrow c} g(x) = 0$
1^∞	$\lim_{x \rightarrow c} f(x) = 1 \text{ and } \lim_{x \rightarrow c} g(x) = \infty \text{ or } -\infty$

Guidelines for Investigating $\lim_{x \rightarrow c} f(x)^{g(x)}$ for the Forms 0^0 , 1^∞ , and ∞^0

- 1 Let $y = f(x)^{g(x)}$
- 2 Take natural logarithms in guideline(1):

$$\ln y = \ln f(x)^{g(x)} = g(x) \ln f(x)$$
- 3 Investigate $\lim_{x \rightarrow c} \ln y = \lim_{x \rightarrow c} [g(x) \ln f(x)]$ and conclude the following:
 - a) If $\lim_{x \rightarrow c} \ln y = L$, then $\lim_{x \rightarrow c} y = e^L$
 - b) If $\lim_{x \rightarrow c} \ln y = \infty$, then $\lim_{x \rightarrow c} y = \infty$
 - c) If $\lim_{x \rightarrow c} \ln y = -\infty$, then $\lim_{x \rightarrow c} y = 0$

Example: Find $\lim_{x \rightarrow 0^+} x^x$

Solution: The indeterminate form is 0^0 . Using Guidelines, we proceed as follows:

Guideline 1 $y = x^x$

Guideline 2 $\ln y = x \ln x$

Guideline 3 This expression form $0 \cdot \infty$. We apply guideline (1)

$$x \ln x = \frac{\ln x}{1/x}$$

Since the quotient on the right has the indeterminate form ∞/∞ at $x = 0$, we may apply l'Hôpital's rule:

$$\lim_{x \rightarrow 0^+} \ln y = \lim_{x \rightarrow 0^+} x \ln x = \lim_{x \rightarrow 0^+} \frac{\ln x}{1/x} = \lim_{x \rightarrow 0^+} \frac{1/x}{-1/x^2} = \lim_{x \rightarrow 0^+} \left(\frac{-x^2}{x} \right) = \lim_{x \rightarrow 0^+} (-x) = 0$$

Since $\lim_{x \rightarrow 0^+} \ln y = 0$, by guideline (3a), we have

$$\lim_{x \rightarrow 0^+} x^x = \lim_{x \rightarrow 0^+} y = e^0 = 1$$

Indeterminate form	Limit form: $\lim_{x \rightarrow c} [f(x) - g(x)]$
$\infty - \infty$	$\lim_{x \rightarrow c} f(x) = \infty$ and $\lim_{x \rightarrow c} g(x) = \infty$

Example: Find $\lim_{x \rightarrow 0^+} \left(\frac{1}{e^x - 1} - \frac{1}{x} \right)$

Solution: The form is $\infty - \infty$; however, if the difference is written as a single fraction, then

$$\lim_{x \rightarrow 0^+} \left(\frac{1}{e^x - 1} - \frac{1}{x} \right) = \lim_{x \rightarrow 0^+} \frac{x - e^x + 1}{xe^x - x}$$

This gives us the indeterminate form 0/0. It is necessary to apply l'Hôpital's rule twice, since the first Application. leads to the indeterminate form 0/0. Thus,

$$\lim_{x \rightarrow 0^+} \frac{x - e^x + 1}{xe^x - x} = \lim_{x \rightarrow 0^+} \frac{1 - e^x}{xe^x + e^x - 1} = \lim_{x \rightarrow 0^+} \frac{-e^x}{xe^x + 2e^x} = -\frac{1}{2}$$

Exercises

In exercises(1 – 20) Find the limit, if it exists.

$$1) \lim_{x \rightarrow 0} \frac{\sin x}{2x} \quad \left(\text{ans: } \frac{1}{2} \right) \quad 2) \lim_{x \rightarrow 25} \frac{\sqrt{x-1}-2}{x^2-5} \quad \left(\text{ans: } \frac{1}{40} \right) \quad 3) \lim_{x \rightarrow 2} \frac{2x^2-5x+2}{5x^2-7x-6} \quad \left(\text{ans: } \frac{3}{13} \right)$$

$$4) \lim_{x \rightarrow 0} \frac{\sin x - x}{\tan x - x} \quad \left(\text{ans: } -\frac{1}{2} \right) \quad 5) \lim_{x \rightarrow 0} \frac{x - \sin x}{x^3} \quad \left(\text{ans: } \frac{1}{6} \right) \quad 6) \lim_{x \rightarrow 0} \lim_{x \rightarrow \frac{\pi}{2}} \frac{1 + \sin x}{\cos^2 x} \quad \left(\text{ans: } \infty \right)$$

$$7) \lim_{x \rightarrow \left(\frac{\pi}{2}\right)^-} \frac{2 + \sec x}{3 \tan x} \quad \left(\text{ans: } \frac{1}{13} \right) \quad 8) \lim_{x \rightarrow 0^+} \frac{\ln \sin x}{\ln \sin 2x} \quad \left(\text{ans: } 1 \right) \quad 9) \lim_{x \rightarrow 0} \frac{x \cos x + e^{-x}}{x^2} \quad \left(\text{ans: } \infty \right)$$

$$10) \lim_{x \rightarrow \infty} \frac{x \ln x}{x + \ln x} \quad \left(\text{ans: } \infty \right) \quad 11) \lim_{x \rightarrow 0} \frac{\sin^{-1} 2x}{\sin^{-1} x} \quad \left(\text{ans: } 2 \right) \quad 12) \lim_{x \rightarrow -\infty} \frac{3-3^x}{5-5^x} \quad \left(\text{ans: } \frac{3}{5} \right)$$

$$13) \lim_{x \rightarrow 1} \frac{x^4 - x^3 - 3x^2 + 5x - 2}{x^4 - 5x^3 + 9x^2 - 7x + 2} \quad \left(\text{ans: } -3 \right) \quad 14) \lim_{x \rightarrow 0} \frac{x - \tan^{-1} x}{x \sin x} \quad \left(\text{ans: } 0 \right) \quad 15) \lim_{x \rightarrow 0^+} x \ln x \quad \left(\text{ans: } 0 \right)$$

$$16) \lim_{x \rightarrow 0} e^{-x} \sin x \quad \left(\text{ans: } 0 \right) \quad 17) \lim_{x \rightarrow \infty} \left(1 + \frac{1}{x} \right)^{5x} \quad \left(\text{ans: } e^5 \right) \quad 18) \lim_{x \rightarrow \left(\frac{\pi}{2}\right)^-} (\tan x)^x \quad \left(\text{ans: } \infty \right)$$

$$19) \lim_{x \rightarrow 0^-} \left(\frac{1}{x} + \frac{1}{\sin x} \right) \quad \left(\text{ans: } -\infty \right)$$

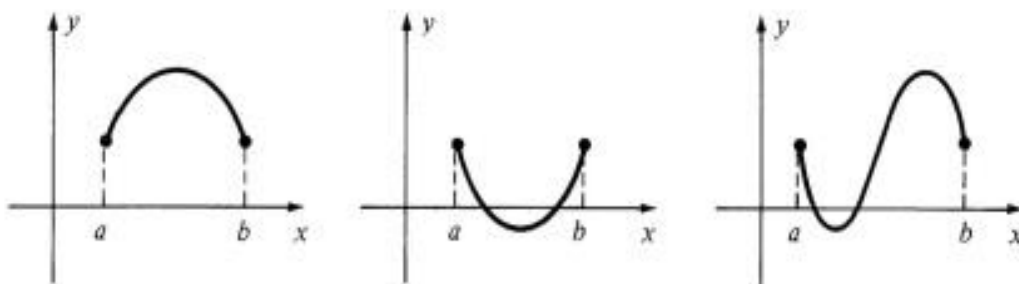
Rolle's Theorem and the Mean Value Theorem**Rolle's Theorem**

If a function f is

- 1) Continuous on a closed interval $[a, b]$
- 2) Differentiable on the open interval (a, b)
- 3) $f(a) = f(b)$

Then there exists a number c in (a, b) such that $f'(c) = 0$

Some typical graphs of functions of this type

**Corollary:**

If a function f is continuous on a closed interval $[a, b]$ and if $f(a) = f(b)$, then f has at least one critical number in the open interval (a, b) .

Example:

Prove that the function f defined by $f(x) = 4x^2 - 20x + 29$ satisfies the hypotheses of the Rolle's Theorem on the interval $[1, 4]$ and find all numbers c in the interval $(1, 4)$ such that $f'(c) = 0$.

Solution:

Since f is a polynomial function it is continuous and differentiable for all real numbers. In particular, It is continuous on $[1, 4]$ and differentiable on $(1, 4)$.

$$f(1) = 4(1)^2 - 20(1) + 29 = 13 \quad \text{and} \quad f(4) = 4(4)^2 - 20(4) + 29 = 13$$

Then $f(1) = f(4)$. Hence f satisfies the hypotheses of the Rolle's Theorem on the interval $[1, 4]$

According to Rolle's Theorem, there exists a number c in $(1, 4)$ such that $f'(c) = 0$.

$$\text{Put } f'(x) = 0 \Rightarrow 8x - 20 = 0 \Rightarrow x = \frac{5}{2} \in (1, 4) \quad \text{Hence } c = \frac{5}{2} \in (1, 4)$$

$$\left[f' \left(\frac{5}{2} \right) = 8 \left(\frac{5}{2} \right) - 20 = 20 - 20 = 0 \right]$$

The Mean Value Theorem

If a function f is

1) Continuous on a closed interval $[a, b]$

2) Differentiable on the open interval (a, b)

Then there exists a number c in (a, b) such that

$$f'(c) = \frac{f(b)-f(a)}{b-a} \quad [\text{or } f(b) - f(a) = f'(c)(b - a)].$$

Example:

Prove that the function f defined by $f(x) = x^3 - 8x - 5$ satisfies the hypotheses of the Mean Value Theorem on the interval $[1, 4]$ and find all numbers c in the interval $(1, 4)$ that satisfies the conclusion of the theorem.

Solution:

Since f is a polynomial function it is continuous and differentiable for all real numbers. In particular, It is continuous on $[1, 4]$ and differentiable on $(1, 4)$.

Hence f satisfies the hypotheses of the Mean Value Theorem on the interval $[1, 4]$

According to Mean Value Theorem, there exists a number c in $(1, 4)$ such that $f'(c) = \frac{f(b)-f(a)}{b-a}$.

$$\text{Since } f'(x) = 3x^2 - 8 \Rightarrow f'(c) = 3c^2 - 8, f(1) = -12, f(4) = 27$$

$$\text{This is equivalent to } 3c^2 - 8 = \frac{27 - (-12)}{4 - 1} \Rightarrow 3c^2 - 8 = \frac{39}{3} \Rightarrow 3c^2 - 8 = 13$$

$$\Rightarrow c^2 = \frac{21}{3} \Rightarrow c^2 = 7 \Rightarrow c = \pm\sqrt{7}$$

Since $-\sqrt{7} \notin (1, 4)$, then $c = \sqrt{7} \in (1, 4)$

Exercises

In Exercises **1-4** show that f satisfies the hypotheses of Rolle's Theorem on the indicated interval $[a, b]$ and find all numbers c in (a, b) such that $f'(c) = 0$.

1 $f(x) = 3x^2 - 12x + 11, [0, 4]$

2 $f(x) = 5 - 12x - 2x^2, [-7, 1]$

3 $f(x) = x^4 + 4x^2 + 1, [-3, 3]$

4 $f(x) = x^3 - x, [-1, 1]$

In Exercises **5-6** determine whether the function f satisfies the hypotheses of the Mean Value Theorem on the indicated interval $[a, b]$ and if so, find all numbers c in (a, b) such that $f(b) - f(a) = f'(c)(b - a)$.

5 $f(x) = x^3 + 1, [-2, 4]$

6 $f(x) = 5x^2 - 3x + 1, [1, 3]$

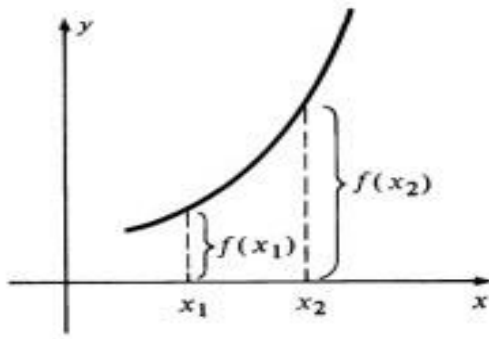
Answers

1 $c = 2$ **2** $c = -3$ **3** $c = 0$ **4** $c = -1/\sqrt{3}$ and $c = 1/\sqrt{3}$ **5** $c = 2$ **6** $c = 1.9$

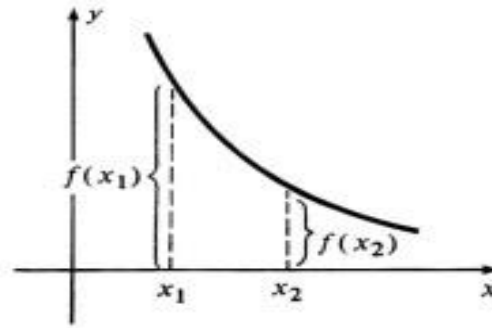
Local extrema of functions

Definition: Let a function f be defined on an interval I and let x_1, x_2 denote numbers in I

- (i) f is **increasing** on I if $f(x_1) < f(x_2)$ whenever $x_1 < x_2$.
- (ii) f is **decreasing** on I if $f(x_1) > f(x_2)$ whenever $x_1 < x_2$.
- (iii) f is **constant** on I if $f(x_1) = f(x_2)$ for every x_1 and x_2 .



(i) Increasing function



(ii) Decreasing function

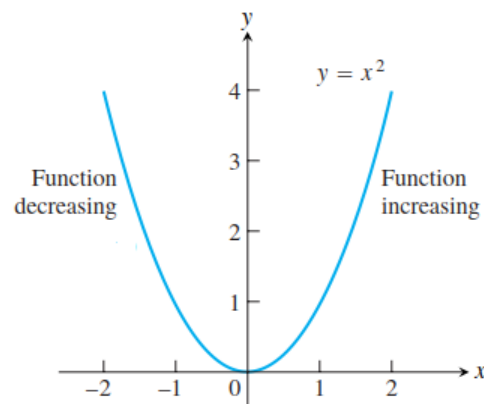
Example: Prove that $f(x) = x^2$ is increasing on the interval $[0, \infty)$ and decreasing on the interval $(-\infty, 0]$

Solution: Let $x_1, x_2 \in [0, \infty)$, $0 \leq x_1 < x_2$ then $x_1^2 < x_2^2$ that is $f(x_1) < f(x_2)$

Then f is increasing on the interval $[0, \infty)$.

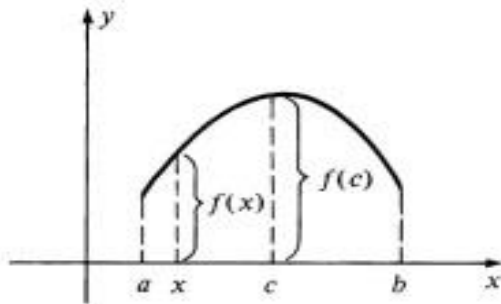
Let $x_1, x_2 \in (-\infty, 0]$, $x_2 > x_1 \geq 0$ then $x_1^2 > x_2^2$ that is $f(x_1) > f(x_2)$

Then f is decreasing on the interval $(-\infty, 0]$

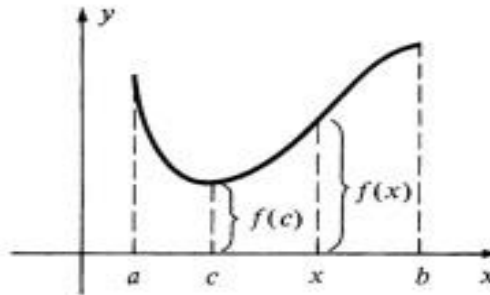


Definition: Let a function f be defined on an interval I and let c be a number in I

- (i) $f(c)$ is the **maximum value** of f on I if $f(x) \leq f(c)$ for every x in I
- (ii) $f(c)$ is the **minimum value** of f on I if $f(x) \geq f(c)$ for every x in I



(i) Maximum value $f(c)$



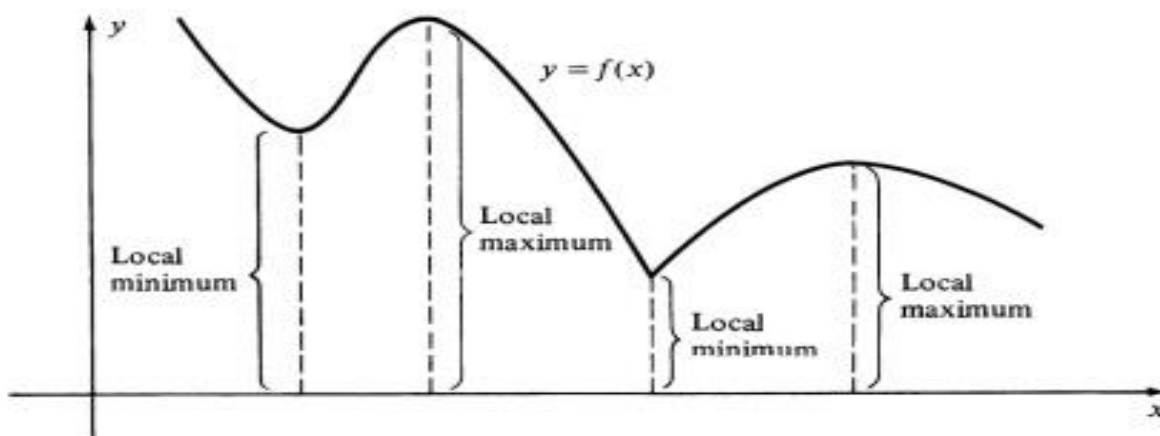
(ii) Minimum value $f(c)$

Theorem: If a function f is continuous on a closed interval $[a, b]$, then f takes on a minimum value or a maximum value at least once in $[a, b]$.

The extrema also called the **absolute minimum** and **absolute maximum values** for f on an interval.

Definition: Let c be a number in the domain of the function f .

- (i) $f(c)$ is a **local maximum** of f if there exists an open interval (a, b) containing c such that $f(x) \leq f(c)$ for all x in (a, b) .
- (ii) $f(c)$ is a **local minimum** of f if there exists an open interval (a, b) containing c such that $f(x) \geq f(c)$ for all x in (a, b) .

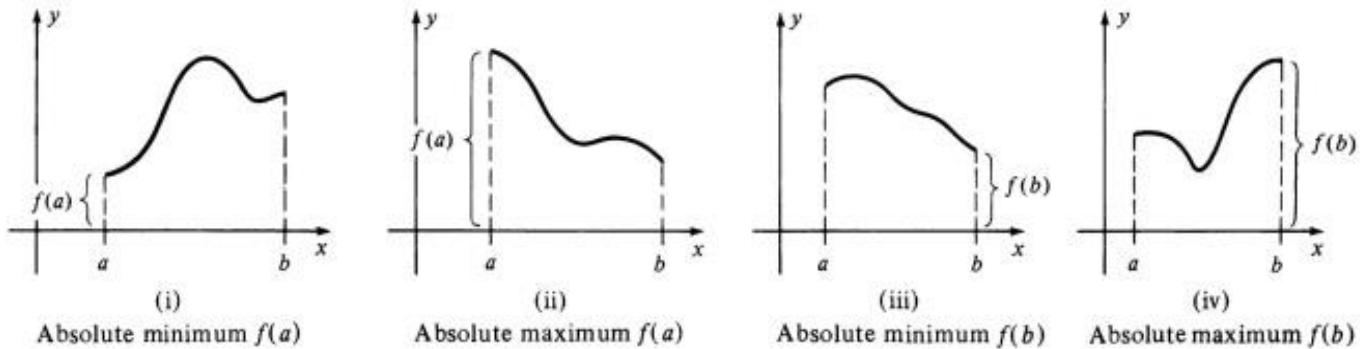


Theorem: If a function f has a local extremum at a number c in an open interval, then either $f'(c) = 0$ or $f'(c)$ does not exist.

Corollary: If $f'(c)$ exists and $f'(c) \neq 0$, then $f(c)$ is not a local extremum of the function f .

Theorem: If a function f is continuous on a closed interval $[a, b]$ and has its maximum or minimum value at a number c in the open interval (a, b) , then either $f'(c) = 0$ or $f'(c)$ does not exist.

Def: A number c in the domain of a function f is a **critical number** of f if either $f'(c) = 0$ or $f'(c)$ does not exist.



Guidelines for finding the Absolute Extrema of a Continuous Function f on a Closed Interval $[a, b]$

- 1 Find all the critical numbers of f .
- 2 Calculate $f(c)$ for each critical number .
- 3 Calculate $f(a)$ and $f(b)$.
- 4 The absolute maximum and minimum of f on $[a, b]$ are the largest and smallest of the functional values calculated in 2 and 3.

Example: If $(x) = x^3 - 12x$, find the absolute maximum and minimum values of f on the closed interval $[-3, 5]$, Sketch the graph of f .

Solution: the endpoints extremum are -3 and 5

$$f'(x) = 3x^2 - 12 = 3(x^2 - 4) = 3(x + 2)(x - 2)$$

$$\text{Put } f'(x) = 0 \Rightarrow 3(x + 2)(x - 2) = 0 \Rightarrow (x + 2)(x - 2) = 0$$

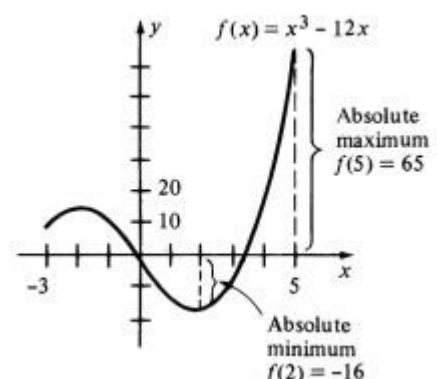
$$\Rightarrow x = \pm 2 \in [-3, 5] \text{ , then the critical numbers is } -2, 2$$

$$f(-2) = (-2)^3 - 12(-2) = -8 + 24 = 16$$

$$f(2) = (2)^3 - 12(2) = 8 - 24 = -16$$

$$f(-3) = (-3)^3 - 12(-3) = -27 + 36 = 9$$

$$f(5) = (5)^3 - 12(5) = 125 - 60 = 65$$



Then the absolute maximum is $f(5) = 65$ and the absolute minimum is $f(2) = -16$

$f(-2)$ is a local maximum and $f(2)$ is a local minimum

Example: Find the critical numbers of f if $f(x) = (x + 5)^2 \sqrt[3]{x - 4}$.

$$\begin{aligned} \text{Solution: } f'(x) &= (x + 5)^2 \frac{1}{3}(x - 4)^{-2/3} + 2(x + 5)(x - 4)^{1/3} \\ &= \frac{(x + 5)^2}{3(x - 4)^{2/3}} + 2(x + 5)(x - 4)^{1/3} \\ &= \frac{(x + 5)^2 + 6(x + 5)(x - 4)}{3(x - 4)^{2/3}} \\ &= \frac{(x + 5)[x + 5 + 6(x - 4)]}{3(x - 4)^{2/3}} = \frac{(x + 5)(7x - 19)}{3(x - 4)^{2/3}} \end{aligned}$$

Consequently, $f'(x) = 0$ if $x = -5$ or $x = \frac{19}{7}$. The derivative $f'(x)$ does not exist at $x = 4$. Thus f has three critical numbers, namely -5 , $\frac{19}{7}$ and 4 .

Example: If $f(x) = 2 \sin x + \cos 2x$, find the critical numbers of f , that are in the interval $[0, 2\pi]$.

Solution:

$$f'(x) = 2 \cos x - 2 \sin 2x = 2 \cos x - 4 \sin x \cos x = 2 \cos x (1 - 2 \sin x)$$

The derivative exists for all x , and $f'(x) = 0$ if either $\sin x = \frac{1}{2}$ or $\cos x = 0$. Hence the critical numbers of f in the interval $[0, 2\pi]$ are $\frac{\pi}{6}$, $\frac{5\pi}{6}$, $\frac{\pi}{2}$ and $\frac{3\pi}{2}$.

Exercises

In Exercises 1-2 find the absolute maximum and minimum of f on the indicated closed interval.

1 $f(x) = 5 - 6x^2 - 2x^3; [-3, 1]$

2 $f(x) = 1 - x^{2/3}; [-1, 8]$

Find the critical numbers of the functions defined in Exercises 3-6

3 $f(x) = 4x^2 - 3x + 2$

4 $s(t) = 2t^3 + t^2 - 20t + 4$

5 $F(w) = w^4 - 32w$

6 $f(z) = \sqrt{z^2 - 16}$

Answers

1 $5; -3$ 2 $1; -3$ 3 $\frac{3}{8}$ 4 $\frac{5}{3}$ and -2 5 2 6 4 and -4 (not 0)

The First Derivative Test**Theorem:**

Let f be a function that is continuous on a closed interval $[a, b]$ and differentiable on the open interval (a, b)

- (i) If $f'(x) > 0$ for all x in (a, b) , then f is increasing on $[a, b]$.
- (ii) If $f'(x) < 0$ for all x in (a, b) , then f is decreasing on $[a, b]$.

Example:

If $f(x) = x^3 + x^2 - 5x - 5$, find the intervals on which f is increasing and the intervals on which f is decreasing, sketch the graph of f .

Solution:

$$f'(x) = 3x^2 + 2x - 5 = (3x + 5)(x - 1)$$

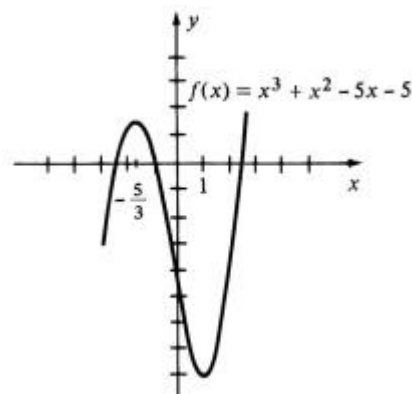
By theorem it is sufficient to find the intervals in which $f'(x) > 0$ and those in which $f'(x) < 0$. The factored form of $f'(x)$ and the critical numbers $-\frac{5}{3}$ and 1 suggest that we consider the open intervals $(-\infty, -\frac{5}{3})$, $(-\frac{5}{3}, 1)$, and $(1, \infty)$

Interval	k	Test value $f'(k)$	Sign of $f'(x)$	Variation of f
$(-\infty, -\frac{5}{3})$	-2	3	+	Increasing on $(-\infty, -\frac{5}{3}]$
$(-\frac{5}{3}, 1)$	0	-5	-	Decreasing on $[-\frac{5}{3}, 1]$
$(1, \infty)$	2	11	+	Increasing on $[1, \infty)$

$$\begin{aligned} f(x) &= x^3 + x^2 - 5x - 5 = x^2(x + 1) - 5(x + 1) \\ &= (x^2 - 5)(x + 1) \end{aligned}$$

$$f\left(-\frac{5}{3}\right) = \frac{40}{27}, \quad f(1) = -8$$

Hence the x - intercepts of the graph are $\sqrt{5}$, $-\sqrt{5}$ and -1 , The y -intercept is $f(0) = -5$



The First Derivative Test: Suppose c is a critical number of a function f and (a, b) is an open interval containing c . Suppose further that f is continuous on $[a, b]$ and differentiable on (a, b) , except possibly at c .

- (i) If $f'(x) > 0$ for $a < x < c$ and $f'(x) < 0$ for $c < x < b$ then $f(c)$ is a local maximum of f .
- (ii) If $f'(x) < 0$ for $a < x < c$ and $f'(x) > 0$ for $c < x < b$ then $f(c)$ is a local minimum of f .
- (iii) If $f'(x) > 0$ or $f'(x) < 0$ for all x in (a, b) except $= c$, then $f(c)$ is not a local extremum of f .

Example:

Find the local extrema of f if $f(x) = x^3 + x^2 - 5x - 5$.

Solution:

The critical numbers of f are $\frac{-5}{3}$ and 1 . We see from the table in previous example that the sign of $f'(x)$ changes from positive to negative as x increases through $\frac{-5}{3}$.

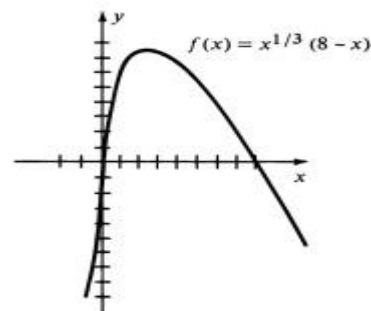
Hence, by the First Derivative Test, f has a local maximum at $\frac{-5}{3}$. This maximum value is $f\left(\frac{-5}{3}\right) = \frac{40}{27}$. Similarly, a local minimum occurs at 1 since the sign of $f'(x)$ changes from negative to positive as x increases through 1 . This minimum value is $f(1) = -8$.

Example: Find the local maxima and minima of f if $f(x) = x^{\frac{1}{3}}(8 - x)$.

Solution:

$$f'(x) = x^{\frac{1}{3}}(-1) + (8 - x)\frac{1}{3}x^{-\frac{2}{3}} = \frac{-3x+8-x}{3x^{2/3}} = \frac{4(2-x)}{3x^{2/3}}$$

The critical numbers of f are 0 and 2



Interval	k	Test value $f'(k)$	Sign of $f'(x)$	Variation of f
$(-\infty, 0)$	-1	4	$+$	Increasing on $(-\infty, 0]$
$(0, 2)$	1	$4/3$	$+$	Increasing on $[0, 2]$
$(2, \infty)$	8	-2	$-$	Decreasing on $[2, \infty)$

By the First Derivative Test, f has a local maximum at 2 since the sign of $f'(x)$ changes from $+$ to $-$ as x increases through 2. This maximum value is $f(2) = 6\sqrt[3]{2} \approx 7.6$. The function does not have an extremum at 0 since the sign of $f'(x)$ does not change as x increases through 0.

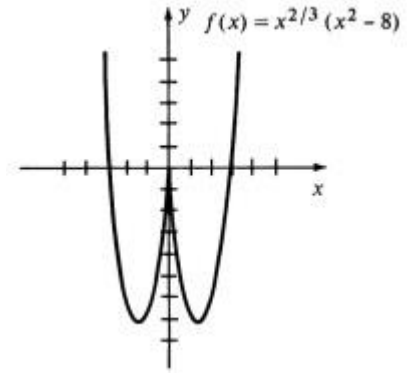
Example: If $f(x) = x^{2/3}(x^2 - 8)$

- 1) Find the local extrema of f , sketch the graph of f .
- 2) Find the absolute maximum and minimum values of f in each of the following intervals: a) $\left[-1, \frac{1}{2}\right]$, b) $[-1, 3]$, c) $[-3, -2]$

Solution:

$$1) f'(x) = x^{2/3}(2x) + \frac{2}{3}x^{-1/3}(x^2 - 8) = \frac{6x^2 + 2(x^2 - 8)}{3x^{1/3}} = \frac{8(x^2 - 2)}{3x^{1/3}}$$

The critical numbers of f are $-\sqrt{2}, 0$ and $\sqrt{2}$



Interval	k	Test value $f'(k)$	Sign of $f'(x)$	Variation of f
$(-\infty, -\sqrt{2})$	-8	$-248/3$	$-$	Decreasing on $(-\infty, -\sqrt{2}]$
$(-\sqrt{2}, 0)$	-1	$8/3$	$+$	Increasing on $[-\sqrt{2}, 0]$
$(0, \sqrt{2})$	1	$-8/3$	$-$	Decreasing on $[0, \sqrt{2}]$
$(\sqrt{2}, \infty)$	8	$248/3$	$+$	Increasing on $[\sqrt{2}, \infty)$

By the First Derivative Test, f has a local minimum at $-\sqrt{2}$ and $\sqrt{2}$ and a local maximum at 0. The corresponding functional values are $f(0) = 0$ and

$$f(\sqrt{2}) = -6\sqrt[3]{2} = f(-\sqrt{2})$$

2)

Interval	Minimum	Maximum
$\left[-1, \frac{1}{2}\right]$	$f(-1) = -7$	$f(0) = 0$
$[-1, 3]$	$f(\sqrt{2}) = -6\sqrt[3]{2}$	$f(3) = \sqrt[3]{9}$
$[-3, -2]$	$f(-2) = -4\sqrt[3]{4}$	$f(-3) = \sqrt[3]{9}$

Exercises

In Exercises 1–8 find the local extrema of f . Describe the intervals in which f is increasing or decreasing.

- 1 $f(x) = 5 - 7x - 4x^2$
- 2 $f(x) = 2x^3 + x^2 - 20x + 1$
- 3 $f(x) = x^4 - 8x^2 + 1$
- 4 $f(x) = x^{4/3} + 4x^{1/3}$
- 5 $f(x) = x^2\sqrt[3]{x^2 - 4}$
- 6 $f(x) = x^{2/3}(x - 7)^2 + 2$
- 7 $f(x) = x^3 + (3/x)$
- 8 $f(x) = 10x^3(x - 1)^2$

In Exercises 9–11 find the local extrema of f on the interval $[0, 2\pi]$, and determine where f is increasing or decreasing on $[0, 2\pi]$.

- 9 $f(x) = \cos x + \sin x$
- 10 $f(x) = \frac{1}{2}x - \sin x$
- 11 $f(x) = 2 \cos x + \sin 2x$

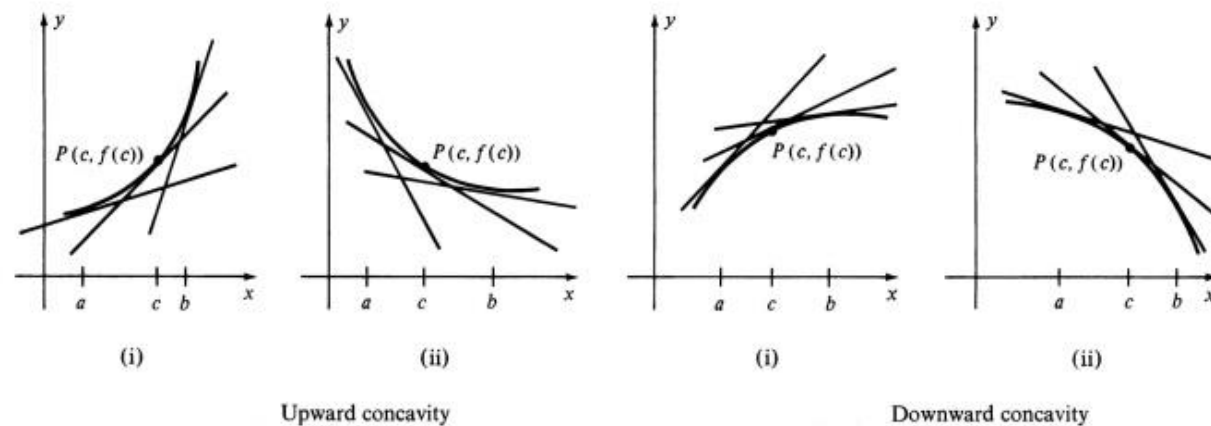
Answers

- 1 Max: $f(-\frac{7}{8}) = \frac{129}{16}$; increasing on $(-\infty, -\frac{7}{8}]$; decreasing on $[-\frac{7}{8}, \infty)$
- 2 Max: $f(-2) = 29$; min: $f(\frac{5}{3}) = -\frac{548}{27}$; increasing on $(-\infty, -2]$ and $[\frac{5}{3}, \infty)$; decreasing on $[-2, \frac{5}{3}]$
- 3 Max: $f(0) = 1$; min: $f(-2) = -15$; min: $f(2) = -15$; increasing on $[-2, 0]$ and $[2, \infty)$; decreasing on $(-\infty, -2]$ and $[0, 2]$
- 4 Min: $f(-1) = -3$; increasing on $[-1, \infty)$; decreasing on $(-\infty, -1]$
- 5 Max: $f(0) = 0$; min: $f(-\sqrt{3}) = f(\sqrt{3}) = -3$; increasing on $[-\sqrt{3}, 0]$ and $[\sqrt{3}, \infty)$; decreasing on $(-\infty, -\sqrt{3}]$ and $[0, \sqrt{3}]$
- 6 Max: $f(\frac{7}{4}) \approx 42$; min: $f(0) = 2$; min: $f(7) = 2$; increasing on $[0, \frac{7}{4}]$ and $[7, \infty)$; decreasing on $(-\infty, 0]$ and $[\frac{7}{4}, 7]$
- 7 Max: $f(-1) = -4$; min: $f(1) = 4$; increasing on $(-\infty, -1]$ and $[1, \infty)$; decreasing on $[-1, 0]$ and $(0, 1]$
- 8 Max: $f(\frac{3}{5}) \approx 0.346$; min: $f(1) = 0$; increasing on $(-\infty, \frac{3}{5}]$ and $[1, \infty)$; decreasing on $[\frac{3}{5}, 1]$
- 9 Max: $f(\pi/4) = \sqrt{2}$; min: $f(5\pi/4) = -\sqrt{2}$; increasing on $[0, \pi/4]$ and $[5\pi/4, 2\pi]$; decreasing on $[\pi/4, 5\pi/4]$
- 10 Min: $f(\pi/3) = (\pi - 3\sqrt{3})/6$; max: $f(5\pi/3) = (5\pi + 3\sqrt{3})/6$; decreasing on $[0, \pi/3]$ and $[5\pi/3, 2\pi]$; increasing on $[\pi/3, 5\pi/3]$
- 11 Max: $f(\pi/6) = 3\sqrt{3}/2$; min: $f(5\pi/6) = -3\sqrt{3}/2$; increasing on $[0, \pi/6]$ and $[5\pi/6, 2\pi]$; decreasing on $[\pi/6, 5\pi/6]$

Concavity and the Second Derivative Test

Def: Let f be a function that is differentiable at a number c .

- (i) The graph of f is **concave upward** at the point $P(c, f(c))$ if there exists an open interval (a, b) containing c such that on (a, b) the graph of f is above the tangent line through P .
- (ii) The graph of f is **concave downward** at the point $P(c, f(c))$ if there exists an open interval (a, b) containing c such that on (a, b) the graph of f is below the tangent line through P .

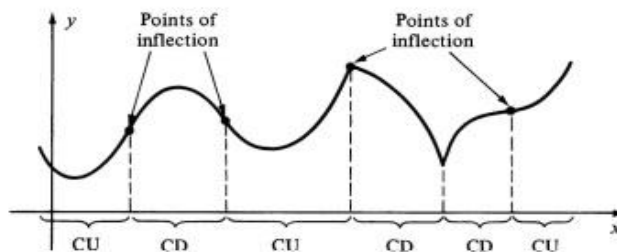


Test for Concavity: Suppose a function f is differentiable on an open interval containing c and $f''(c)$ exists.

- (i) If $f''(c) > 0$ the graph is concave upward at $P(c, f(c))$.
- (ii) If $f''(c) < 0$ the graph is concave downward at $P(c, f(c))$.

Def: A point $P(k, f(k))$ on the graph of a function f is a **point of inflection** if there exists an open interval (a, b) containing k such that one of the following statements holds.

- (i) $f''(x) > 0$ if $a < x < k$ and $f''(x) < 0$ if $k < x < b$; or
- (ii) $f''(x) < 0$ if $a < x < k$ and $f''(x) > 0$ if $k < x < b$



Example: if $f(x) = x^3 + x^2 - 5x - 5$ determine intervals on which the graph of f is concave upward and intervals on which the graph of f is concave downward.

Solution:

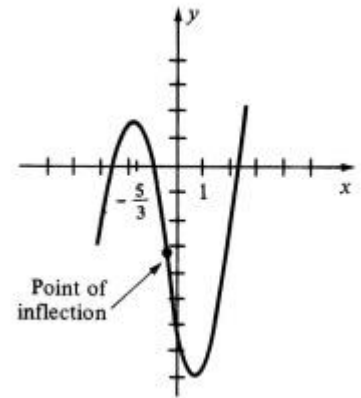
$$f''(x) = 6x + 2 = 2(3x + 1)$$

Hence $f''(x) < 0$ if $3x + 1 < 0$, that is, if $x < -\frac{1}{3}$. it follows from the Test for

Concavity that the graph is concave downward on the infinite interval $(-\infty, -\frac{1}{3})$.

Similarly $f''(x) > 0$ if $x > -\frac{1}{3}$. and, therefore the graph is concave upward on $(-\frac{1}{3}, \infty)$.

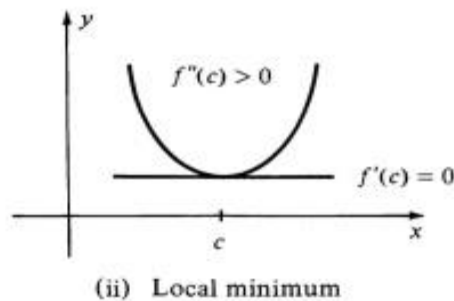
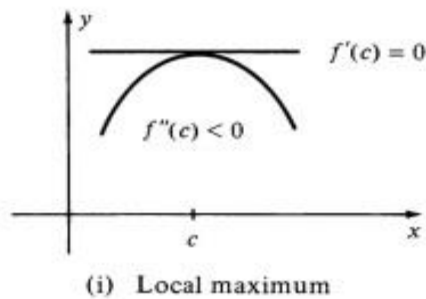
The point $P(-\frac{1}{3}, \frac{88}{27})$ at which $f''(x)$ changes sign (and the concavity changes from downward to upward) is a point of inflection.



The Second Derivative Test: Suppose a function f is differentiable on an open interval containing c and $f'(c) = 0$.

(i) If $f''(c) < 0$, then f has a local maximum at c .

(ii) If $f''(c) > 0$, then f has a local minimum at c .



Example: If $f(x) = 12 + 2x^2 - x^4$, use the Second Derivative Test to find local maxima and minima of f . Discuss concavity, find the points of inflection, and sketch the graph of f .

Solution:

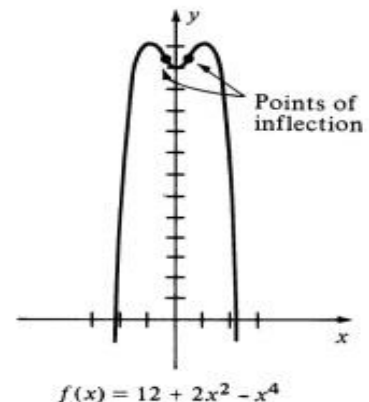
$$f'(x) = 4x - 4x^3 = 4x(1 - x^2)$$

$$f''(x) = 4 - 12x^2 = 4(1 - 3x^2)$$

The expression for $f'(x)$ is used to find the critical numbers 0, 1, and -1. The values of f'' at these numbers are

$$f''(0) = 4 > 0 \quad , \quad f''(1) = -8 < 0 \quad , \quad f''(-1) = -8 < 0$$

Hence, by the Second Derivative Test there is a local minimum at 0 and local maxima at 1, and -1. The corresponding functional values are $f(0) = 12$, and



$$f(1) = 13 = f(-1)$$

Critical number c	$f''(c)$	Sign of $f''(c)$	Conclusion
-1	-8	-	Local max: $f(-1) = 13$
0	4	+	Local min: $f(0) = 12$
1	-8	-	Local max: $f(1) = 13$

To locate the possible points of inflection we solve the equation $f''(x) = 0$, that is, $4(1 - 3x^2) = 0$. Evidently, the solution are $-\sqrt{3}/3$ and $\sqrt{3}/3$

Interval	k	Test value $f''(k)$	Sign of $f''(x)$	Concavity
$(-\infty, -\sqrt{3}/3)$	-1	-8	-	Downward
$(-\sqrt{3}/3, \sqrt{3}/3)$	0	4	+	Upward
$(\sqrt{3}/3, \infty)$	1	-8	-	Downward

The points $(\pm\sqrt{3}/3, 113/9)$ are points of inflection

Example: If $f(x) = 2 \sin x + \cos 2x$, find local extrema of f on the interval $[0, 2\pi]$.

Solution:

$$f'(x) = 2 \cos x - 2 \sin 2x$$

$$f''(x) = -2 \sin x - 4 \cos 2x$$

The critical numbers of f in the interval $[0, 2\pi]$ are $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{\pi}{2}$ and $\frac{3\pi}{2}$.

$$f''\left(\frac{\pi}{6}\right) = -3, \quad f''\left(\frac{5\pi}{6}\right) = -3, \quad f''\left(\frac{\pi}{2}\right) = 2, \quad f''\left(\frac{3\pi}{2}\right) = 6$$

The local maxima are $\frac{\pi}{6}$ and $\frac{5\pi}{6}$, and the local minima are $\frac{\pi}{2}$ and $\frac{3\pi}{2}$.

Exercises

In Exercises 1–9, use the Second Derivative Test (whenever applicable) to find the local extrema of f . Discuss concavity, find x -coordinates of points of inflection, and sketch the graph of f .

1 $f(x) = x^3 - 2x^2 + x + 1$

2 $f(x) = 3x^4 - 4x^3 + 6$

3 $f(x) = 2x^6 - 6x^4$

4 $f(x) = (x^2 - 1)^2$

5 $f(x) = \sqrt[5]{x} - 1$

6 $f(x) = x^2 - (27/x^2)$

7 $f(x) = x/(x^2 + 1)$

8 $f(x) = \sqrt[3]{x^2}(3x + 10)$

9 $f(x) = 8x^{1/3} + x^{4/3}$

Answers

In Exercises 1–9 the notations CU and CD mean that the graph is concave upward or downward, respectively, in the interval that follows. PI denotes point(s) of inflection.

1 Max: $f(\frac{1}{3}) = \frac{31}{27}$; min: $f(1) = 1$. CD on $(-\infty, \frac{2}{3})$; CU on $(\frac{2}{3}, \infty)$; x -coordinate of PI is $\frac{2}{3}$.2 Min: $f(1) = 5$; CU on $(-\infty, 0)$ and $(\frac{2}{3}, \infty)$; CD on $(0, \frac{2}{3})$; x -coordinates of PI are 0 and $\frac{2}{3}$.3 Max: $f(0) = 0$ (by first derivative test); min: $f(-\sqrt{2}) = f(\sqrt{2}) = -8$; CU on $(-\infty, -\sqrt{\frac{6}{5}})$ and $(\sqrt{\frac{6}{5}}, \infty)$; CD on $(-\sqrt{\frac{6}{5}}, \sqrt{\frac{6}{5}})$; x -coordinates of PI are $\pm\sqrt{\frac{6}{5}}$.4 Max: $f(0) = 1$; min: $f(-1) = f(1) = 0$. CU on $(-\infty, -1/\sqrt{3})$ and $(1/\sqrt{3}, \infty)$; CD on $(-1/\sqrt{3}, 1/\sqrt{3})$; x -coordinates of PI are $\pm 1/\sqrt{3}$.5 No local extrema; CU on $(-\infty, 0)$; CD on $(0, \infty)$; PI $(0, -1)$ 6 No max or min. CU on $(-\infty, -3)$ and $(3, \infty)$; CD on $(-3, 0)$ and $(0, 3)$; x -coordinates of PI are ± 3 .7 Min: $f(-1) = -\frac{1}{2}$; max: $f(1) = \frac{1}{2}$. CU on $(-\sqrt{3}, 0)$ and $(\sqrt{3}, \infty)$; CD on $(-\infty, -\sqrt{3})$ and $(0, \sqrt{3})$; x -coordinates of PI are $0, \pm\sqrt{3}$.8 Max: $f(-\frac{4}{3}) \approx 7.27$; min: $f(0) = 0$; CD on $(-\infty, 0)$ and $(0, \frac{2}{3})$; CU on $(\frac{2}{3}, \infty)$; PI is $(\frac{2}{3}, 10\sqrt[3]{12}/3)$.9 Min: $f(-2) \approx -7.55$; CU on $(-\infty, 0)$ and $(4, \infty)$; CD on $(0, 4)$; x -coordinates of PI are 0 and 4.

Strategy for Graphing $y = f(x)$

1. Identify the domain of f and any symmetries the curve may have.
2. Find y' and y'' .
3. Find the critical points of f , and identify the function's behavior at each one.
4. Find where the curve is increasing and where it is decreasing.
5. Find the points of inflection, if any occur, and determine the concavity of the curve.
6. Identify any asymptote.
7. Plot key points, such as the intercepts and the points found in Steps 3-5, and sketch the curve.

Example: Sketch the graph $(x) = \frac{2x^2}{9-x^2}$. (using the graphing strategy)

Solution:

1. $D_f = \mathbb{R}/\{-3,3\}$

$f(-x) = \frac{2(-x)^2}{9-(-x)^2} = \frac{2x^2}{9-x^2} = f(x)$ the graph is symmetric with respect to the y - axis.

2. $f'(x) = \frac{4x(9-x^2)-2x^2(-2x)}{[9-x^2]^2} = \frac{36x-4x^3+4x^3}{[9-x^2]^2} = \frac{36x}{[9-x^2]^2}$

$f''(x) = \frac{36(9-x^2)^2 - (36x)(2)(9-x^2)}{[9-x^2]^4} = \frac{(9-x^2)[36(9-x^2) + 144x^2]}{[9-x^2]^4} = \frac{324 + 108x^2}{[9-x^2]^3} = \frac{108(x^2 + 3)}{[9-x^2]^3}$

3. The critical points of f is 0, -3 and 3 are not the critical numbers because $-3, 3 \notin D_f$

Since $f''(0) = \frac{324}{729} = \frac{4}{9} > 0$, then the local minima at 0

4.

Interval	k	Test value $f'(k)$	Sign of $f'(x)$	Variation of f
$(-\infty, 0)$	-1	$-9/16$	$-$	Decreasing on $(-\infty, 0]$
$(0, \infty)$	1	$9/16$	$+$	Increasing on $[0, \infty)$

5. Since $f''(0) = \frac{4}{9} > 0$, then the graph is concave upward at $P(0, \frac{4}{9})$

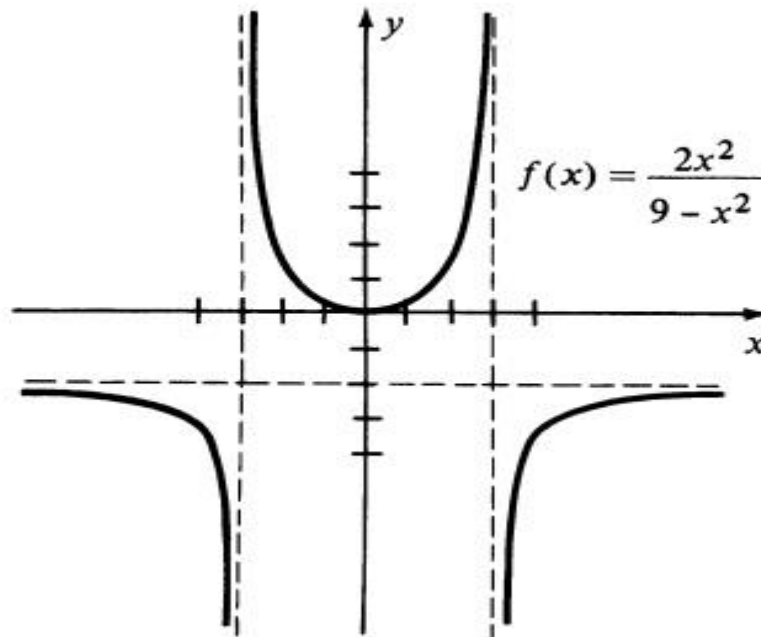
There is no inflection points because $-3, 3 \notin D_f$

6. Since $\lim_{x \rightarrow \pm\infty} \frac{2x^2}{9-x^2} = -2$, then the line $y = -2$ is a horizontal asymptote.

Since $\lim_{x \rightarrow 3^-} \frac{2x^2}{9-x^2} = \infty$, $\lim_{x \rightarrow 3^+} \frac{2x^2}{9-x^2} = -\infty$

$\lim_{x \rightarrow -3^-} \frac{2x^2}{9-x^2} = -\infty$, $\lim_{x \rightarrow -3^+} \frac{2x^2}{9-x^2} = \infty$

Then the lines $x = 3$ and $x = -3$ are a vertical asymptotes.



Exercises:

In exercises (1- 3) Sketch the following graph. (using the graphing strategy)

1) $f(x) = \frac{x^2-2x+4}{x-2}$

2) $f(x) = \frac{x^2}{x^2-x-2}$

3) $f(x) = \frac{x^2-9}{2x-4}$

